اتحاد طلبة هندسة الطاقة والاتصالات

pce

أساسيات القيادة الكهربائية

الطبعة الاول

Dont miss the change.. be part of it

رئيس الجمعية الطلابية :قصي حيالات

Groups/pce.fet

PCE_FET

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بسم الله الرحمن الرحيم

Subject: Electric drive system

- Definitions (page "1").
- The block diagram and components (2-8).
- Functions and advantages (9).
- Classification (10).

Subject: Electric drive sytem characteristics

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Mechanical char for industrial machineries (Load char) (13-15).
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Subject: Dynamic of electric drive system

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- Transient periods (23-24)
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↙ drive system

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Electric drive system

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Electric drive system

Subject: Automatic control of DC & AC motors

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SUBJECT Electric drive system

SECTION Definitions

* Definetions of drive system :-

- A system use electrical power to produce mechanical power.

- A system used to generat, transmit & control the mechanical power to perform a usful work.

- Definetion in respect to energy conversion ⇒ is an electro mechanical system used to transmit or supply motion to working mechanism & control the operation of the machine (Load)

* If the prime mover is an electrical motor then, the drive system is called "electrical drive system"

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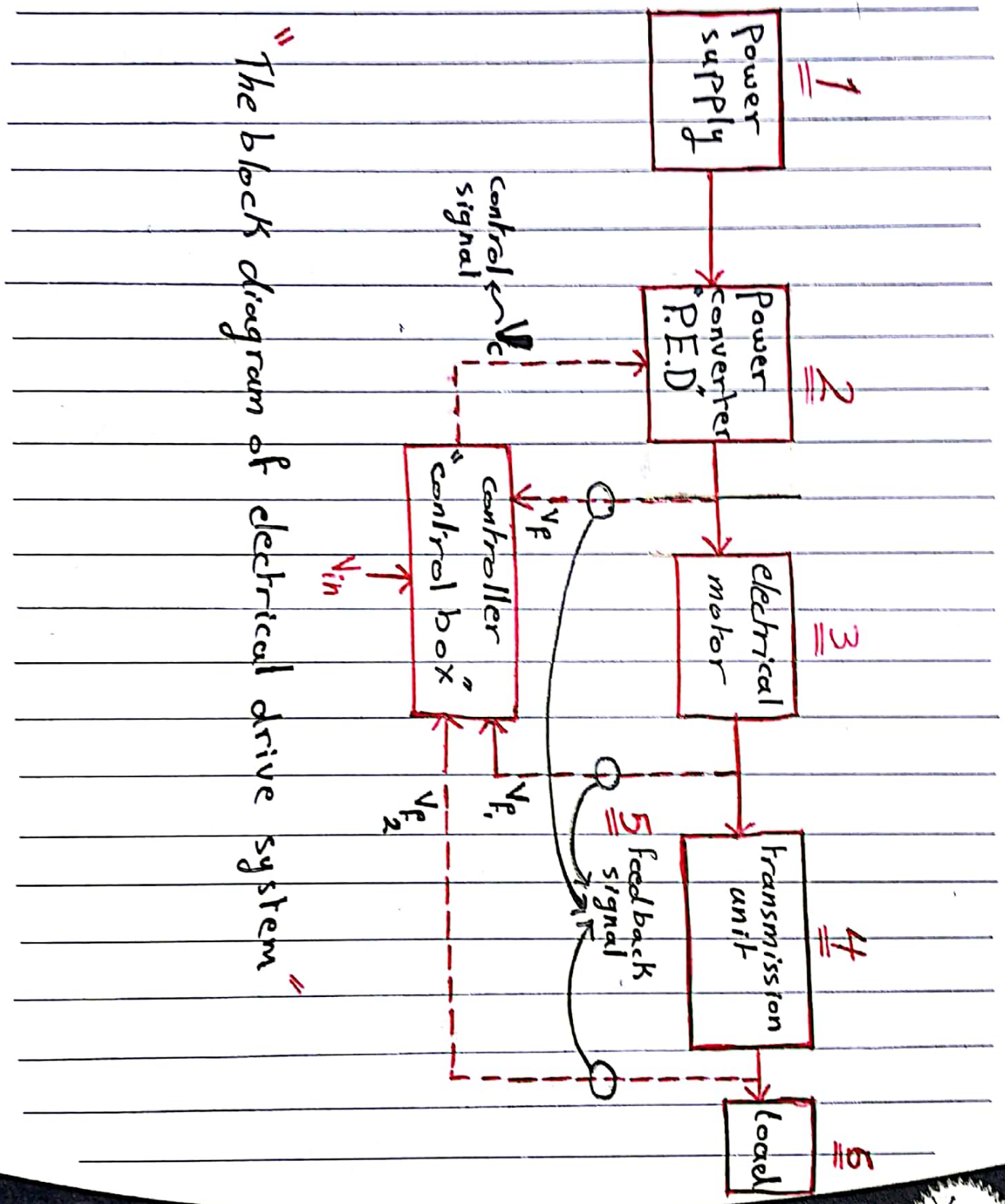
SUBJECT Electric drive system

SECTION The block diagram & components.



"The block diagram of electrical drive system"

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SUBJECT Electric drive system

SECTION The block diagram & components

[1] power supply

supply the system with electrical power

تختلف نوع المحرك باختلاف نوع الطاقة المستخدمة:-

1- DC power supply.
2- AC power supply. → 1 phase
                     → 3 phase
3- Solar energy.
4- Wind energy.
5- Accumulator. "بطاريات"
6- Diesel generator

[2] Controller

* to control the electrical drive system and do the work of start, break, and revers.
* Control system contains:
  - Diodes.
  - Thyristors.
  - Transistors.
  - Relays.
  - Contactors.
  - push buttons.
  - Transformers.
  - Signal lamps.
  - Push buttons
  - بالإضافة الى وسائل حماية مختلفة (مثل حماية من U/V & O/C)
    U/V ← under voltage
    O/C ← over current

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SUBJECT Electric drive system

SECTION The block diagram & components

[3] Power Converter

تحويل الطاقة الكهربائية إلى طاقة كهربائية بمعطيات جديدة.

*Types:

1- Ac voltage controller (Ac → Ac). $V_1 \leq V_{in}$ & $f_1 = f_{in}$

يتغير الجهد ويبقى التردد ثابت.

2- Frequency converter (Ac → Ac). V_1 → variable.
f_1 → variable.

3- Rectifier (Ac → Dc) → controlled or uncontrolled.
→ single phase or 3 phase.
→ half or full wave.

4- Inverter (Dc → Ac).

5- Chopper (DC → Dc).

[4] Electrical motor

$P_1 = VI$ → "converts electrical power into mechanical power" ← $P_2 = T\omega = FV$

*Types of electrical motor is what difines the type of the "drive system"

① DC motors → A) seperately motors.
→ B) series motors.
→ C) compound motors.

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SUBJECT Electric drive system

SECTION The block diagram & components.

② AC motors → A) Synchronous motors

- $\omega_s = \omega_L = \frac{2\pi f_1}{P/2}$ ⇒ constant
- T_L ⇒ variable

→ B) Asynchronous motors

- ↳1) squiral cage
- ↳2) wound - slip rings
- ↳3) stepper motor
- ↳4) linear motor

تقل سرعة الروتر مع زيادة الحمل ← سرعة المجال ≠ سرعة الحمل.

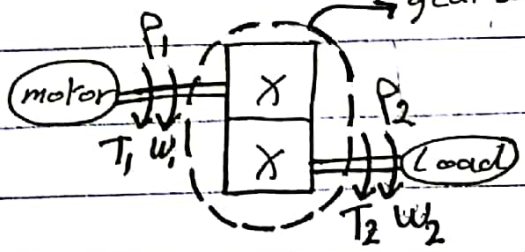
تقسم حسب عدد الأطوار إلى أحادية وثلاثية الطور ← كلما زاد عدد الأطوار تكون جودة الماكينة أحسن

5 Transmission unit [MTS]

* mechanical transmission system (MTS).

"Convert mechanical power to mechanical power with stepping up or down the speed"

- If there is MTS in the system ⇒ the system is called "gear system"
- If there isn't MTS " " " ⇒ " " " " "gearless system"



$$\eta = \frac{P_2}{P_1} < 1$$

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SUBJECT Electric drive system

SECTION The block diagram & components

* في أنظمة القيادة يتم استخدام MTS لخفض السرعة فقط.
اي زيادة العزم ← وبذلك يمكن اختيار محرك كهربائي بسرعة عالية وعزم قليل أي بمقاييس وسعر أقل.

power = torque * rotational speed

$P = T * \omega$ ⇒ (when $\omega \downarrow \Rightarrow T \uparrow$) & vice versa

* أنواع النواقل:-

1) Gear Box (ناقل يستخدم المسننات)

$$i = \frac{Z_2}{Z_1} = \frac{n_1}{n_2}$$

where ⇒ (i) is the gear ratio, يجب أن يكون عدد صحيح
(Z) is no of tooths.
(n) is the speed.
$(n_1 > n_2)$ ⇝ to step down speed

2) sheaves (يستخدم البكرات)
يستخدم لنقل الحركة، على مسافات متباعدة بين أعمدة الدوران
باستخدام Belt (سير) or chains (جنازير) drive ⇐

⇒

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SUBJECT Electric drive system

SECTION The block diagram & components.

وتكون السرعة الخطية واحدة على جميع أجزاء السير أو الجنزير ($V_1 = V_2$)

وذلك حتى لا ينقطع السير أو الجنزير

$V_1 = V_2$ & $V = \omega R$ ← وهذا يعني

so $\Rightarrow \omega_1 R_1 = \omega_2 R_2 = constant$

وهذا يعني ← زيادة نصف قطر البكرة (R) يؤدي إلى تقليل السرعة الزاوية (ω)

* Functions of MTS in drive systems:-

1- step down the speed.
2- multiply the torque "increas load".
3- reverse the direction of motion.
4- easy to connect the load.

6 Feed backs

- To reduce overshoot.
- // // error.
- // // the effect of noise.
- // make the system more stable.

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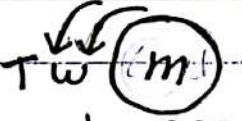
SUBJECT Electric drive system

SECTION The block diagram & Components

* Rotational and translational motion:-

Rotational motion



$P = T\omega$; ω is the angula speed (rad/sec)

$T = \frac{1}{2} g \omega^2$; g is the moment of inertia ($Kg.m^2$)
(the gain of Kinetic energy)

Translational motion or ⟺ m

$P = Fv$; $F = m\ddot{x}$, $\ddot{x}$ is the acceleration

$T = \frac{1}{2} m v^2$

[7] Loads (Mechanical loads)

* Categories in two types, depend on direction:

A- undirectional loads (active)

the load torque is independant of the rotating direction or speed.

B- Passive loads (directional loads)

of Industrial machineries: page 13 & 14 & 15 mech char

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SUBJECT Electric drive system

SECTION Functions & advantages

* Main functions of electric drive system :-

1- provide transisio. operation (start, brak, reverse) for electrical motors.

2- speed control (control system can be close or open) التحكم بالسرعة مع المحافظة عليها وبنسبة خطأ ستاتيكي معين.

3- Tracking (servo drive) - positioning drive control

4- Adaptive control ⇒ "the drive system changes it's structure or algorithm to achieve maximum efficiency and best operation"

5- protection ⇒ "shut down ~~of~~ the system in emergencies or short circuit faults"

* Advantages of electric drive system:-

1- Flexible control.
2- High efficiency.
3- low noise.
4- low maintainance.
5- wide rang of speed, torque & power.

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SUBJECT Electric drive system

SECTION classification

classification

* of drive system due to "power supply" :-

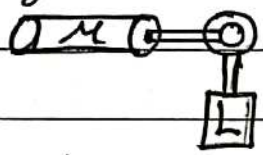
- If the motor is AC motor ⇒ "AC drive system"

- If the motor is DC motor ⇒ "DC drive system"

classification

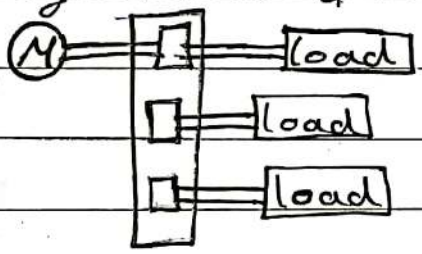
* of drive system due to "load" :-

- single motor & single load ⇒ "individual DS"

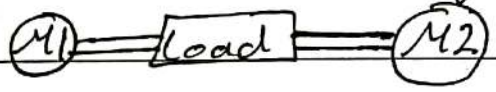


drive system

- single motor & mor than one load ⇒ "line-shaft DS"



- Multi motor & single load ⇒ "group DS"



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SUBJECT E.D.S characteristics

SECTION Motor characteristics

* خواص وعمل خواص المحرك هي التي تحدد نوع نظام القيادة.

(A) Motor characteristics:

↳ I) Static char (steady stat) → natural, regulated → mechanical char, electr mechanical "

↳ Dynamic char

II) Static characteristics:

{n = f(T) or ω = f(T)} ⇒ speed is a function of torque

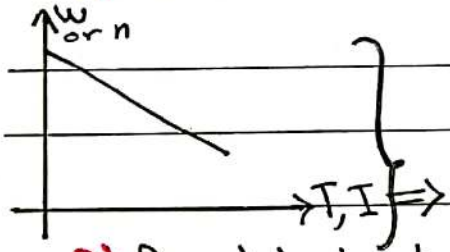
"mechanical char"

mechanical char ≡ torque-speed char

{n = f(I) or ω = f(I)} ⇒ speed is a function of current

"electro mechanical char"

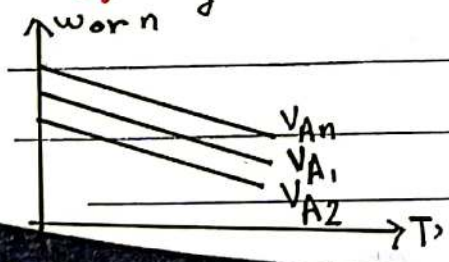
A) Natural char



"دون إضافة مقاومات أو أي تغيير معتمد لمعطيات المحرك"

⇒ separatly excited DC motor :مثال

B) Regulated char



مع التغيير المتعمد لإحدى معطيات المحرك أو إضافة مقاومات مادية لدارات المحرك.

$V_{AN} > V_{A1} > V_{A2}$

مثال تغيير V ولكن ليس أثناء عمل التنظيم.

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SUBJECT E.D.S characteristics

SECTION Motor characteristics.

[2] Dynamic characteristics :

تعبر عن عمل نظام المحرك في الحالات العابرة مثل

starting, braking, reversing

$w = f(t)$ & $T = f(t)$ & $I = f(t)$

(w, T, & I ⇒ are functions of time)

يتم دراسة الحالة العابرة باستخدام معادلة الحركة :-

$$T_M - T_L = J_\varepsilon \frac{dw}{dt}$$

T_M ⇒ motor torque

T_L ⇒ load torque

J_ε ⇒ total moment of inertia

* Modes of operation :

[1] when $T_M = T_L \Rightarrow \frac{dw}{dt} = \text{zero} \Rightarrow$ w is constant

(system is in steady state)

[2] when $T_M < T_L \Rightarrow \frac{dw}{dt} < \text{zero} \Rightarrow$ deaccelaration

[3] when $T_M > T_L \Rightarrow \frac{dw}{dt} > \text{zero} \Rightarrow$ accelaration

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SUBJECT E.D.S char

SECTION Mechanical char for industrial machineries

* خواص الحمل وخواص المحرك هي التي تحدد نوع النظام.

(B) mechanical char for industrial machineries:

$T_L = T_0 + (T_n - T_0)\left(\frac{w}{w_n}\right)^x$ ⇒ For typical industrial load

T_L ⇒ العزم الساكن للحمل

T_0 ⇒ العزم الابتدائي للاحتكاك في الآلة (constant)

T_n ⇒ القيمة الاسمية للعزم الساكن (constant)

w ⇒ السرعة الزاوية للآلة

w_n ⇒ السرعة الاسمية للآلة (constant)

x ⇒ معامل تحدد العلاقة بين تغير كل من السرعة والعزم حسب نوع الآلة.

* Industrial machineries are classified (due to value of x) into:

I. For practical cases ⇒ $T_0 \approx 0$ $(T_0 \ll T_n)$

so ⇒ $T_L = C \cdot T_n w^x$ ⇒ $\boxed{T_L \propto w^x}$ (C ↓ constant)

$P = T w \Rightarrow P = w^{(x+1)}$

II. العزم معتمد على السرعة / constant load drives / active loads

when $x = 0 \Rightarrow T_L \propto w^0 \Rightarrow T_L$ is constant

$\Rightarrow P \propto w$ ⇒

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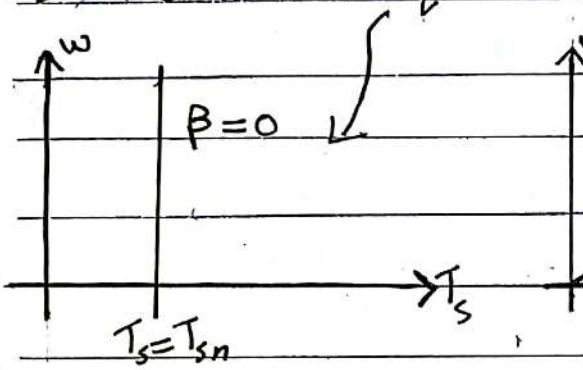
SUBJECT E.D.S char

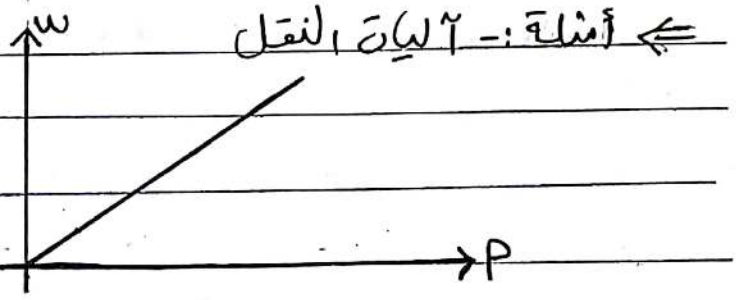
SECTION Mech char for industrial machineries

(تكون قيمة العزم ثابتة ولا تعتمد على "قيمة واتجاه" السرعة)

→ "the active torque"



⇐ أمثلة :- آليات النقل



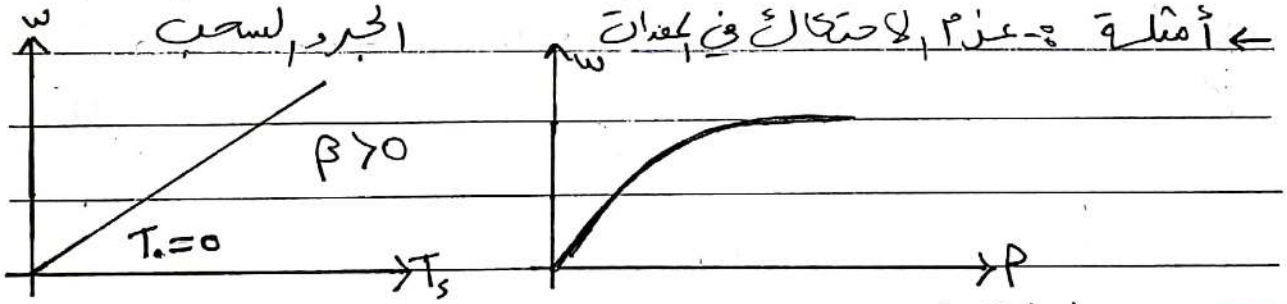
2 الخطية الصاعدة :-

when $x=1 \Rightarrow T \propto (w)^1 \Rightarrow P \propto w^2$

⇒ "reactive torque"

(تتغير قيمة واتجاه العزم بتغير قيمة واتجاه السرعة)

← أمثلة :- عزم الاحتكاك في المحامل

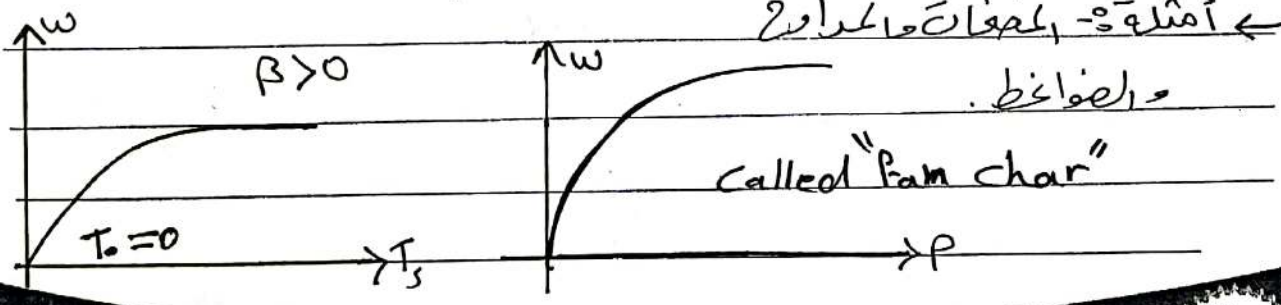


3 الغير خطية الصاعدة :-

when $x=2 \Rightarrow T \propto w^2 \Rightarrow P \propto w^3$

← الخاصية المروحية (النوع الثاني من reactive Torque)

← أمثلة :- المضخات والمراوح والضواغط.



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SUBJECT E.D.S chear

SECTION Mech char for industrial machineries

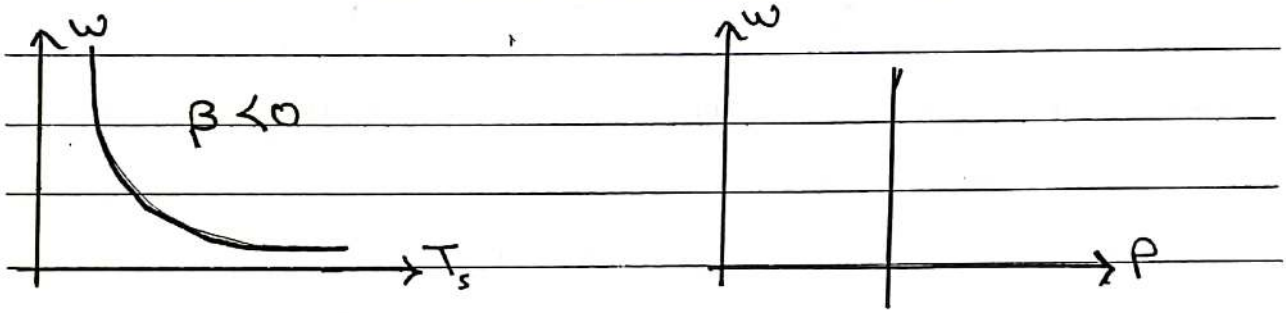
4 عند الخطية الطبيعية.

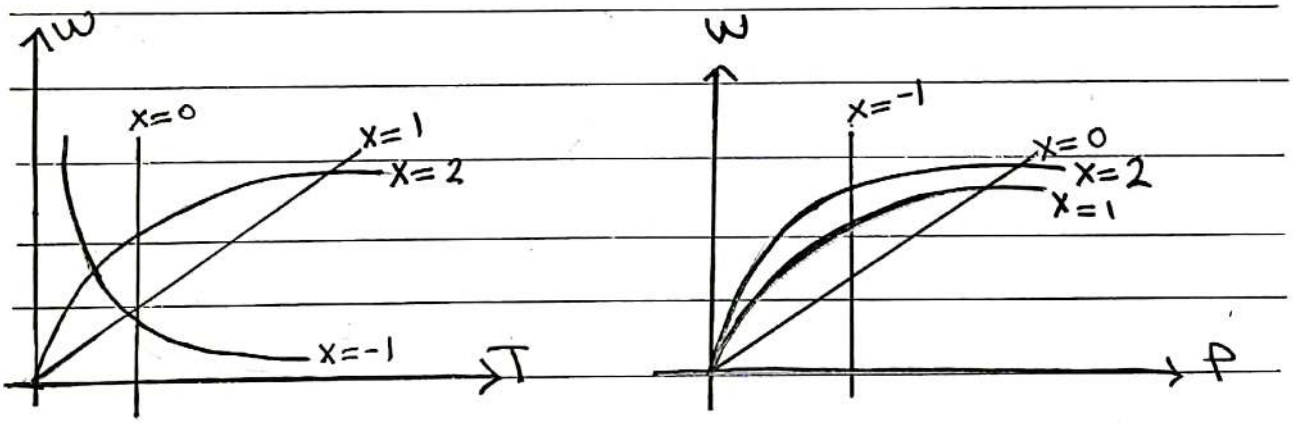
when $x=-1 \Rightarrow T \propto \frac{1}{w} \Rightarrow P(\text{constant}) \equiv \alpha w^0$

$\Rightarrow$ also areactive torque

العزم يتناسب عكسياً مع السرعة.

← أمثلة :- القاسط والمخارط والطاحون (تحتاج لعزم عالي وسرعات قليلة)





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SUBJECT E.D.S char

SECTION Hardness

* Hardness (β): is the ratio between the change of torque (ΔT) to chang of speed (Δw)

$$\left[\beta = \frac{\Delta T}{\Delta w}\right]$$ وتختلف هذا التغير من نوع آخر

β For motors → (always -), but β for loads (- or +)

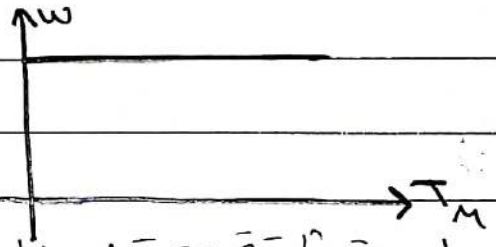
* Mechanical (torque-speed) char of motors have 3 types "due to hardness":

تقسم الخواص الميكانيكية للمحركات حسب الصلادة إلى:-

1- Absolutely hard mechanical char

Ex⇒ Synchronous motors

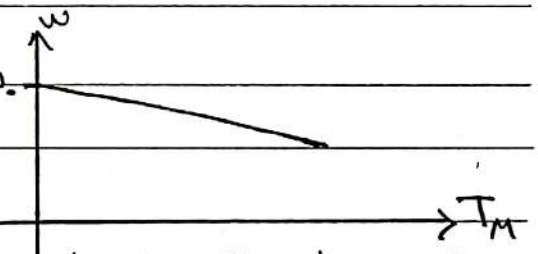
$$\beta = \frac{\Delta T_M}{\Delta w} = \infty \quad \Delta w \to \Delta w = 0$$



السرعة ثابتة مع تغير العزم.

2- Hard mechanical char

Ex⇒ DC shunt or separately motors



$$\Delta w < 0 \Rightarrow \beta = \frac{\Delta T_M}{\Delta w} < 0$$

تناقص السرعة بشكل خطي مع تغير العزم.

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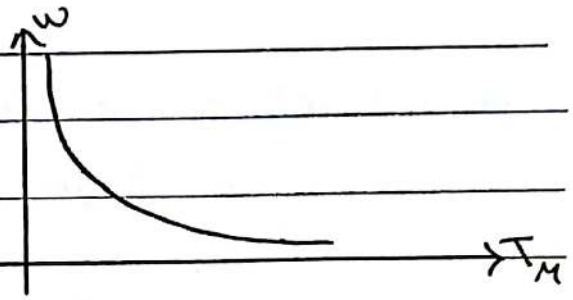
SUBJECT Dynamic of E.D.S

SECTION stability & stead state operating point

3- Soft mechanical char

EX⇒ series & compound motors

تتناقص السرعة بشكل كبير مع ازياد العزم.



$\Delta w << 0 \Rightarrow \beta < 0$

* كلما كان الهبوط في السرعة أقل (Dw small) ← تزداد قيمة β ↑↑

$$T_d = T_M - T_L = \frac{J}{\varepsilon} \frac{dw}{dt}$$

when dynamic torque (T_d) = zero ⇒ [the system is in steady state]

$T_d = 0 \Rightarrow T_M = T_L \Rightarrow$ change in speed due time is zero $\left(\frac{dw}{dt} = 0\right)$

* conditions of steady state $\left(T_M = T_L \ \& \ w = \underset{\text{(constant)}}{w_{ss}}\right)$

* dynamic torque $\left(T_d = J \frac{dw}{dt}\right)$ ⇒ only present during the transient operations

* we can find steady state operation point througe mechanical char for both motor ($T_M = F(w)$), and load ($T_L = F(w)$).

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SUBJECT Dynamic of E.D.S

SECTION Stability & S.S.O.P

* Stability :-

the system is said to be "stable", if the system is able to return to the SSOP (steady state operation point).

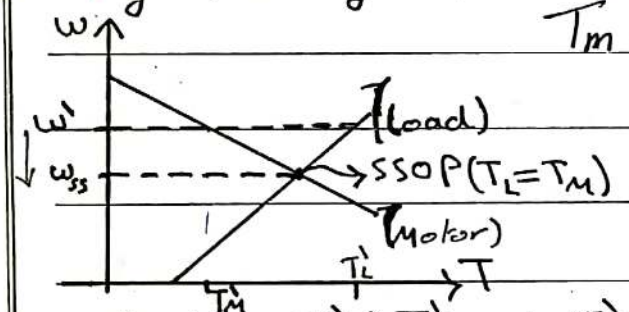
* For stable SSO $\Rightarrow$ $\boxed{\beta_L > \beta_M}$ شرط

لقيادة المحركات دائماً سالبة أما الحمل يمكن أن تكون سالبة أو موجبة. لكن حتى يكون النظام مستقر ($\beta_L > \beta_M$)

* To find SSOP & check the stability of the system:

1) graphically

* نقطة التقاطع هي SSOP $\leftarrow T_m = T_L$



at $w' \Leftrightarrow T'_L > T'_M \Rightarrow T'_M - T'_L < 0 \Rightarrow \frac{dw}{dt} < 0 \Rightarrow$ deaccelerating

أي أن السرعة تنخفض من w' إلى w_{ss} أي أن النظام مستقر (Stable)

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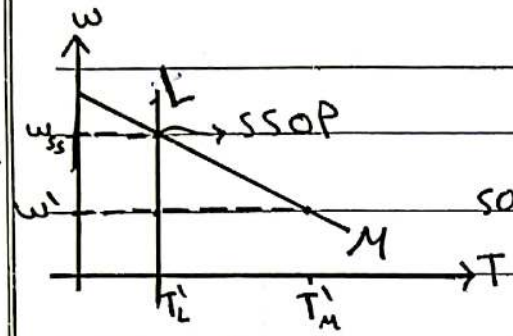
SUBJECT Dynamic of E.D.S

SECTION stability & SSOP

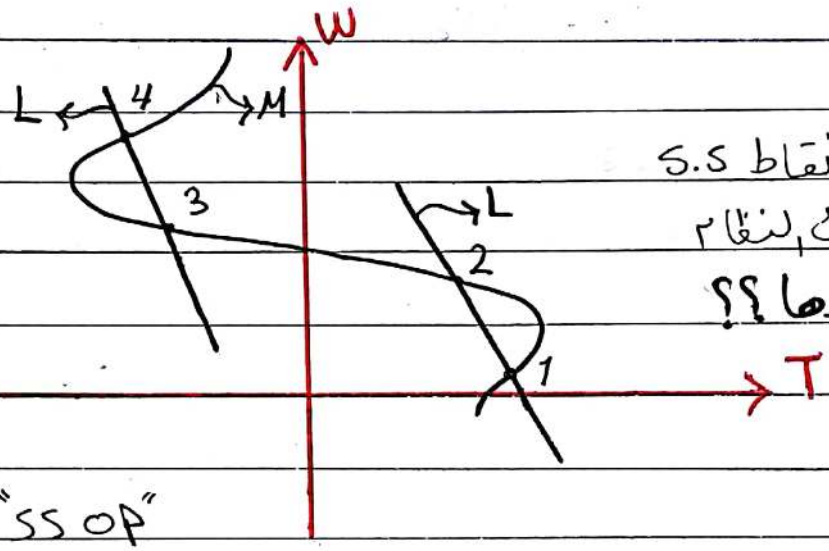


at $w' \Leftrightarrow T'_M > T'_L \Rightarrow \frac{dw}{dt} > 0$

so ⇒ the system is accelarating

بتسارع للنظام ← أي تزداد سرعته

حتى تصل إلى w_{ss} أي أنه النظام مستقر.



يوجد 4 نقاط S.S ★

هل سلوك النظام نفسه عندها ؟؟

$\beta = \frac{\Delta T}{\Delta w}$

* we have 4 "SSOP"

at (1): $\beta_M > \beta_L$ ⇒ not stable

(+) (−)

at (2): both β_M & β_L are (−), but $|\beta_L| > |\beta_M|$ ⇒ stable

at (3): " " " " " " $|\beta_L| > |\beta_M|$ ⇒ stable

at (4): $\beta_M > \beta_L$ ⇒ stable

+ −

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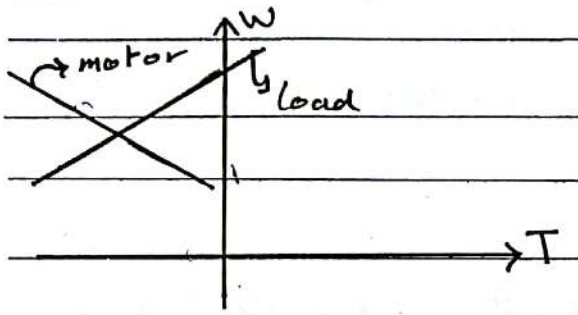
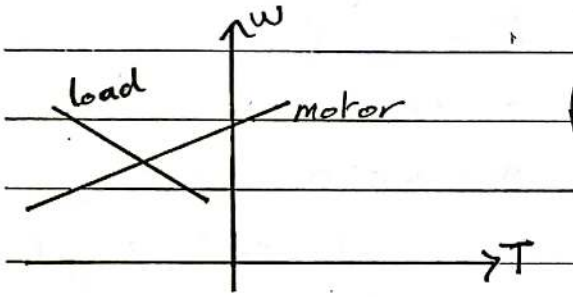
Examples:

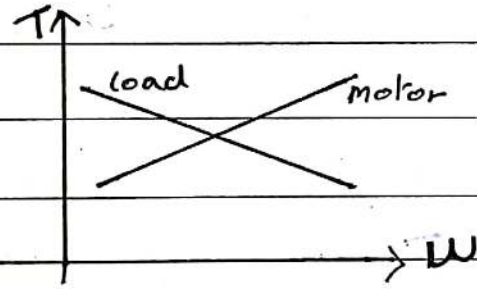


$\beta_L > \beta_M \Rightarrow$ stable

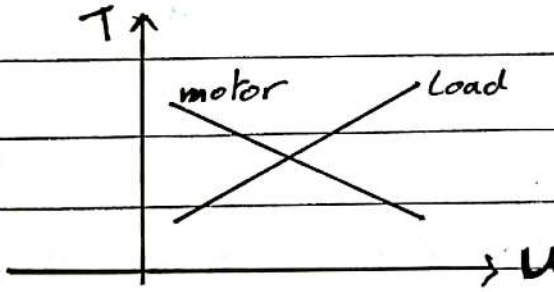


$\beta_L < \beta_M \Rightarrow$ unstable

مهم: يمكن عكس المحاور لكن تبقى القدرة لاستقرار النظام نفسها وهي $\beta_L > \beta_M$ (حيث β هي مقدار تغير $\underline{T}$ بالنسبة لـ $\underline{w}$)



$\beta_L < \beta_M \Rightarrow$ stable



$\beta_L > \beta_M \Rightarrow$ stable

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2) Analytically

EX: $T_M = 100 - 0.5\omega$ (N.m)

$T_L = 10 + 0.4\omega$ (N.m)

Sol:

*To find SSOP $\Rightarrow$ find (ω_{ss} & T_{ss})

at SS $\Rightarrow T_M = T_L$

$100 - 0.5\omega = 10 + 0.4\omega \Rightarrow \omega_{ss} = 100$ rad/sec

$T_{M(ss)} = 100 - 0.5 * 100 = 50$ N.m

$T_{L(ss)} = 50$ N.m

*To check the stability $\Rightarrow$ find (β_M & β_L), $\beta = \frac{\Delta T}{\Delta \omega}$ slope

$\beta_M = 0.5$
$\beta_L = 0.4$
$\Rightarrow \beta_M > \beta_L \Rightarrow$ not stable

EX: $T_M = 1 + 2\omega$ & $T_L = 3\sqrt{\omega}$

SS $\Rightarrow T_M = T_L$

$1 + 2\omega = 3\sqrt{\omega} \Rightarrow \omega = 1$, $\omega = 0.25$

$\Downarrow$ $T_{ss} = 3$, $\Downarrow$ $T_{ss} = 1.5$

$\beta_M = (1 + 2\omega)' = 2$ & $\beta_L = \frac{3}{2} * \frac{1}{\sqrt{\omega}}$

at $\omega = 1 \Rightarrow \beta_M = 2$ & $\beta_L = \frac{3}{2} \Rightarrow \beta_M > \beta_L \Rightarrow$ not stable at $\omega = 1$

at $\omega = 0.25 \Rightarrow \beta_M = 2$ & $\beta_L = 3 \Rightarrow \beta_L > \beta_M \Rightarrow$ stable at $\omega = 0.25$

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Ex: $T_M = 25 + 0.05n$

$T_L = 5 + 0.06n$

$T_M = T_L$

$25 + 0.05n_{ss} = 5 + 0.06n_{ss} \Rightarrow n_{ss} = 2000$ rpm

$\omega_{ss} = \frac{2\pi n_{ss}}{60} = 209.44$ rad/sec

$T_{ss} = 25 + 0.05 n_{ss} = 125$ N.m

للتحقق من استقرارية النظام نجيب اتجاه B أي اشتقاق T بالنسبة لـ (ω) وليس (n) لذلك ←

$\omega = \frac{2\pi n}{60} \Rightarrow n = \omega \frac{60}{2\pi}$

So $\Rightarrow T_M = 25 + 0.05 * \frac{60}{2\pi} \omega = 25 + 0.48\omega$

$T_L = 5 + 0.06 * \frac{60}{2\pi} \omega = 5 + 0.52\omega$

* The use of studing stability:
1- To difined steady stat operation points.
2- To check the stability of the drive system at SSOP.
3- To study the dynamic char of the motor.
4- To difined hardness (β).

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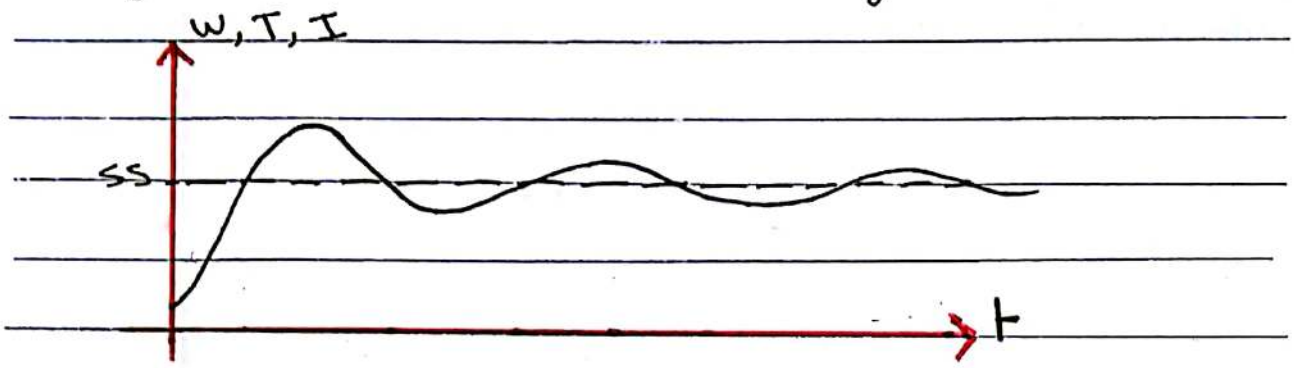
SUBJECT Dynamic of E.D.S

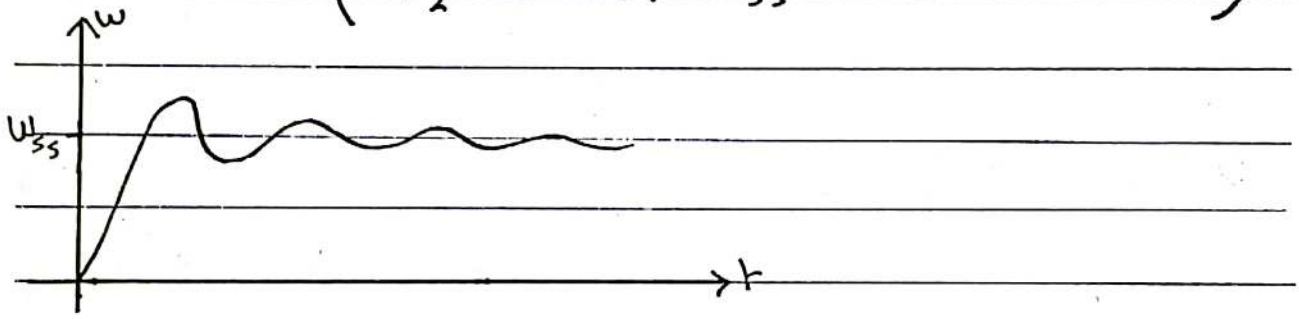
SECTION Transient periods

* Dynamic (char) of electric drive system *

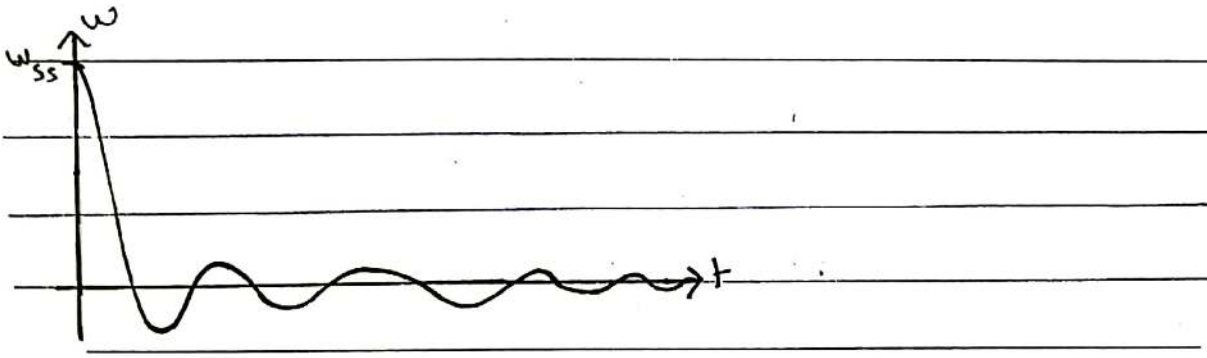


* Transient periods :

1 starting (transtion from stand stele ($w_1 = 0$), to $w_2 = 97.5\%\ w_{ss}$)



2 braking (from $w_1 = w_{ss}$ to $w_2 = 2.5\%\ w_{ss}$)



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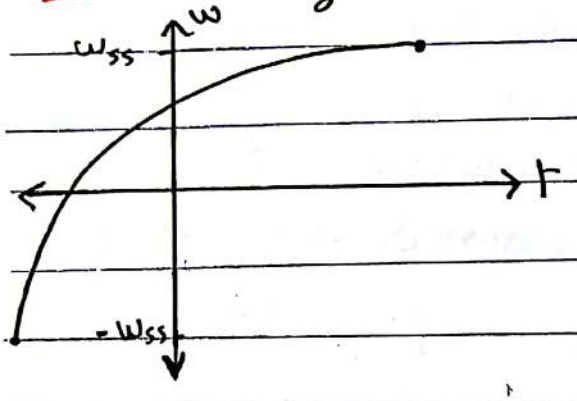
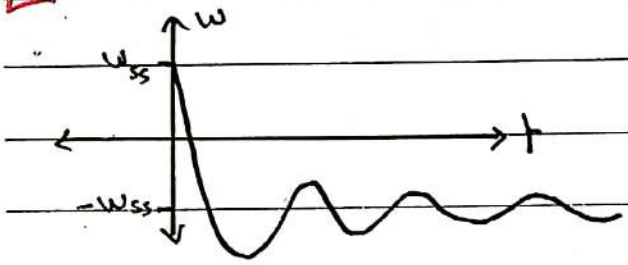
3 reversing (from $w_1 = w_{ss}$ to $w_2 = -w_{ss}$ or $w_2 = -97.5\% w_{ss}$)



4 Due to disturbance or speed control



*Calculation of time in transient operations:

-To find time of transient operations, from equation:

$T_M - T_L = J_{eq} \frac{dw}{dt}$, J_{eq} : equivalent moment of inertia

$$dt = \frac{J_{eq}}{T_M - T_L} dw \Rightarrow \boxed{t = J_{eq} \int_{w_1}^{w_2} \frac{1}{T_M - T_L} dw}$$

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SUBJECT Dynamic of E.D.S

SECTION Calculations of time in transient operations.

EX: a drive has following parameters:

$J_{eq} = 10\ (Kg.m^2)$, $T_M = 100 - 0.1n\ (N.m)$, $T_L = 0.05n\ (N.m)$

Find the time of reverse if:

- initial state is ⇒ steady state operation ($T_M = T_L$ & $n = n_{ss}$)
- required state ⇒ revers the motion ($n_2 = -97.5\%\ n_1$)
- motor char changed to $T = -100 - 0.1n\ (N.m)$

sol:

* For steady state ($T_M = T_L$)

$100 - 0.1 n_{ss} = 0.05 n_{ss} \Rightarrow n_{ss} = 666.67$ rpm

the initial speed $(n_1) = n_{ss} = 666.67$ rpm

* required speed $(n_2) = -n_{ss} = -666.67 * \frac{97.5}{100} = -650$ rpm

$\omega = \frac{2\pi}{60} n$ rad/sec

$T_M - T_L = J_{eq} \frac{d\omega}{dt}$

$T_M - T_L = J_{eq} \frac{2\pi}{60} \frac{dn}{dt}$

$-100 - 0.1n - 0.05n = 10 * \frac{2\pi}{60} \frac{dn}{dt}$

$-95.49 - 0.143n = \frac{dn}{dt}$

$t = \int_{n_1 = 666.67}^{n_2 = -650} \frac{1}{-95.49 - 0.143n}\, dn \Rightarrow t = 25.58$ sec

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SUBJECT Dynamic of EDS

SECTION Calculations of time in transient operations.

EX: $T_M = 150 - 0.3\,\omega\ (N.m)$, $J_{eq} = 0.25\ Kg.m^2$

- initial state is ⇒ no load state ($T_L = 0$) & ($\omega_1 = 0$)
- required state is ⇒ steady state ($\omega_2 = \omega_{ss}$) & ($T_M = T_L$)

Find the steady state time (t_{ss})

sol: $T_L = 0$

$150 - 0.3\,\omega_{ss} = 0 \Rightarrow \omega_{ss} = 500$ rad/sec

$$t_{ss} = J_{eq} \int_{\omega_1=0}^{\omega_2=500} \frac{d\omega}{T_M} = 3.132 \text{ sec}$$

EX: $n_{ss} = 1375$ (rpm), $T_L = 3$ (N.m), $J_{eq} = 0.25$ ($Kg.m^2$)

The motor has been disconnected from the main supply ⇒ ($T_M \to 0$) because it has stopped, find t_{stop}.

sol: $T_M^{\to 0} - T_L = J_{eq} \frac{d\omega}{dt}$

$$-3 = 0.25 \frac{d\omega}{dt} \Rightarrow t = \frac{-0.25}{3} \int_{\omega_1}^{\omega_2} d\omega$$

* تم فصل المحرك عن المصدر وانخفض عزم المحرك لكي أنّ وصل الى الصفر أي أن سرعته النهائية [$\omega_2 = 2.5\%\omega_{ss}$]

وسرعته الابتدائية : $\omega_1 = \omega_{ss}$

$$\omega_{ss} = \frac{2\pi}{60} * 1375 = 144 \text{ rad/sec} \Rightarrow t = 11.7 \text{ sec}$$

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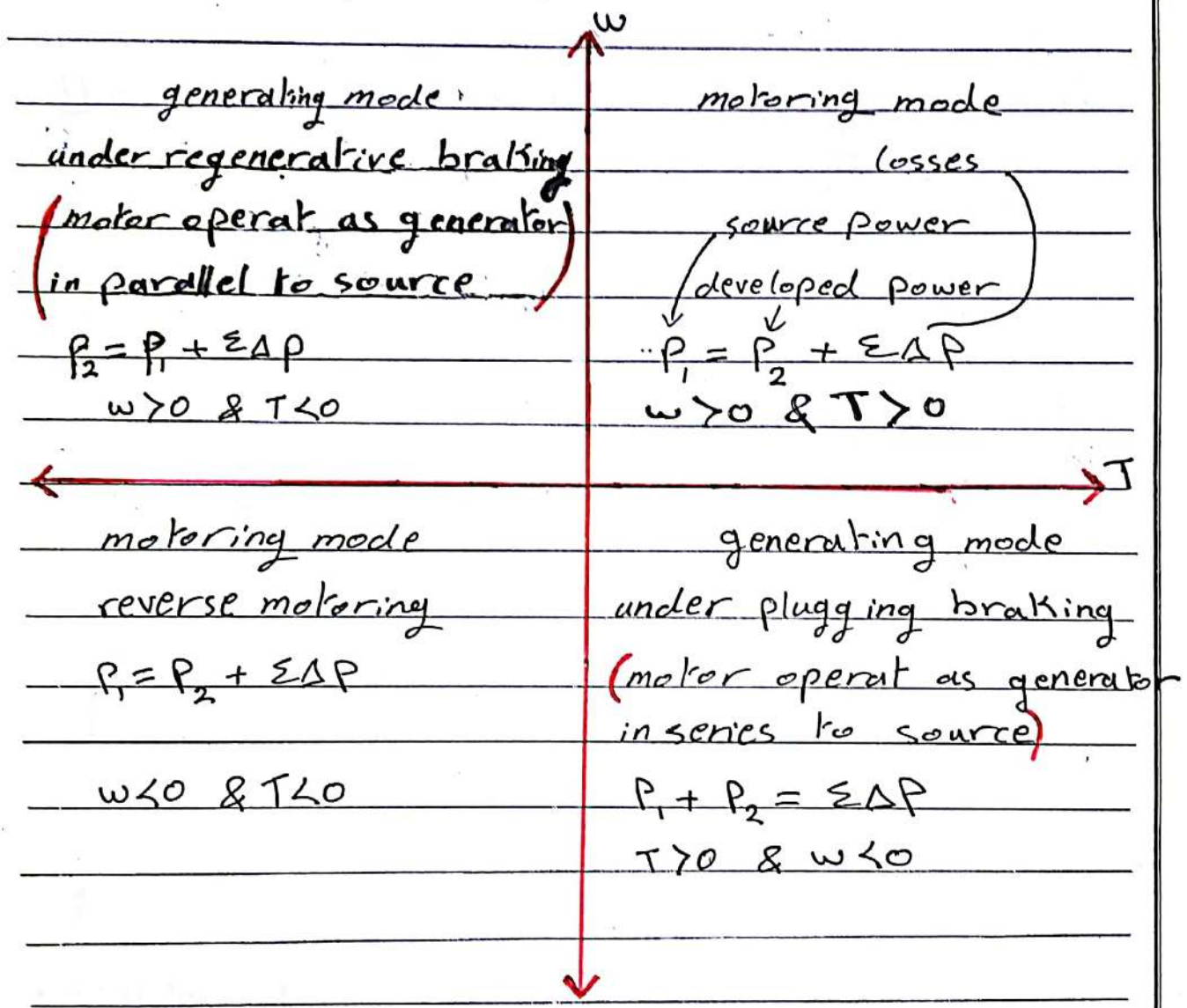
SUBJECT Dynamic of EDS

SECTION Quadrants operation

- Motoring mode ⇒ (T_M & w are in the same direction)

- generating mode ⇒ (T_M & w " " " opposite ")

* Quadrants operation :



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SUBJECT Dynamic of EDS

SECTION Equivalent values of D.S parameters

* equivalent values of drive system parameters:

* حساب عزم العطالة المكافئ للعمود المحرك (J_{eq})

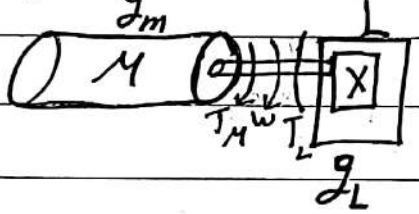
* حساب عزم المقاومة المكافئ للعمود المحرك (T_{eq})

مع الأخذ بعين الاعتبار نوع الحركة (خطية أو دورانية)

II Rotating load connected directly to motor shaft (without MTS).

هنا حالة الوحيدة التي يكون فيها

equivalent load = physical load.



* To find the total moment of inertia:

وفق قانون حفظ الطاقة $\frac{1}{2} J_m \omega_m^2 + \frac{1}{2} J_L \omega_m^2 = \frac{1}{2} J_{eq} \omega_m^2$

* بالتالي يوجد MTS فيدور المحرك والحمل على نفس السرعة (ω_m)

* بالضرب على ($\frac{1}{2} \omega m^2$) ←

$$J_m + J_L = J_{eq}$$

* J_m عزم العطالة الذاتي للمحرك

* J_L عزم العطالة الذاتي للحمل

* J_{eq} عزم العطالة الذاتي المكافئ.

في حالة وجود أكثر من حمل (J_{Leq}) $J_{Leq} = J_{L1} + J_{L2}$

* total load torque ⇒ $T_{Leq} = T_L$

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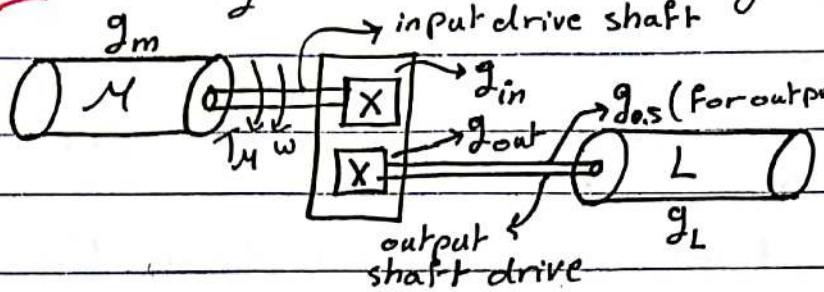
SUBJECT Dynamic of EDS

SECTION Equivalent values of DS parameters.

2 Rotating load connected to gear or (MTS).



$\eta_{MTS} = \frac{J_{out}}{J_{in}}$

J_{in}: moment of inertia of the input of MTS.

J_{out}: moment of inertia of the output of MTS.

1- moment of inertia of input shaft:

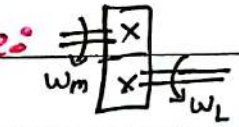
$$J_{in\Sigma} = J_m + J_{in}$$

2- moment of inertia of output shaft:

$$J_{out\Sigma} = J_{out} + J_{o.s} + J_L$$

$$\frac{1}{2} J_{eq} \omega_m^2 = \frac{1}{2} J_m \omega_m^2 + \frac{1}{2} J_{in} \omega_m^2 + \frac{1}{2} J_{out} \frac{\omega_L^2}{\eta} + \frac{1}{2} J_{o.s} \frac{\omega_L^2}{\eta} + \frac{1}{2} J_L \frac{\omega_L^2}{\eta}$$

$$J_{eq} = J_m + J_{in} + J_{out}\left(\frac{\omega_L}{\omega_m}\right)^2 \cdot \frac{1}{\eta} + J_{o.s}\left(\frac{\omega_L}{\omega_m}\right)^2 \cdot \frac{1}{\eta} + J_L\left(\frac{\omega_L}{\omega_m}\right)^2 \cdot \frac{1}{\eta}$$

note: 

- gear ratio $(r_g) = \frac{D_o}{D_i}$
- velocity ratio $(a) = \frac{\omega_L}{\omega_m}$

$\Rightarrow r_g = \frac{1}{a} = \frac{\omega_m}{\omega_L}$

$$\left(J_m + J_{in}\right) + \left(J_o \frac{1}{r_g^2 \eta} + J_{o.s} \frac{1}{r_g^2 \eta} + J_L \frac{1}{r_g^2 \eta}\right) = J_{eq}$$

input

J_{Leq}

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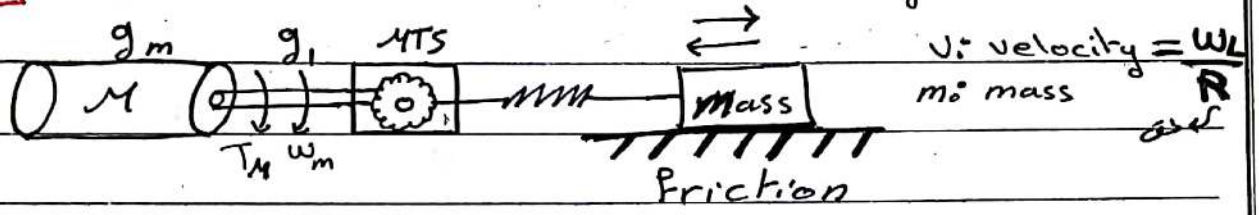
SUBJECT Dynamic of EDS

SECTION Equivalent values of DS parameters

من قانون حفظ الطاقة $T_{Leq}\,\omega_m = T_L \frac{\omega_L}{\eta_g}$

$T_{Leq} = T_L \frac{\omega_L}{\omega_m}\cdot\frac{1}{\eta} \Rightarrow \boxed{T_{Leq} = T_L \frac{1}{r_g \eta}}$

[3] Transtional motion, load only:



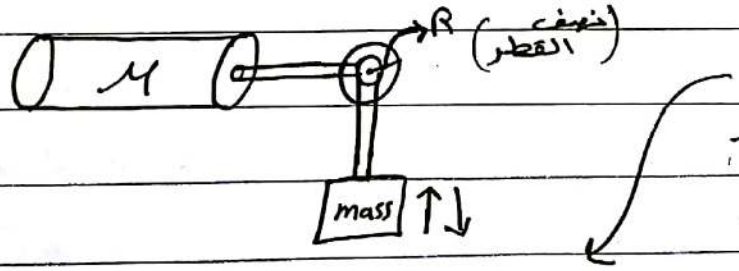
$\frac{1}{2} g_{eq}\,\omega_m^2 = \frac{1}{2} g_m\,\omega_m^2 + \frac{1}{2} g_1\,\omega_m^2 + \frac{1}{2} m \frac{v^2}{\eta}$

$g_{eq} = g_m + g_1 + \underbrace{\left(m\left(\frac{v}{\omega_m}\right)^2 \frac{1}{\eta}\right)}_{g_{Leq}}$

$T_{eq}\,\omega_m = F\,v \Rightarrow$ في حالة رفع الحمل $\Rightarrow T_{eq} = \frac{F\,\omega_L\,R}{\omega_m}\cdot\frac{1}{\eta}$

$T_{eq}\,\omega_m = F\,v\,\eta \Rightarrow$ في حالة انزال الحمل $\Rightarrow T_{eq} = F\frac{R}{r_g}\cdot\frac{1}{\eta}$

[4] Load in compaination motion



$g_{Leq} = m\left(\frac{v}{\omega_m}\right)^2 \cdot \frac{1}{\eta}$

$T_{Leq} = F\left(\frac{v}{\omega_m}\right)\cdot\frac{1}{\eta}$

$T_{Leq} = F\frac{R}{r_g}\cdot\frac{1}{\eta}$

$g_{eq} = m\left(\frac{v}{\omega_m}\right)^2 \cdot \frac{1}{\eta} = m\left(\frac{R}{r_g}\right)^2 \cdot \frac{1}{\eta}$

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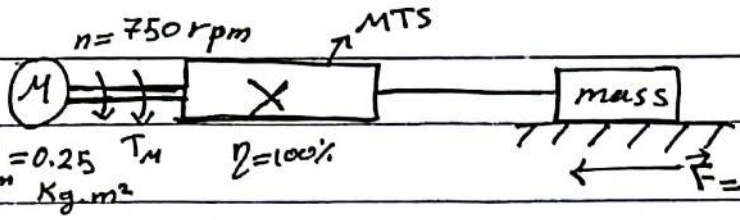
SUBJECT Dynamic of- EDS

SECTION Equivalent values of- DS parameters

EX:



$\omega_m = \frac{2\pi}{60} * 750 = 78.5$ rad/sec $T_F = 2$ N.m

* $g_{eq} = g_m + g_{Leq}$

$= 0.25 + m\left(\frac{V}{\omega_m}\right)^2 \frac{1}{\eta}$

$= 0.25 + 250\left(\frac{0.5}{78.5}\right)^2 = 0.26$ Kg.m²

* mechanical power developed by the motor (output)?

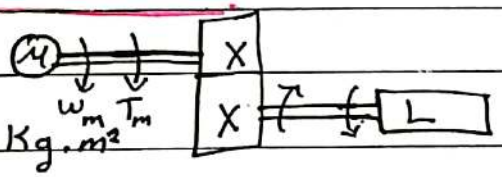
$P_{out} = T_{eq}\,\omega_m$

$T_{eq} = T_F + T_{Leq}$

$T_F + F\left(\frac{V}{\omega_m}\right)\frac{1}{\eta}$

$2 + 1000 * \frac{0.5}{78.5} = 8.37$ N.m $\Rightarrow P_{out} = 657$ Watt

EX:



$g_m = 0.1$ Kg.m² $T_L = 100$ N.m

$n_m = 750$ rpm $g_L = 1.5$ Kg.m²

$T_o = 5$ N.m $n_L = 250$ rpm

$r_g = \frac{n_m}{n_L} = \frac{750}{250} = 3$ $\Longrightarrow$

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SUBJECT Dynamic of EDS

SECTION Equivalent Values of DS parameters

* $J_{eq} = J_m + J_{Leq}$

$J_{eq} = J_m + J_L \frac{1}{r_g^2} \cdot \frac{1}{\eta}$

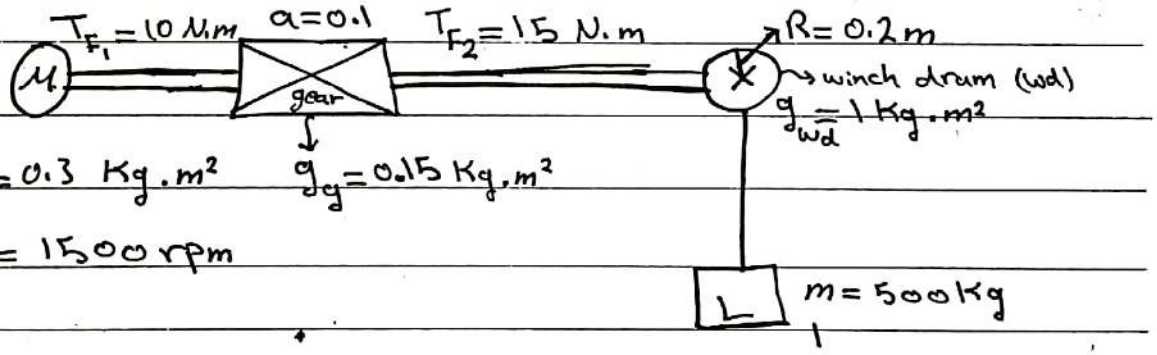
$J_{eq} = 0.1 + 1.5 \frac{1}{9 * 0.95} = 0.27 \ Kg.m^2$

* $T_{Leq} = T_L \frac{1}{r_g \eta} = 100 * \frac{1}{3 * 0.95} = 35.08$

$T_{tot} = T_. + T_{Leq} = 5 + 35.08 = 40.08$

$P_2 = T_{tot} \omega_m = 3148.08$ watt.

EX



$J_m = 0.3 \ Kg.m^2$ $J_g = 0.15 \ Kg.m^2$

$n_m = 1500$ rpm

T_{F_1} is the friction torque at motor shaft.

T_{F_2} is the friction torque at winch shaft.

Find J_{eq} & T_{eq} ?

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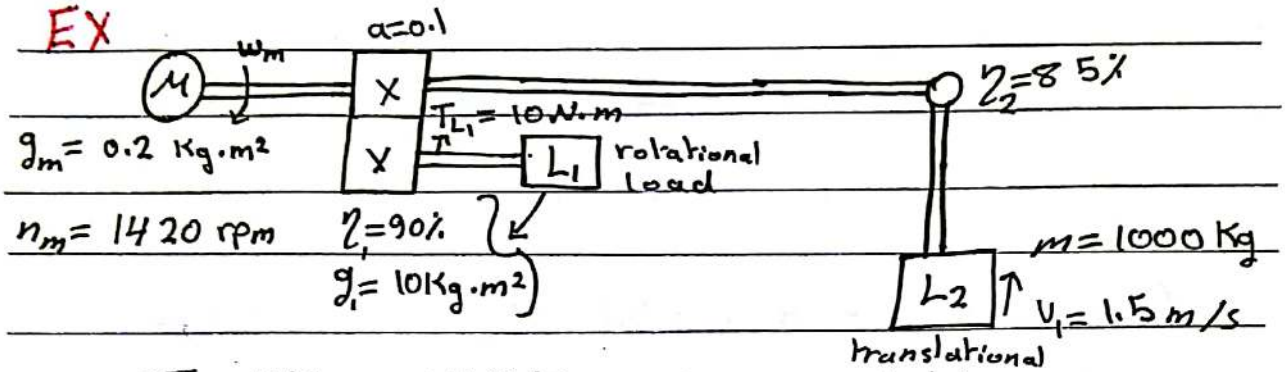
SUBJECT Dynamic of EDS

SECTION Equivalent values of DS parameters

EX



$\omega_m = \frac{2\pi}{60} * 1420 = 148.7$

$* J_{eq} = J_m + J_{L_1} + J_{L_2}$

$J_{eq} = J_m + J_1 \frac{a^2}{\eta_1} + m\left(\frac{V}{\omega_m}\right)^2 \frac{1}{\eta_2}$

$J_{eq} = 0.2 + 10\frac{(0.1)^2}{0.9} + 1000\left(\frac{1.5}{148.7}\right)^2 * \frac{1}{0.85} = 0.43 \text{ Kg.m}^2$

$* T_{Leq} = T_{L_1} + T_{L_2}$

$\boxed{r = \frac{V}{\omega}}$, $F = mg$

$= T_{L_1} * \frac{a}{\eta} + F\left(\frac{V}{\omega}\right) \cdot \frac{1}{\eta}$

$= 10 * \frac{0.1}{0.9} + (1000 * 9.81) * \left(\frac{1.5}{148.7}\right) * \frac{1}{0.85} = 117.53$

$* P_2 = P_{out} = P_{mech} = T_{Leq}\, \omega_m = 117.53 * 148.7$

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SUBJECT Motor selection

SECTION Introduction

* Motor selection:

any motor can be selected if we know it's rated parameters

(Ratings ≡ name plate data ≡ nominal values)

if the motor operates under this values ⇒ it will operates in it's age without any failures.

Selection depends on:

↳ operating conditions.

↳ operating invironment.

Heating causes:

↳ current generation or current flow

* heating effects the isolation of winding

وانهيار العازلية ← يؤدي إلى تماس بالفان

لذلك شرط ← $\theta_c^\circ \leq \theta_{permissible}$

$\Sigma \Delta P$ (cause of heating) $\Rightarrow \theta^\circ \uparrow$

Load disconnection (cause of cooling) $\Rightarrow \theta^\circ \downarrow$

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SUBJECT Motor selection

SECTION Introduction

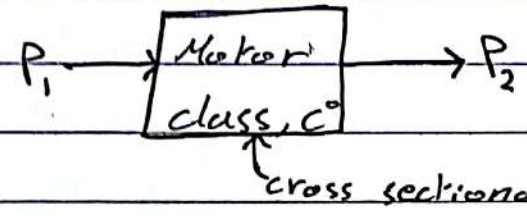
Thermal model of motor for heating and cooling :

class A : 105° C°

class F : 155 C°

class H : 180 C°

class C : above 180 C° ⇒ تتحمل درجات حرارة عالية لكنها مكلفة



$P_1 dt - P_2 dt = m c^\circ d\theta$

this difference leads to motor heating

m : mass

c° : temperature

θ : increase in temp

d : thermal coeff of heat transfere

$P_1 dt - P_2 dt = G d\theta$, $G = mc^\circ$

G : thermal capacitance

$P_1 - P_2 = G \frac{d\theta}{dt}$

$P_2 = \theta d A$ ↓ cooling surface

$P_1 - \theta D = G \frac{d\theta}{dt}$

$P_2 = \theta D$ ↓ head dissipation constant

$$\boxed{P_1 = G \frac{d\theta}{dt} + \theta D}$$ ⇒ represents heating or cooling taking into consideration the initial state.

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Solution of the last equation :

$$\theta = \theta_{ss} + K e^{-t/\tau}$$

K: by initial condition

$\theta_{ss} = \frac{P_1}{D}$ (steady state change in tempreture)

$\tau = \frac{C}{D}$ (heating & cooling time constant)

* cases of operation :

- case 1 ⇒ (During heating) :

$t=0$, $\theta = \theta_1$, K = from intial condition

so ⇒ $\theta = \theta_{ss}(1 - e^{-t/\tau}) + \theta_1 e^{-t/\tau}$

- case 2 ⇒ (during cooling for connected motor) :

$t=0$, $\theta = \theta_2$

$\theta = \theta_{ss}'(1 - e^{-t/\tau'}) + \theta_2 e^{-t/\tau'}$

where $\theta_{ss}' = \frac{P_1'}{D'}$ & $\tau = \frac{C}{D'}$

- case 3 ⇒ (case of supply disconnection from supply) :

$P_1' = 0$, $\theta_{ss}' = 0$

$\theta = \theta_2 \cdot e^{-t/\tau}$ during cooling

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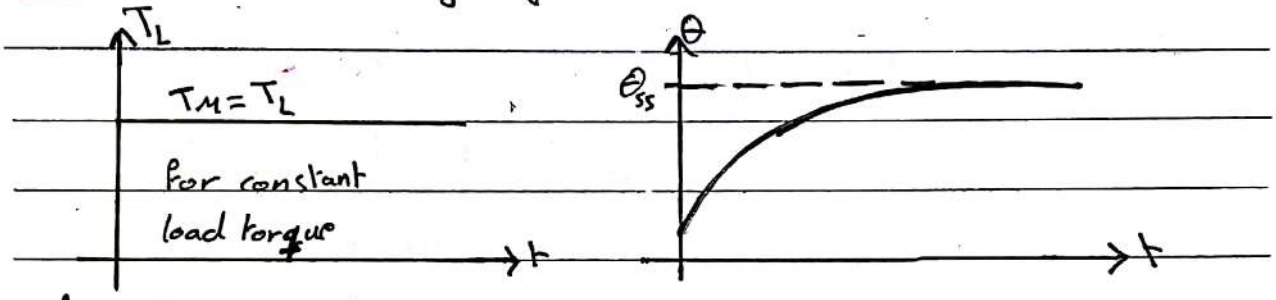
SUBJECT Motor selection

SECTION Duty cycles

* Heating & cooling processes depend on the "duty cycle" (loading diagram) of the motor.

* Duty cycles: (standardized)

[1] Continuous duty cycle

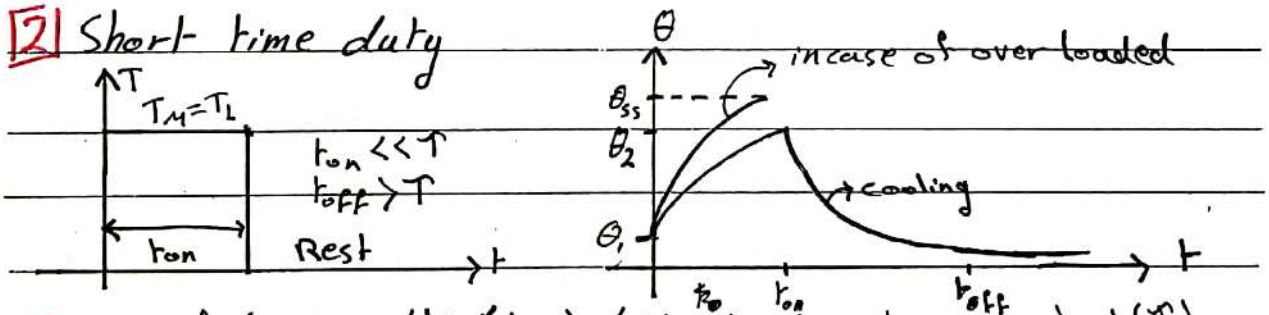


* $\Sigma \Delta P \approx$ constant

* running time is long enough for the motor temp to reach SS value.

EX: (Pumps, Fans, Compresors, Conveyers, & paper mill drive)

[2] Short time duty



* time of drive operation (t_{on}) < the heating time constant (τ).

* machine is allowed to cool off to ambient temp.

* motor can be over loaded until temp at the end of loading time reaches the permissible limit (θ_{ss}).

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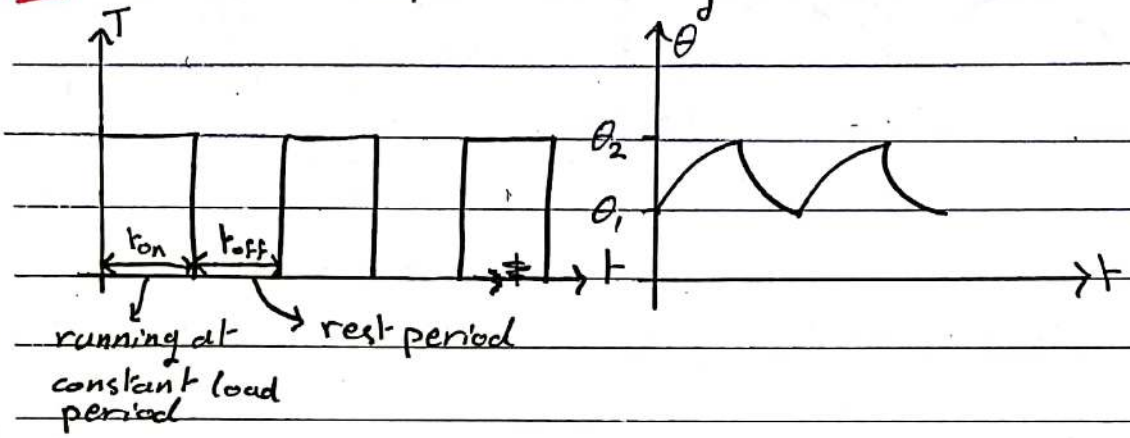
SUBJECT Motor selection

SECTION Duty cycles

Ex on short time duty cycle:

Crane drive, positioning control system

[3] Intermittent periodic duty



* running time (t_{on}) is not long enough to reach the ~~temp~~ the S.S. value of temp.
* rest time (t_{off}) is not long enough to cool off to ambient temp.
* heating of machine during starting & braking is negligable.

Ex: ~~metal~~ Metal cutting machines, Driling machines

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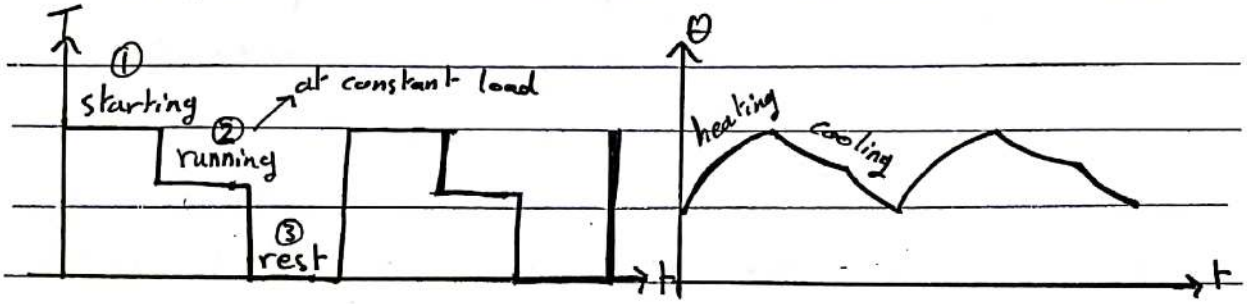
SUBJECT Motor selection

SECTION Duty cycles

4 Intermittent period duty with starting

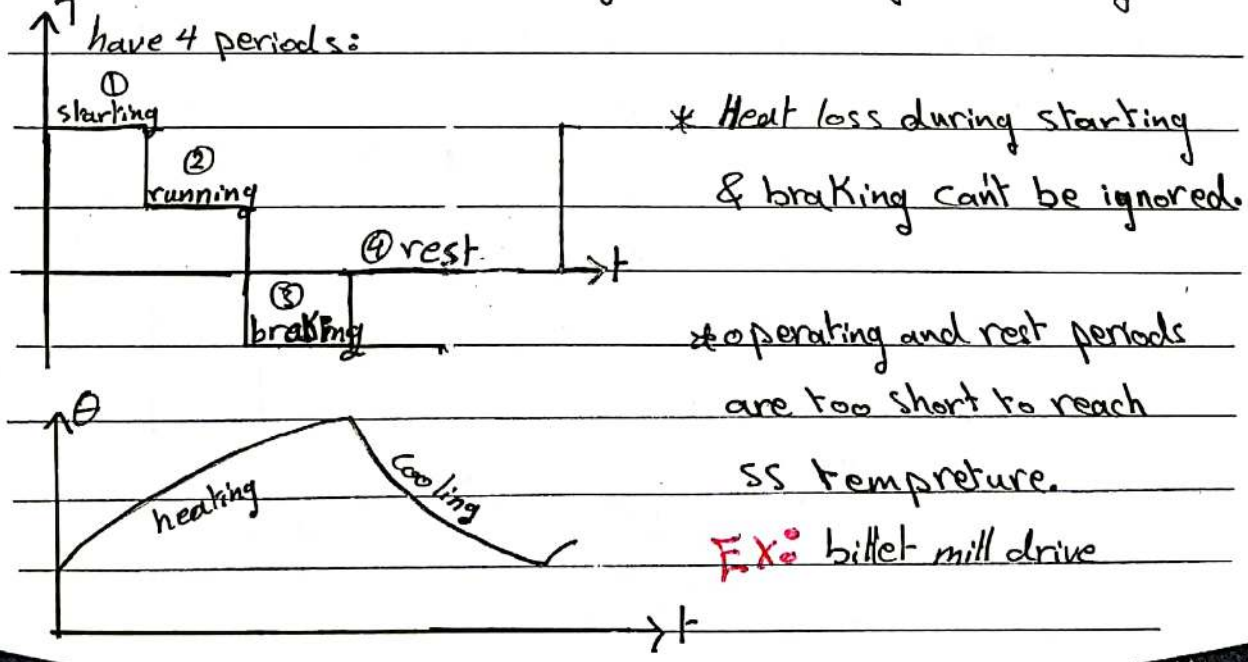
* have 3 periods:



* operating & rest periods are too short to reach ss tempreture.
* heat losses during starting can't be ignored.
* " " " braking is negligible.

EX: metal cutting, driling tool drives, & drives for fork lift trucks

5 Intermittent period duty with starting & braking



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SUBJECT Motor selection

SECTION Determination of motor ratings

Determination of motor ratings:

motor rating various duty cycle classified in 3 categorises:

1- Continuous duty.
2- Fluctuating and Intermittent loads.
3- short time duty.

Continuous duty

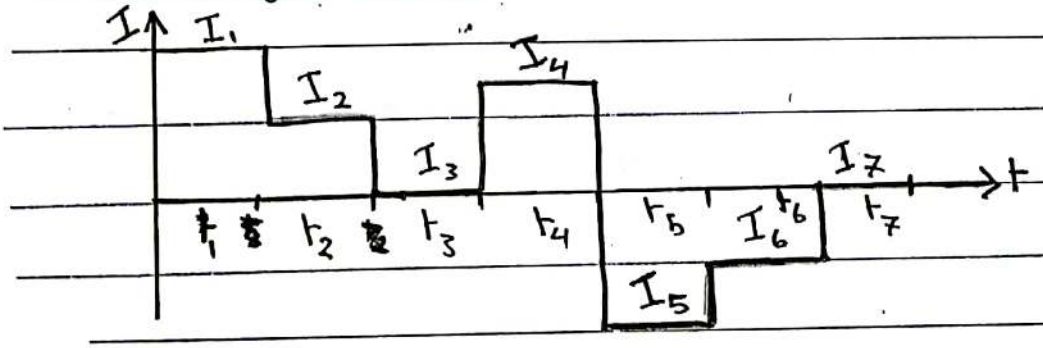
$T_M = T_L$ كيف نحسب الـ rating ؟

$P_2 = T_M \omega_L = T_L \omega_L$

ثم نختار الـ rating اللي قيمته أعلى من P_2 ← $P_{max} > P_2$ rating

A motor with next higher power rating from comercially available ratings is selected.

Fluctuating and Intermittent loads.



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SUBJECT Motor selection

SECTION Determination of motor ratings

① ⇒ To calculate equivalent current

* during periode the current has the same heating losses

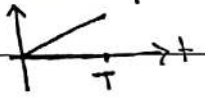
$$I_{eq} = \sqrt{\frac{I_1^2 t_1 + I_2^2 t_2 + \cdots I_n^2 t_n}{(t_1 + t_2 + \cdots t_n)}}$$

الفترة الكاملة $T \Rightarrow = (t_1 + t_2 + t_3 + \cdots t_n)$

لعمل حساب I_{eq} لابد من وجود I_{eq} في Data sheet أو في الكتالوج

if the current varies smoothly over a period T

then $I_{eq} = \sqrt{\frac{1}{T}\int_0^T i^2 t^2 dt}$



② To calculate equivalent torque

$$T_{eq} = \sqrt{\frac{T_1^2 t_1 + T_2^2 t_2 + \cdots + T_n^2 t_n}{T}}$$

For DC sep or shunt motor $\Rightarrow \phi$ = constant

$T \propto K\phi I_a \Rightarrow T \propto I$

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SUBJECT Motor selection

SECTION Determination of motor ratings

③ ⇒ To calculate equivalent power

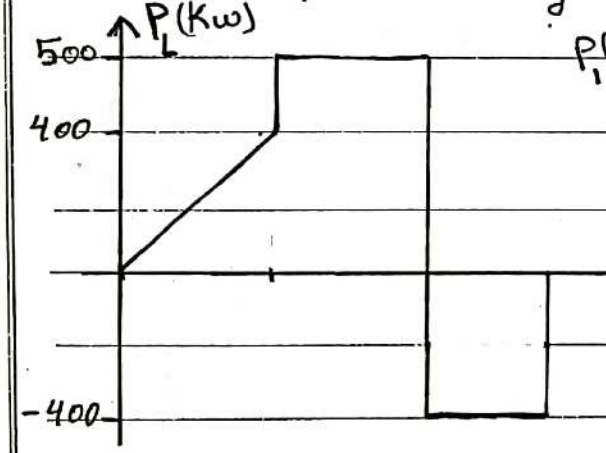
For constant ϕ & constant ω

$$P_2 = \frac{K\phi I}{T} \cdot \omega \propto I \Rightarrow P_{2_{eq}} = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + \cdots P_n^2 t_n}{T}}$$

Ex: ω is constant (constant speed drive)

1) Load rising from 0 → 400 Kw, for 5 min.
2) uniform load of 500 Kw, for 5 min.
3) regenerative power of 400 Kw, returned to the supply, for 4 min.
4) remains idle (Rest), for 2 min.

Find the power rating of the motor



$$P_1(t) = \frac{400}{5} t \Rightarrow P_{1rms} = \frac{1}{5}\int_0^5 \left(\frac{400}{5}\right)^2 \cdot t^2 \, dt$$

$$P_{1rms} = \frac{400}{\sqrt{3}} \text{ Kw}$$

$$P_{eq_{rms}} = \sqrt{\frac{\left(\frac{400}{\sqrt{3}}\right)^2 \cdot 5 + (500)^2 \cdot 5 + (400)^2 \cdot 4}{16}} = 367 \text{ Kw}$$

⇒

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SUBJECT Motor selection

SECTION Determination of motor ratings

Rated power = rms value of power = 3.67 Kw

Since $P_{max} = 500$ Kw $< 2 * P_{rms} \Rightarrow$ the motor rating = 3.67 Kw

Short time duty

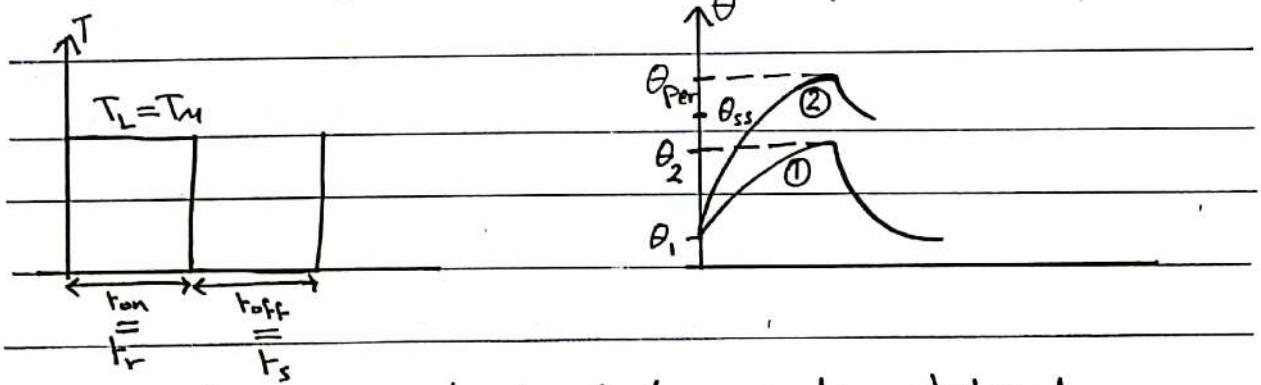
Remember, in short time duty:

↳ $t_{on} < T$ (heating time constant)

↳ $\theta_2 < \theta_{ss}$

↳ motor is allowed to cool down to the ambient temp, befor it's required to operat again.

* motor can be over loaded (by a factor "K" >1), so the temp rise to reach $\theta_{permissible}$.



① ⇒ motor is under loaded or under utilized.

② ⇒ motor is over loaded ⇒ $\theta_2' = \theta_{per}$ ⇒ fully utilized motor

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SUBJECT Motor selection

SECTION Determination of motor ratings

$$\theta_{per} = \theta_{ss}\left(1 - e^{-t/T}\right)$$

$$K = \sqrt{\frac{1+\alpha}{1-e^{-t/T}} - \alpha} \quad > 1$$

$$\alpha = \frac{\Sigma \Delta P_{(constant)}}{\Sigma \Delta P} \simeq \text{zero} \Rightarrow \boxed{K = \sqrt{\frac{1}{1-e^{-t/T}}}} \quad > 1$$

$\Rightarrow$ For intermittent periodic duty

$$K = \sqrt{\frac{1 - e^{-\left[(t_r/T_r)+(t_s/T_s)\right]}}{1 - e^{-t_r/T_r}}}$$

T_r : thermal time constant of motor for working interval.

T_s : " " " " " standstill interval.

EX: a motor has :

- a heating time constant (T_r) = 60 min
- a cooling time constant (T_s) = 90 min
- $P_{2(rated)}$ = 20 Kw
- final temp (θ_2) = 40°C

$\Rightarrow$

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SUBJECT Motor selection

SECTION Determination of motor ratings

1) what load motor can be delivered for 10 min [t_r] if this followed by a shut down period long enough for it to cool [$(t_s >> T_s) \Rightarrow$ short time duty]?

Solution:

for short time duty $\Rightarrow K = \sqrt{\frac{1}{1 - e^{-t_r/T_r}}} = \sqrt{\frac{1}{1 - e^{-10/60}}} = 2.55$

permitted load $= 2.55 * 20 = 51$ Kw

2) if it's on an intermittent load of 10 min [t_r], followed by 10 min shut down [t_s], whats is the maximum value of load it can supply during the on load period?

$$K = \sqrt{\frac{1 - e^{-\left(\frac{10}{60} + \frac{10}{90}\right)}}{1 - e^{10/60}}} = \sqrt{\frac{0.2425}{0.1535}} = 1.257$$

permitted load $= 1.257 * 20 = 25.14$ Kw

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SUBJECT DC motors

SECTION Static char & configurations

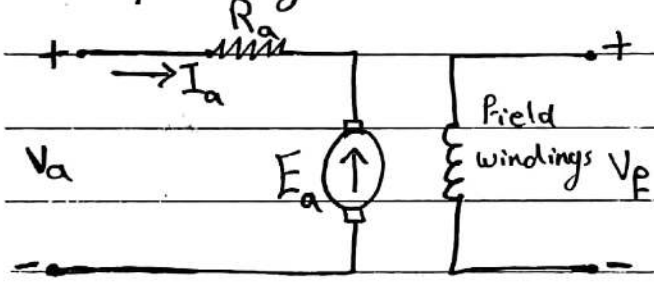
* Static char of DC motors:

[A] mechanical (torque-speed) char ($T = f(\omega)$).

[B] electro mechanical char ($I_a = f(\omega)$).

char of the DC motors depends on the excitation type (configuration of the motor).

1- Separately excited DC motor



• $R_a \Rightarrow$ total resistance of Armature circuit.

$R_a = r_a + r_p + r_c$

armature coil — pole coil — commutater brushes

$R_a \approx r_a$

• $E_a \Leftarrow$ قوة دافعة عكسية تتولد حسب قانون (فاراداي) لكهرومغناطيسي

بسبب قطع الموصل (أو) تدوير بسرعة (ω) وجال مغناطيسي (ϕ)

$E_a = K \phi \omega$, $K = \frac{(P/2)N}{2\pi a}$

P: عدد أقطاب المحرك

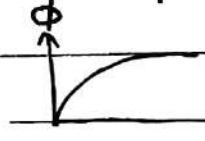
N: عدد الموصلات النشطة

a: no of parallel poles.

$\phi = f(I_a)$

but in sep motors $\Rightarrow I_f \neq f(I_a) \Rightarrow \phi \neq f(I_a)$

ϕ لا تعتمد على تيار المنتج I_a



$\Rightarrow$

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SUBJECT DC motors

SECTION static char & configurations

$V_a = E_a + I_a R_a$

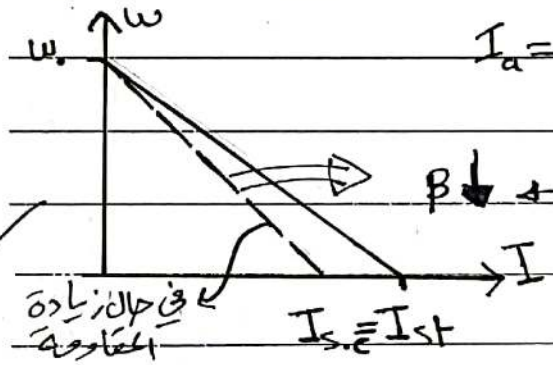
$V_a = K\phi\omega + I_a R_a$

$\Rightarrow \left(\omega = \frac{V_a}{K\phi} + \frac{I_a R_a}{K\phi}\right) \Rightarrow$ "describes the electro mech. char"

($\omega_\circ$ the no-load speed I_a = zero) ($\Delta\omega$ drop in speed due to load $\Delta\omega = f(I_a)$) where I_a is variable and $K\phi$ is constant

linear because "$K\phi$" is const

$\Rightarrow \omega = \omega_\circ - \Delta\omega$



* عند الإقلاع $\omega = 0 \Leftarrow I_a = I_{st} = \frac{V_a}{R_a}$ (starting)

كلما زاد الهبوط في السرعة ($\Delta\omega\uparrow$) $\beta\downarrow$

(elect mech char of sep DC motor)

* At steady state $\Rightarrow T_L = T_M = T = K\phi I_a \Rightarrow \boxed{I_a = \frac{T}{K\phi}}$

$\Rightarrow \left(\omega = \frac{V_a}{K\phi} - \frac{T R_a}{(K\phi)^2}\right) \Rightarrow$ "describes the mech char of sep"

$T = T_{st} = \frac{V_a K\phi}{R_a} \Leftarrow \omega = 0 \Leftarrow$ at starting

$T_{st} \equiv I_{st} K\phi$

- $I_a = \frac{V_a - E_a}{R_a}$

عند زيادة الحمل فإنه ليتغلب على $I_a \uparrow \Leftarrow E_a \downarrow$

و $\Delta\omega \uparrow$ (الهبوط في السرعة يزداد أي أنه السرعة تقل)

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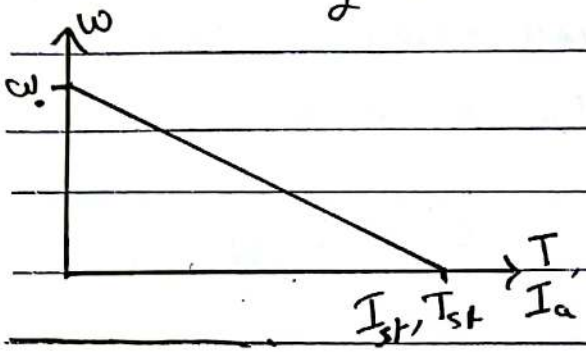
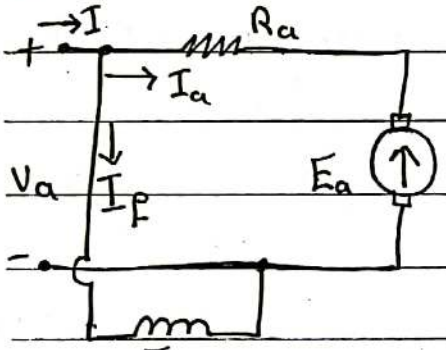
SUBJECT DC motors

SECTION Static char & configurations

*For searatly DC motors ⇒ mech & electro-mech char are semilar



2- Shunt motors



$I = I_f + I_a$

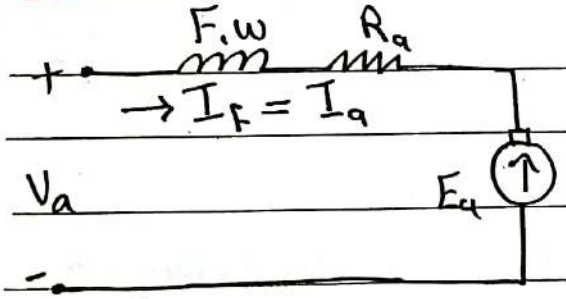
$R_f >> R_a \Rightarrow I \simeq I_a$

I_f (constant) $\Rightarrow \phi$ constant

أي أن Arm يؤثر على المجال ولكن بشكل ضئيل

* ولذلك يكون لـ shunt motors نفس منحنى خواص لـ separatly motor.

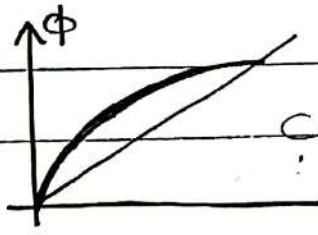
3- Series field motors



$I_f = I_a$

and $\phi = f(I_f) \Rightarrow \phi = f(I_a)$

$\Sigma R_a = R_a + R_{SF}$



R_{SF} ⇒ resistance of series field

تعتبر العلاقة جيدة لتسهيل عمليات الحساب ⇒

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* R_f is so small
* series motors have hige starting torque and high speed at low loads.

$$V = E_a + I_a R_a$$

$$\omega = \frac{V_a}{K\phi} - \frac{I_a R_\varepsilon}{K\phi}$$

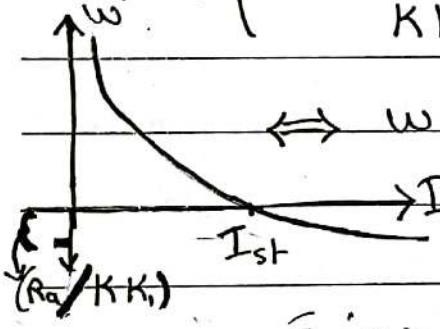
in series motors $\Rightarrow (I_a \& \phi)$ are variabl

$$\phi = f(I_a)$$

$$\phi = K_1 I_a \quad , K_1 = \frac{\phi_{n\,(rated)}}{I_{an(rated)}}$$

بالتعويض $\Rightarrow \left(\omega = \frac{V}{K K_1 I_a} - \frac{R_\varepsilon}{K K_1} \right) \Rightarrow$ electro mech char

$\Leftrightarrow \omega = f\left(\frac{1}{I_a}\right) \Rightarrow$ non linear



من التعبير بالسرعة عالية جدا عند الاحمال الصغيرة لذلك يجب أنه لا يعمل المحرك إلا تحت الحمل

minimum load = (10 - 15)% of full load..

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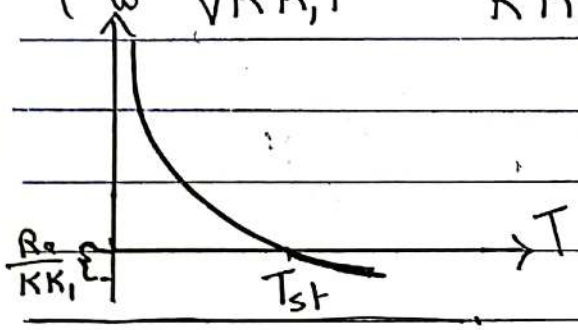
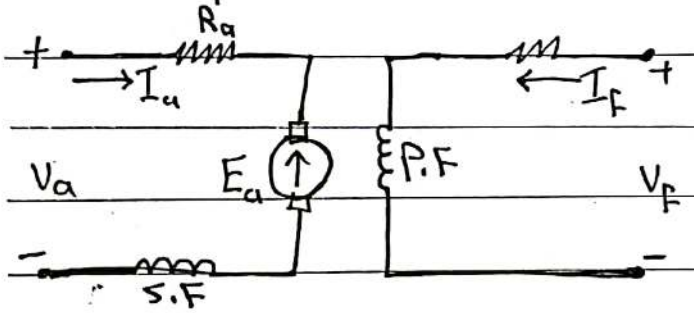
* $T = K\phi I_a = K K_1 I_a^2$, $\phi = K K_1 I_a$

$\Rightarrow I_a = \sqrt{\frac{T}{K K_1}}$

$\left(\omega = \frac{V_a}{\sqrt{K K_1 T}} - \frac{R}{K K_1}\right) \Rightarrow$ mechanical char (non linear)

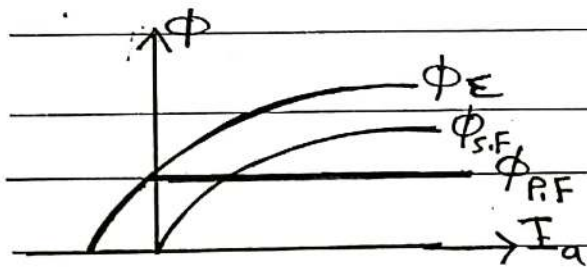


4- Compound motors



S.F ⇒ series Field windings

P.F ⇒ parallel Field windings



$\Phi_E = \Phi_{P.F} + \Phi_{S.F}$

$\Phi_{P.F}$ ⇓ Constant $\neq F(I_a)$

$\Phi_{S.F}$ ⇓ Variable $= F(I_a)$

يوجد فيهان معناطيسيان, أحدهما ينتج من (P.F) وهو لا يتأثر بتيار المنتج (I_a) والآخر ينتج من (S.F) ويتأثر بتيار المنتج (I_a)

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for compound motors:

$\omega = \frac{V_a}{K\Phi_E} - \frac{I_a R_E}{K\Phi_E}$ ⇒ electro mech char

$\omega = \frac{V_a}{K\Phi_E} - \frac{T R_E}{(K\Phi_E)^2}$ ⇒ mech char

if $\Phi_{S.F} > \Phi_{P.F}$ ⇒ motor is more of a series motor.

if $\Phi_{P.F} > \Phi_{S.F}$ ⇒ motor is more of a sep or shunt motor.

E.X: If a drive has the following parameters:

$V_{an} = 200\,V$, $I_{an} = 10.5\,A$, $n_n = 200$ rpm, $R_a = 0.47\,\Omega$

Find n^* if $V_a^* = 175\,V$ n(rated values)

Sol: $\Phi \propto I_f \Rightarrow \propto V_f \Rightarrow \propto V_a$

$\frac{\Phi_n}{\Phi^*} = \frac{V_{an}}{V_a^*} = \frac{200}{175} \Rightarrow \Phi^* = 0.875\,\Phi_n$

*since that $T_L = T_n$ = constant

~~$K\Phi_n I_{an} = K\Phi^* I_a^*$~~

$I_a^* = \frac{\Phi_n}{\Phi^*} \cdot I_{an} = 11.4\,A$ ⇒ should be 12 A

⇒

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SUBJECT DC motors

SECTION operating modes of sep

$E_a = V_a - I_a R_a$

$E_{an} = V_{an} - I_{an} R_a = 195\ V$

$E_a^* = V_a^* - I_a^* R_a = 169.3\ V$

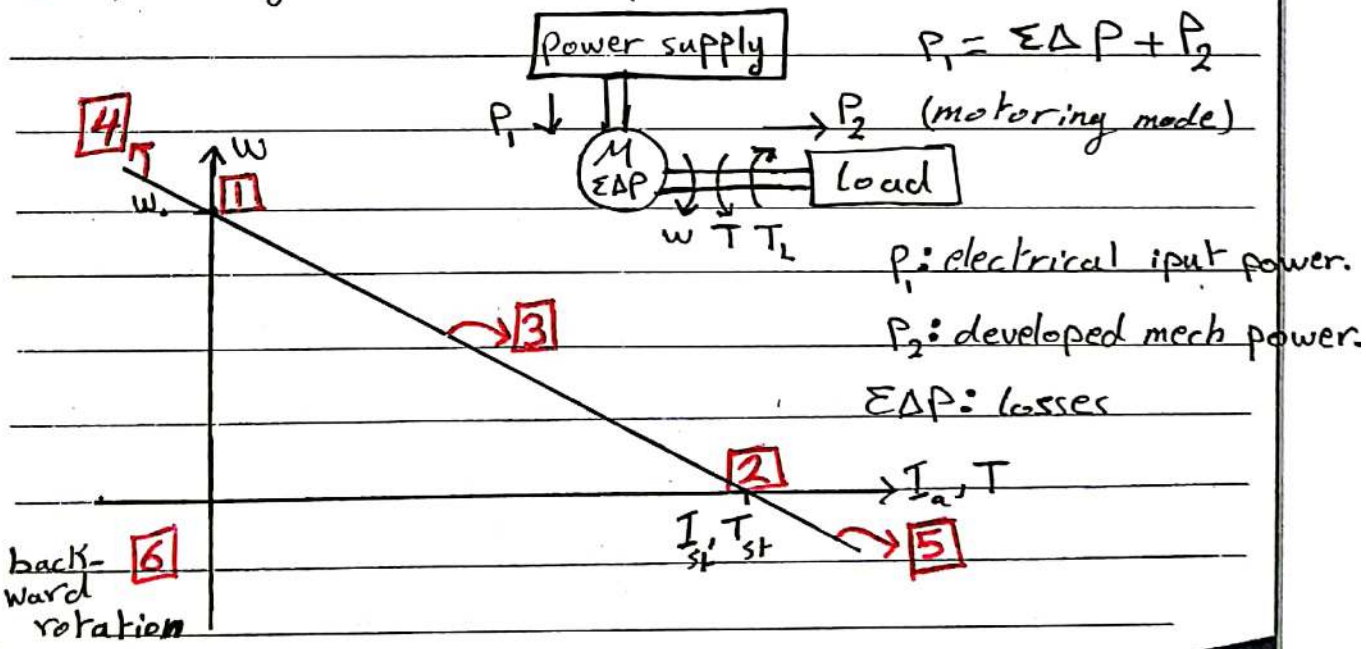
$E_a = K\phi\omega \Rightarrow E_a \propto \phi\omega \Rightarrow E_a \propto \phi n$, $\omega = \frac{2\pi n}{60}$

$E_{an} \propto \phi_n n_n$ and $E_a^* \propto \phi^* n^*$

$$\frac{E_{an}}{E_a^*} = \frac{K\phi_n\omega_n}{K\phi^*\omega^*} = \frac{\phi_n\, n_n}{\phi^* n^*}$$

$$\Rightarrow n^* = \frac{E_a^*}{E_{an}} \cdot \frac{\phi_n}{\phi^*} \cdot n_n = 198.5\ rpm$$

* operating modes of sep DC motors :



$P_1 = \Sigma\Delta P + P_2$ (motoring mode)

P_1: electrical input power.

P_2: developed mech power.

$\Sigma\Delta P$: losses

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SUBJECT DC motors

SECTION operating modes of- separatly

1 At point 1 :

(Ideal no-load mode)

$\omega = \omega_o$ & T_L = zero & I_a = zero

عملياً يوجد عزم تيار قليل ناتج عن الاحتكاك .

* $P_2 = T_L \omega_L = T_L \omega_o \cong 0 \Rightarrow \eta = 0$

$P_1 = P_2 + \Sigma \Delta P \Rightarrow \boxed{P_1 = \Sigma \Delta P}$ ومعظم الضياعات (احتكاك

* $I_a = 0 \Rightarrow \frac{V_a - E_a}{R_a} = 0 \Rightarrow V_a = E_a = K\phi\omega_o$

(Friction & copper losses) ← الضياعات

ضياعات كهربائية في المنتج (قليلة) ضياعات ميكانيكية تشكل معظم الضياعات

2 At point 2 :

(short circuit mode)

$\omega_L = 0$ & $I_a = I_{st} = I_{s.c} = \frac{V_a}{R_a}$ & $T_L = T_{st} = T_{s.c}$

$P_2 = T_L \omega_L$ = zero $\Rightarrow \eta$ = zero لايدور المحرك ، ولا يعطي

قدرة للحمل

$P_1 = \cancel{P_2^{\to 0}} + \Sigma \Delta P = \Sigma \Delta P \cong I_{s.c}^2 R_a$ بالتالي (P_1) تذهب للخسائر

الضياعات وبعضها كهربائية حرارية ناتجة عن ($I_{s.c}$) العالي

ولا يسمح للمحرك بالعمل لفترة طويلة عند هذه الحالة لأنه يأخذ تيار المنتج

العالي يؤدي إلى تلف ملفات المنتج ..

$\omega = 0 = \frac{V}{K\phi} - \frac{I_{s.c} R_a}{K\phi} \Rightarrow I_{a_{s.c}} = \frac{V_a}{R_a}$

upnormal operational mode and very dangerous.

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SUBJECT DC motors

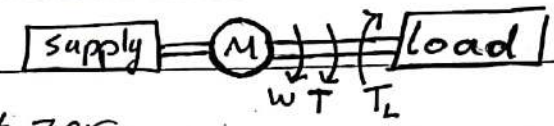
SECTION operating modes of separately

3 At point 3 :

(Motoring mode) between no-load & S.C points

$0 < T < T_{s.c}$
$\omega_o > \omega > 0$
$\Rightarrow T > 0 \ \& \ \omega > 0 \Rightarrow$ الربع الأول

$0 < I_a < I_{s.c}$



η (high) $\quad P_2 = T_L \omega_L \neq$ zero

$P_1 = P_2 + \Sigma \Delta P$

المقدرة الداخلة تذهب لتغطية الضياعات ومقدرة الحمل معاً

أي كهربائية + ميكانيكية

ملاحظة:- زيادة الحمل عند المقدرة ثابتة $\Leftarrow \uparrow I_a \Leftarrow$ خطر على الملفات

الحرك لارتفاع حرارتها $\Leftarrow$ تلف الملفات.

4 At point 4 :

(Regenerating mode)

$\omega_L > \omega_o$
$T_L < 0$
$\Rightarrow (P_2 = T_L \omega_L) < 0 \Leftrightarrow$ في الربع الثاني

$$I_{a_4} = \frac{V_a - E_a}{R_a} = \frac{V_a - K\phi\omega_4}{R_a} < 0$$

لأن $\omega_o < \omega_4$

$V_a = E_a = K\phi\omega_o$ & $\omega = \omega_o$ at no-load

$(P_1 = V_a I_a < 0) \Leftarrow I_a (-ve) \Leftarrow V_a < E_a \Leftarrow \omega_4 > \omega_o$ عند

أي أنه

$P_2 < 0 \Rightarrow$ motor operat as generator in parallel to the source.

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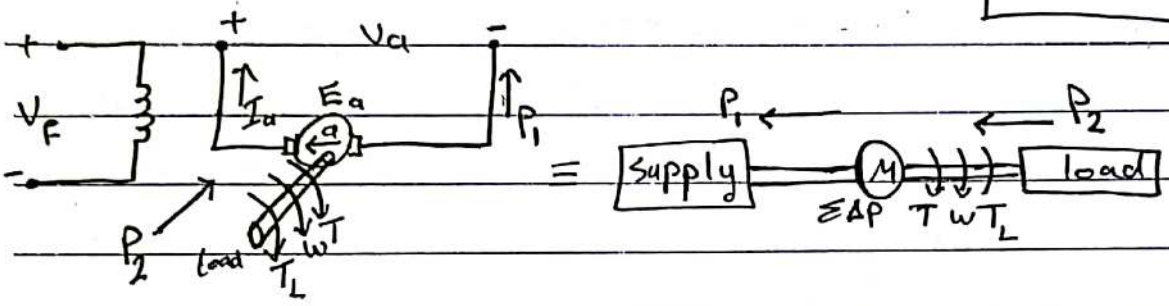
$(P_2 > 0) \Leftarrow$ قدرة، لحل تتجه للمنبع، تتحول القدرة الكهربائية

حزء يذهب للعنصر — حزء ضياعات

(المحرك لا يستهلك قدرة بل يزود العنصر بها) ← في هذه الحالة.

$(P_1 \& P_2) < 0 \Rightarrow -P_1 = -P_2 + \Sigma\Delta P \Rightarrow \boxed{P_2 = P_1 + \Sigma\Delta P}$



5 At point 5 :

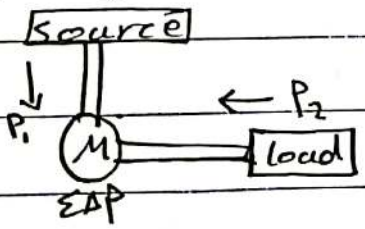
$T_L > T_{sc}$
$w < 0$
$I_a > I_{s.c}$

} الوجه 4 (generating mode under plugging braking)

$w < 0 \Rightarrow (P_2 = T w) < 0 \Rightarrow$ conversion of mechanical power (P_2) into electrical power

$P_1 = -P_2 + \Sigma\Delta P \Rightarrow \boxed{\Sigma\Delta P = P_1 + P_2}$

$\eta = 0$



ضياعات حرارية كبيرة.

lost inside the motor

The motor operat as generator in series to source

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SUBJECT DC motors

SECTION construction of the char

* Construction of characteristic:

Natural char from name plate (rated, nominal) values.

V_a, n, I, η, P_{out}

R_a might not be given.

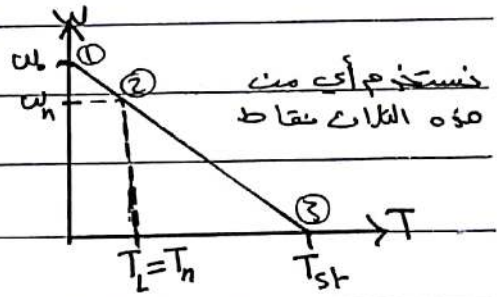
* For separatly excited or shunt motors:

- linear char ⇒ requires 2 points

- For nominal point:

↳ ω_n or n_n is given

↳ $T_n = \frac{P_{2n}}{\omega_n}$



نستخدم أي من هذه النقاط

- For no-load point:

↳ $T = 0$

↳ $\omega = \omega_0 = \frac{V_a}{(K\phi)_n}$, $\omega_n = \frac{V_{an}}{(K\phi)_n} - \frac{R_a I_a}{(K\phi)_n}$

$\eta_n \approx \eta_{max} \Rightarrow \Delta P_{max} = \Delta P_{cons} \cong 0.5 \Sigma \Delta P$

$R_a = 0.5 \frac{V_{an}}{I_{an}} (1 - \eta_n)$

- For starting (short circuit) point:

↳ $\omega = 0$

↳ $T = T_{st} \Rightarrow \omega = 0 = \frac{V_a}{K\phi} - \frac{T_{st} R_a}{(K\phi)^2} \Rightarrow T_{st} = \frac{V_a \cdot K\phi}{R_a}$

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SUBJECT DC motors

SECTION construction of the char

Ex: Find the mech char for DC sep excited motor, which has the following ratings:

$V_{an} = 220\ V$, $P_{2n} = 10\ Kw$, $I_{an} = 52.2\ A$, $n_n = 2250$ rpm

sol:

$P_{2n} = T_n \omega_n$

$10 * 10^3 = T_n * 2250 * \frac{2\pi}{60} \Rightarrow T_n = 42.44\ N.m$

$T_n = K\phi I_{an}$

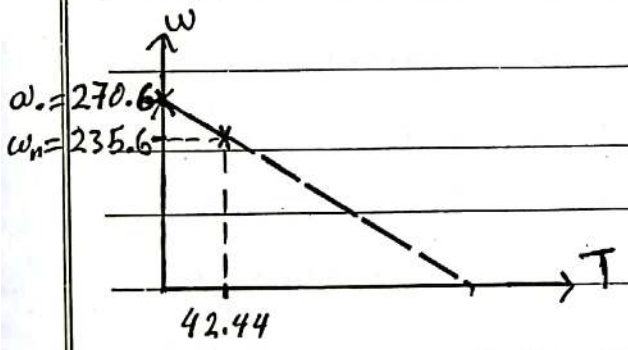
$42.44 = K\phi * 52.2$

$\Rightarrow K\phi = 0.813$

- nominal point $(T_n, \omega_n) \Rightarrow (42.44, 235.62)$

$\omega_\circ = \frac{V_a}{K\phi} = \frac{220}{0.813} = 270.6$ rad/sec

- no-load point $(0 \overset{T}{\nearrow}, \omega_\circ) \Rightarrow (0, 270.6)$



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SUBJECT DC motors

SECTION Construction of the char

Ex: $P_{2n} = 4.8\,Kw$, $V_{an} = 220\,V$, $I_{an} = 24.2A$

$n_n = 1500\ rpm$, $R_a = 0.38$, $w = 90\ rad/sec$ ↳ means will Radd

, $T_1 = 25\,N.m$, Find $R_{additional}$ ↳ with Radd

Sol: $w_n = \frac{2\pi}{60} * 1500 = 157 \Rightarrow T_n = \frac{P_{2n}}{w_n} = 30.57\,N.m$

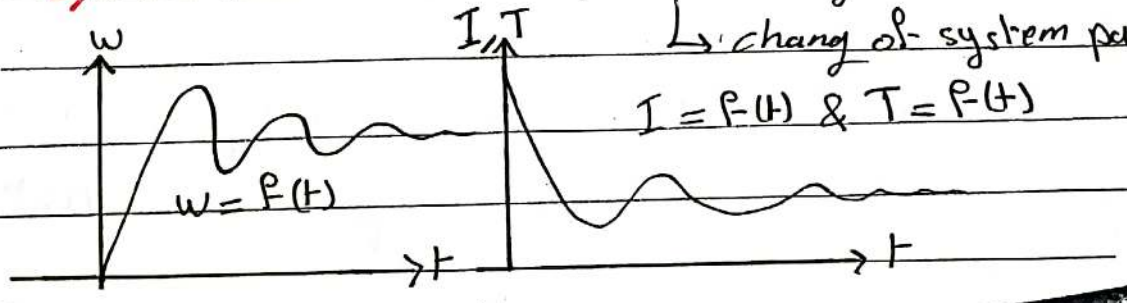
$K\phi = \frac{T_n}{I_{an}} = \frac{30.57}{24.2} = 1.263$

$w_0 = \frac{V_a}{K\phi} = \frac{220}{1.263} = 174.19\ rad/sec$

$w_1 = w_0 - \frac{T(R_a + R_{add})}{(K\phi)^2} \Rightarrow R_{add} = 4.99$

*Dynamic analysis of DC motors:

- Transient operations
  - ↳ starting: from 0 ro w_{ss} .
  - ↳ braking: from w_{ss} ro 0 .
  - ↳ reverse: from w_{ss} ro $-w_{ss}$.
  - ↳ disturbance: for example → chang of load.
    - ↳ chang of system parameters.



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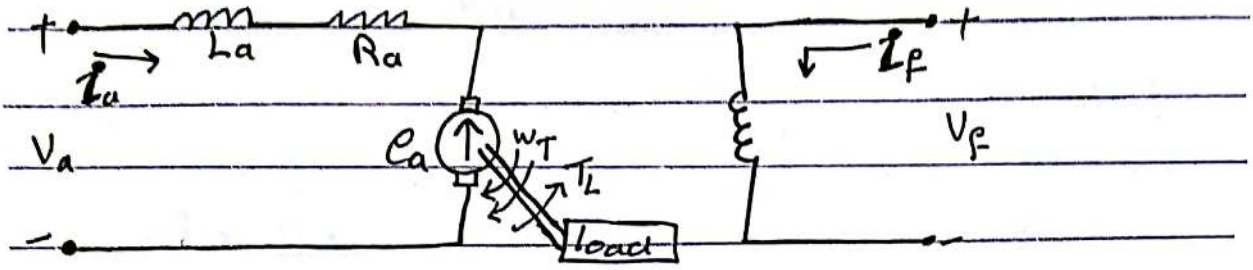
SUBJECT DC motor

SECTION Dynamic analysis

- Dynamic modle for DC sep or shunt excited motor :



* $V_a = e_a + L_a \frac{di}{dt} + i_a R_a$ ①

* $T - T_L = J_{eq} \frac{d\omega}{dt}$ ② ⇒ main equations

$T = K\phi I_a$ & $E_a = K\phi\omega$ & $K\phi$ constant

*taking laplace transformation for eq ① :

$V_a(s) = E_a(s) + I_a(s) R_a + sI_a(s) L_a$

$V_a(s) - E_a(s) = I_a(s) R_a \left(1 + s\left(\frac{L_a}{R_a}\right)\right)$ → T_e (electro magnetic time constant)

$$\boxed{W_1(s) = \frac{I_a(s)}{V_a(s) - E_a(s)} = \frac{1/R_a}{1 + sT_e}}$$ ⇒ "electrical part of the motor"

* taking Laplace transformation for eq ② :

$T(s) - T_L(s) = J_{eq}\, s\, \Omega(s)$ ⇒

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SUBJECT DC motor

SECTION Dynamic analysis

$$I_a(s) - I_L(s) = \frac{J_{eq}}{K\phi} s\,\Omega(s)$$

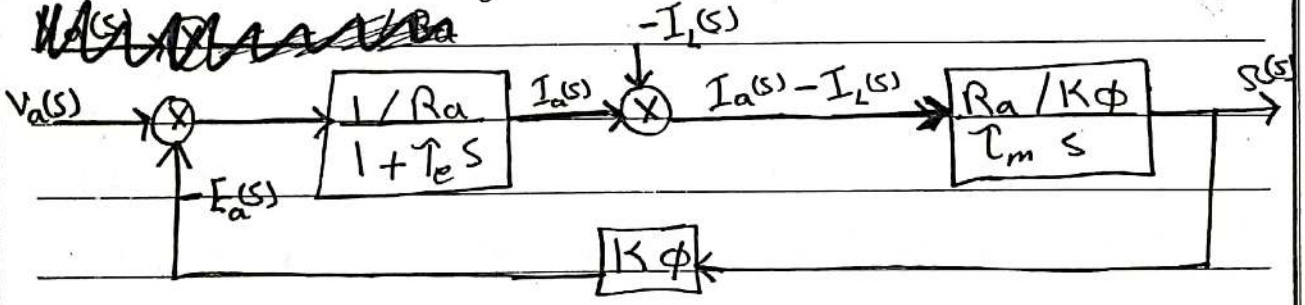
$$W_2(s) = \frac{\Omega(s)}{I_a(s) - I_L(s)} = \frac{R_a / K\phi}{T_m s} \Rightarrow \text{mechanical part}$$

$$T_m = J_{eq} \frac{R_a}{(K\phi)^2} \Rightarrow \text{mechanical time constant}$$

$$T_m = J_{eq} \frac{\omega_\circ}{T_{ss}} \Rightarrow \text{For calculations}$$

$W_1(s)$, $W_2(s)$ $\Rightarrow$ dynamic modle of the DC motor (represented in the form of transfer function)

* The block diagram of DC sep excited motor:



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SUBJECT DC motors

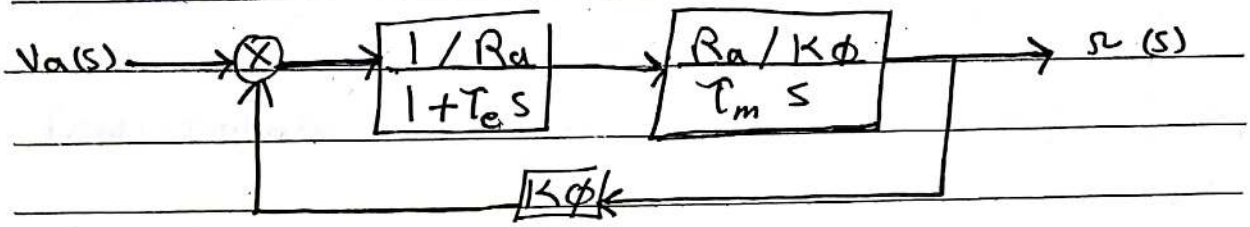
SECTION Dynamic analysis

* Since the system is linear.

⇒ we can apply super position principle

A case 1 : (no-load)

$V_a(s) \neq$ zero , $T_L(s) =$ zero $\Leftrightarrow I_L(s) =$ zero



$$W_A(s) = \frac{1/K\phi}{T_e T_m s^2 + T_m s + 1} = \frac{\Omega(s)}{V_a(s)}$$

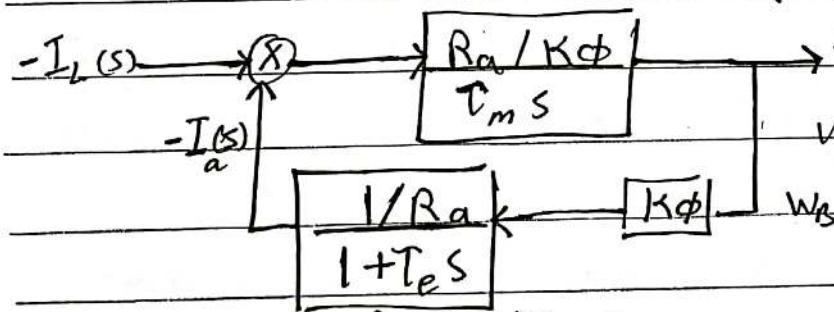
at steady state $\Rightarrow \frac{d}{dt} = 0 \Rightarrow s = 0 \Rightarrow W_A(s) = \frac{1}{K\phi}$

$$W_A(s) = \frac{W(s)}{V_a(s)}$$

$$\Rightarrow \boxed{W(s) = \frac{V_a(s)}{K\phi} = w_o}$$

B Case 2 :

$V_a(s) = 0$, $T_L(s) \neq$ zero (motor disconnected "mech braking")



when $s = 0 \Rightarrow W_B(s) = \frac{R_a}{K\phi}$

$$W_B(s) = \frac{W(s)}{-I_L(s)}$$

$$W_B(s) = \frac{\Omega(s)}{-I(s)} = \frac{\frac{R_a}{K\phi}(1 + T_e s)}{T_e T_m s^2 + T_m s + 1}$$

$$\Rightarrow \boxed{W(s) = \frac{-I_L R_a}{K\phi}}$$

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SUBJECT DC motors

SECTION Dynamic analysis

Characteristic equation for the system:

$\tau_e \tau_m s^2 + \tau_m s + 1 = 0$

$As^2 + Bs + C = 0$

$s_{1,2} = \frac{-\tau_m}{2\tau_e\tau_m} \pm \frac{\sqrt{\tau_m - 4\tau_e\tau_m}}{2\tau_e\tau_m} = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$

Discriminator $(D) = (\tau_m^2 - 4\tau_e\tau_m)$ المميز الذي يحدد نوع الجذور

$D = \tau_m^2 - 4\tau_e\tau_m$

↳ $D = 0$

↳ $D > 0$

↳ $D < 0$

Case 1 $(D > 0)$:

$s_1, s_2 \Rightarrow$ negative, real, and unequal roots

over damped response $\left(\frac{\tau_M}{\tau_E} > 4\right)$

Case 2 $(D = 0)$:

$s_{1,2} = \frac{-1}{2\tau_e} = s \Rightarrow$ two real, neg, and equal roots

critically damped response $\left(\frac{\tau_M}{\tau_E} = 4\right)$

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SUBJECT DC motors

SECTION Dynamic analysis

Case 3 ($D<0$):

$s_{1,2} = -\sigma \pm j\Omega$, $\sigma = \frac{-1}{2\tau_e}$, $\Omega_o = \frac{\sqrt{\tau_m^2 - 4\tau_m\tau_e}}{2\tau_e\tau_m}$

(two imaginary - conjugate roots)

$\sigma \Rightarrow$ damping factor

$\Omega_o \Rightarrow$ frequency of free oscillation

Oscillatory process (under damped process)

$\frac{\tau_M}{\tau_E} < 4$

* Cases of oscillation:-

1 oscillation with decaying (damping). $\Rightarrow$ acceptable

2 // // constant amp.

3 // // increasing amp.

special case $\Rightarrow$ over damping & $\tau_M >> \tau_E$

$\Rightarrow \tau_E = 0 \Rightarrow \tau_E\tau_M s^2 + \tau_M s + 1 = 0 \Rightarrow \boxed{s = \frac{-1}{\tau_M}}$

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SUBJECT DC motors

SECTION Methods of transient operation

* Methods of transient operations:

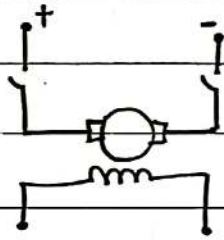
(starting, braking, reverse) ⇒ transient operation

A) Methods of starting (from stand stel. to steady state):

(For sep excited DC motors)

1) Direct starting:

"used in small motors, when the power source is unlimited"



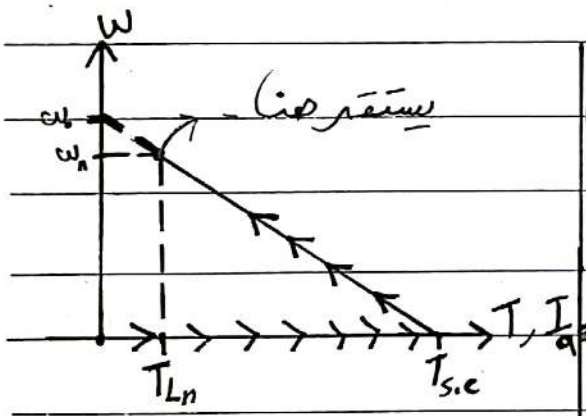
At starting $\Rightarrow I_a = I_{s.c} = \frac{V_a}{R_a}$

$P_i = V_a I_{s.c} \Rightarrow \varepsilon\Delta P >> \varepsilon\Delta P_n$

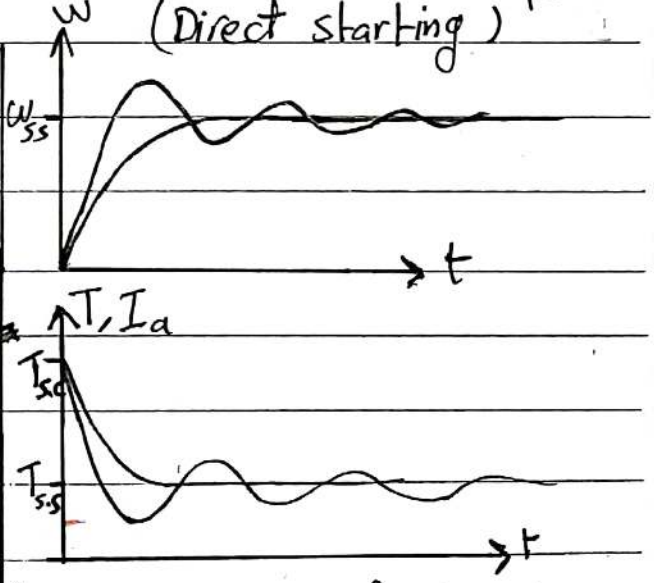
($I_{s.c}$ عالي & لمشاعاة كبيرة (أكبر من اللازم) لهيم الجزء

$\Delta P = R_a I_{s.c}^2$

يتم استخدام أي وسائل وسيطة لتقليل تيار البدء في حالة (Direct starting)



"Static char of direct starting for sep DC moter"



"Dynamic char of direct starting for sep DC motor"

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SUBJECT DC motors

SECTION methods of transient operation

2) Electrical (Indirect) starting:

"used for meduim & large power motors"

* electrical methods use an additional device to reduce short circuit current

How to reduce $I_{s.c}$?:

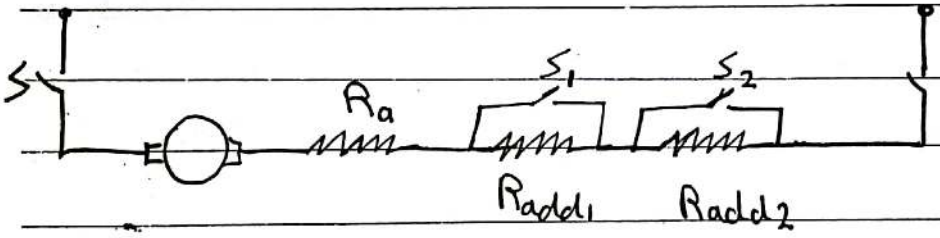
↳ by increasing R_{add}

↳ by reducing V_a (or by gradual the increas of $-V_a$)

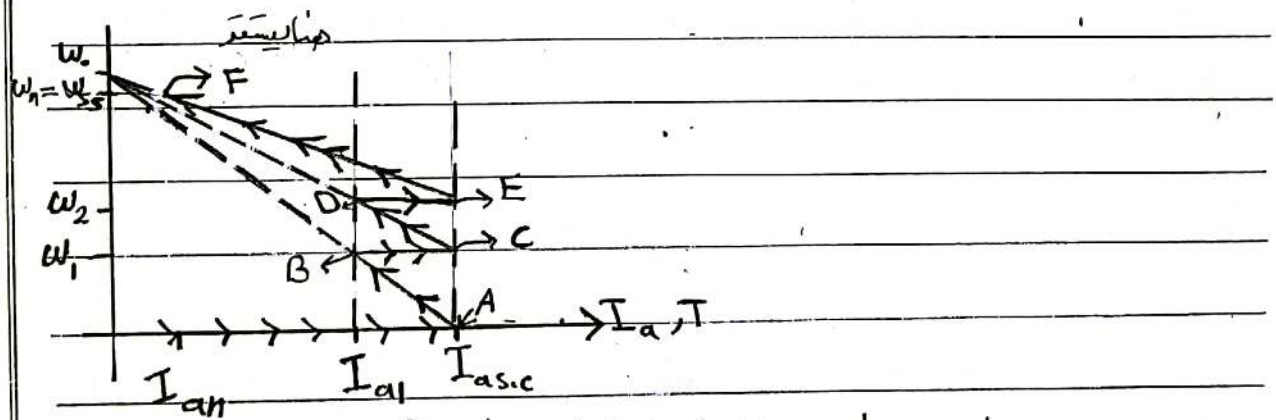
زيادة V_a ولكن بالسالب

① starting by inserting R_{add} in to the armature circuit:

$$I_{sc} = \frac{V_a}{R_a} \Rightarrow I_{s.c} = \frac{V_a}{R_a + R_{add}} \quad (R\uparrow \Rightarrow I_{s.c}\downarrow)$$



* $I_{as.c}$ should be $\leq (2-2.3)\, I_{an}$



static char of Rheostatic start in two steps

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SUBJECT DC motors

SECTION methods of transient operations

* At first S is closed only

* تبدأ الحركة من السكون (I = Is.c) & (ω = 0) point A

* بما أن كل المقاومات على التوالي ⇐ مفتوحان (S_1 & S_2)

$$\sum R_a = R_a + R_2 + R_1$$

$$I_{s.c} = \frac{V_a}{R_a + R_1 + R_2}$$ ⇒ تم حساب Radd من هنا لعلاقة

$$R_{add} = R_1 + R_2$$

* تنطلق الحركة باكتساب سرعة إلى أن يصل point B

$$A \to B \Rightarrow I = I_1 \;\&\; \omega = \omega_1 = \frac{V_a - I_1 (R_a + R_1 + R_2)}{K\phi}$$

* Then S_1 is closed ⇒ $R_\Sigma = R_a + R_2$

* عند إغلاق (S_1) ينتقل المحرك فوراً من B إلى C :-

$$A \to C \Rightarrow I = I_{s.c} \;\&\; \omega = \omega_1 = \frac{V_a - I_{s.c} (R_a + R_2)}{K\phi}$$

ملاحظة :- سرعة المحرك عند كل من C & B هي نفسها لكنه لفرق في المعادلتين أنه عند B يكون التيار منخفض والمقاومة عالية.

أما عند C فيكون التيار عالي والمقاومة تقل بسبب إغلاق (S_1)

$$(I_a < I_{s.c}) \;\&\; (R_a + R_1 + R_2) > (R_a + R_2)$$

* تنطلق الحركة باكتساب سرعة إلى أن يصل D :-

$$A \to D \Rightarrow I = I_1 \;\&\; \omega = \omega_2 = \frac{V_a - I_1 (R_a + R_2)}{K\phi}$$ ⇒

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* Then S_2 is closed $\Rightarrow \varepsilon R_a = R_a$

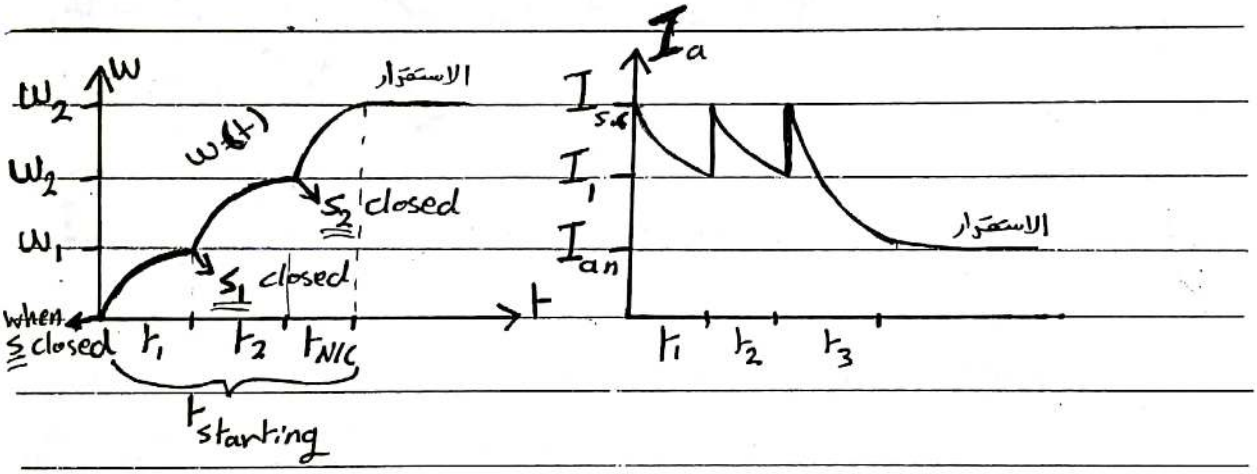
* عند اغلاق S_2 ينتقل المحرك فوراً من D إلى F :-

At E $\Rightarrow I = I_{s.c}$ & $w = w_2 = \frac{V_a - I_{s.c}(R_a)}{K\phi}$

* ينتقل المحرك بهذه الخاصية حتى يصل النقطة (الاستقرار) F :-

$I = I_{an}$ & $w = w_n = w_{ss} = \frac{V_a - I_{an} R_a}{K\phi}$

عادة يتم ايجاد $(K\phi)$ من العلاقة السابقة واثبت في معطيات:



Dynamic char During starting for 2 rheostat steps

يمكن ايجاد الـ Dynamic char بسهولة عبر تتبع I & w في الـ static char من بدء التشغيل إلى الاستقرار.

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Ex: A DC sep excited motor has following parameter: $P_{2n} = 32\,KW$, $I_{an} = 172\,A$, $V_{an} = 220V$, $R_a = 0.062\,\Omega$, $n_n = 1000\,rpm$, $I_{st} = I_{s.c} = 350\,A$, $I_1 = 250\,A$

Find R_{add} and plot the running (starting) char.

Sol:

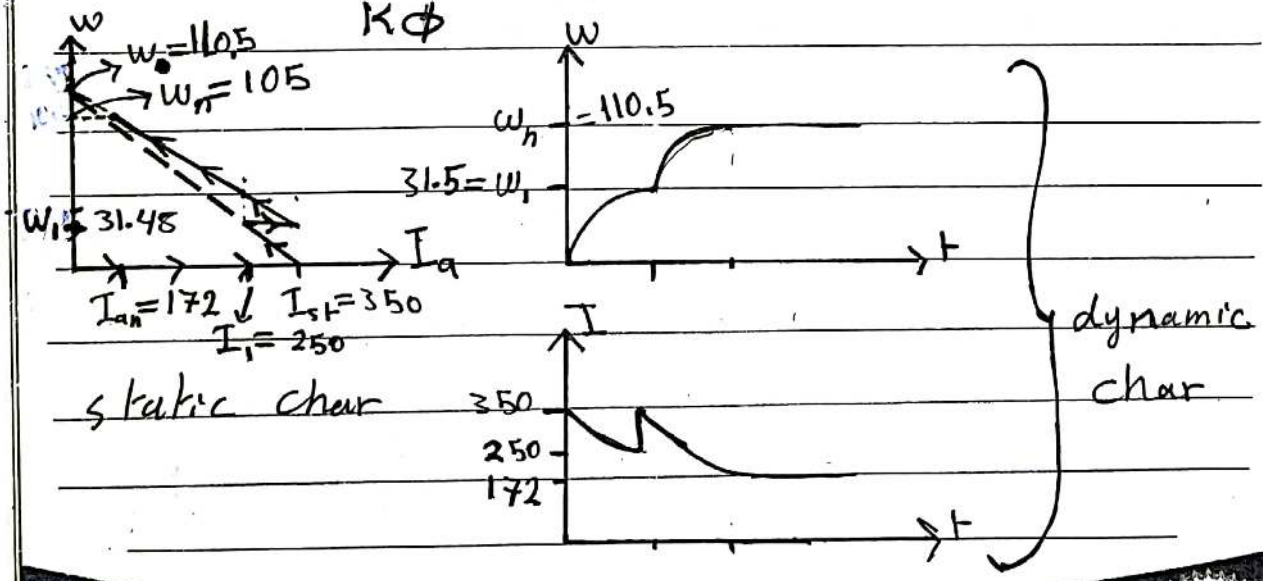
$$\omega_n = \frac{2\pi}{60} n_n = 105\ rad/sec$$

$$\omega = \frac{V_{an} - I_{an} R_a}{K\phi} \Rightarrow K\phi = 1.99$$

$$\omega_\circ = \frac{V_{an}}{K\phi} = \frac{220}{1.99} = 110.5\ rad/s$$

$$I_{st} = \frac{V_a}{R_a + R_{add}} \Rightarrow R_{add} = 0.567\,\Omega$$

$$\omega_1 = \frac{V_a - I_1 (R_a + R_{add})}{K\phi} = 31.48\ rad/s$$



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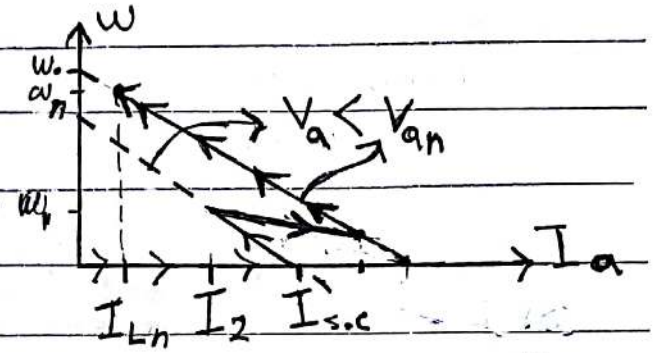
(2) Starting by reducing Voltag :

$I_{s.c} = \frac{V_a}{R_a} \Leftrightarrow (V_a \downarrow \Rightarrow I_{s.c} \downarrow)$

Natural char (N/C)

$I_2 < I_{s.c}$ (N/C)



* يقلع المحرك على الجهد المنخفض ثم يتم رفعه.

B Methods of Braking (stopping) of DC motor:

1) Mechanical braking
2) Electrical braking

1) Mechanical braking / For sep motor :

يتم فصل المنبع عن العضو وجعل المحرك يتوقف نتيجة تأثير الحمل.

$(I_a \& T_M) = 0$ ← يفصل العضو

ويحتاج هذا التوقف لوقت كبير لذلك يستخدم كابح ميكانيكي لتقليل زمن التوقف عبر زيادة طاقة الاحتكاك.

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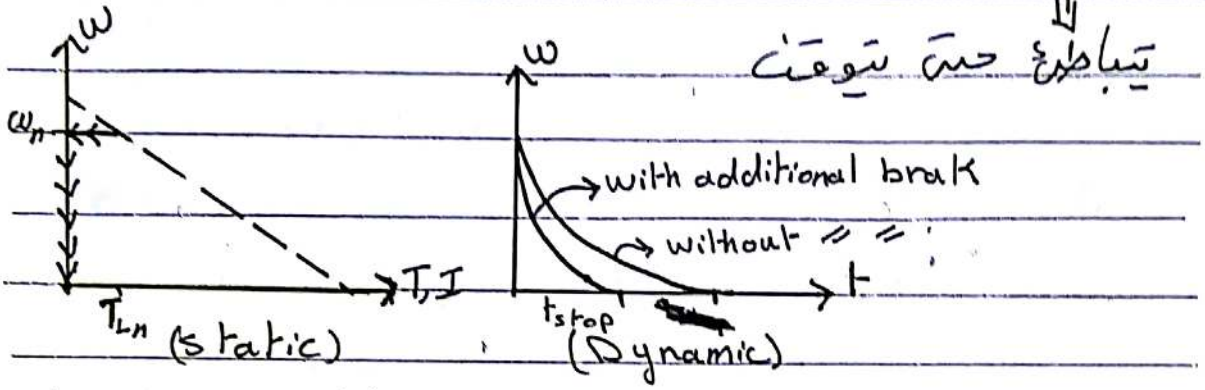
$T_M - T_L = J_\varepsilon \frac{dw}{dt} \Rightarrow -T_L = J_\varepsilon \frac{dw}{dt} \Rightarrow \frac{dw}{dt} < 0$

⇓

تباطؤ حتى تتوقف



Kinatic energy $(E_K) = \frac{1}{2} J_e w_{ss}^2$

E_K goes to friction and windings

2) electrical braking methods:

(in this method the motor torque should be (-ve))

-ve ← $T - T_L = J_\varepsilon \frac{dw}{dt} \Rightarrow \frac{dw}{dt} < 0$

* the torque becomes negative if I_a changes it's direction.

1) Dynamic braking of sep DC motor:

يتم فصل المنتج عن الحمل مع إغلاق دارة عبر إضافة مقاومة خارجية على التوالي مع المنتج ± اتجاه ملفات المنتج (T-w) موصلة مع الحمل.

جاي غلق دارة المنتج ← $V_a = 0$ ← $I_a = \frac{\vec{V_a} - E_a}{R_a + R_{stop}}$ ← $I_a < 0$

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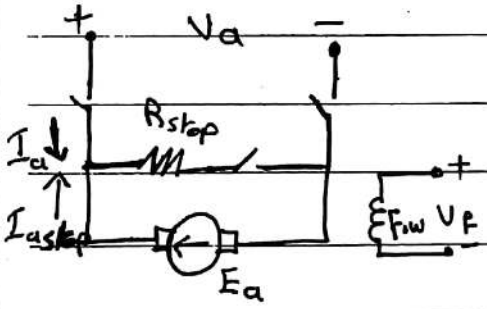
SUBJECT DC motors

SECTION method of transient operation

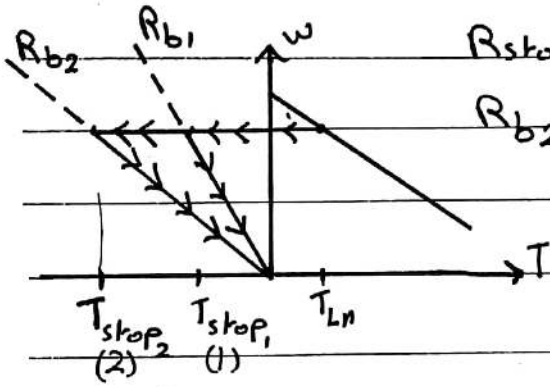
* R_{stop} يكون بتيار التوقف عالي جداً لذلك يتم اضافة مقاومة للحد من قيمته

$$I_{stop} = \frac{-E_a}{R_a + R_{stop}}$$

* سرعة الدوران بالاتجاه العكسي فيكون عزم التوقف ($I_a < 0 \Rightarrow T_{stop} = K\phi I_{stop} < 0$) كمولد وقتي طبيعي



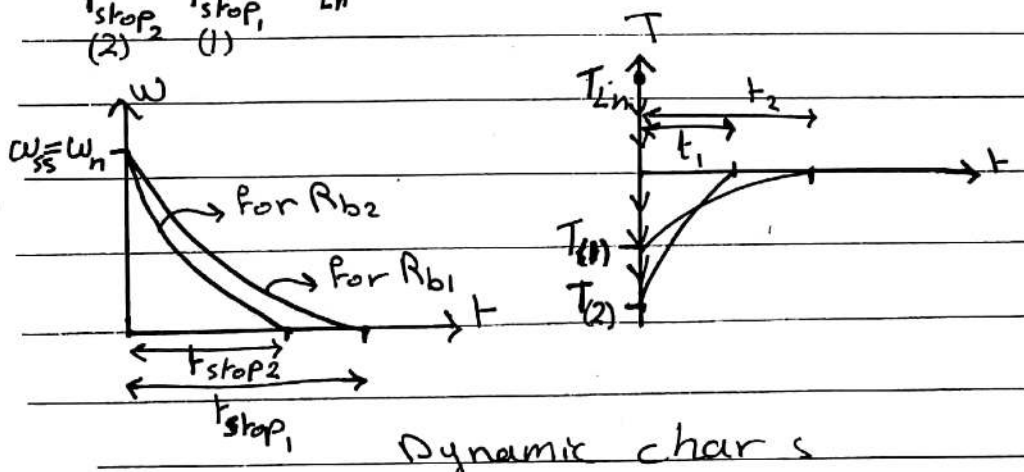
$R_{stop} \uparrow\uparrow \Rightarrow I \& T \downarrow\downarrow \Rightarrow t_{stop} \uparrow\uparrow$



$R_{stop} \equiv R_{braking}$

$R_{b2} < R_{b1} \Leftarrow T_{stop_2} > T_{stop_1}$

$\Rightarrow$ static char



Dynamic chars

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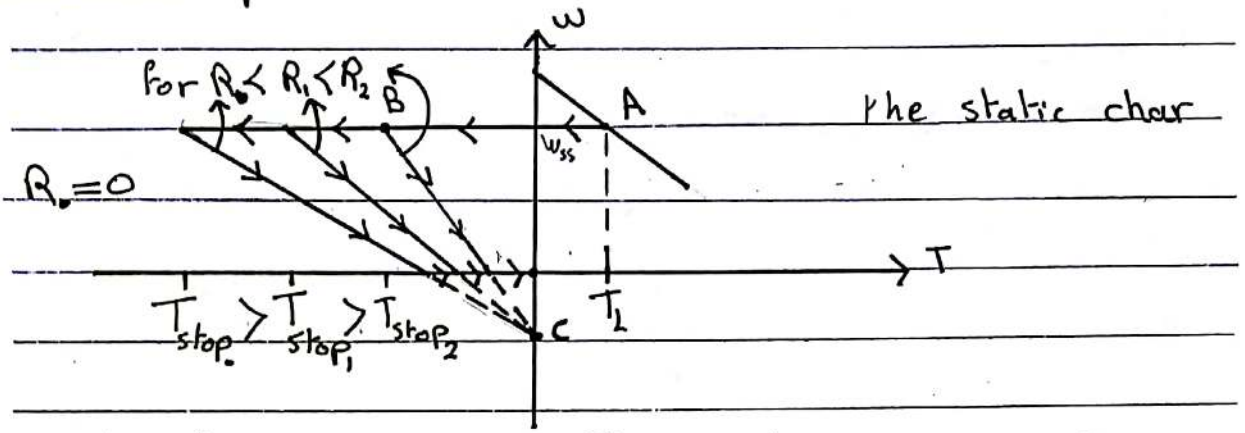
SUBJECT DC motors

SECTION method of transient operation

(2) plugging braking (change in polarity):

* أسرع أنواع التوقف (عالي T_{stop})

for sep motor:



the static char

* يتم عكس دوران المحرك عبر عكس قطبية المنتج ($-V_a$) حيث ينتقل المحرك للعمل في الربع الثاني (From A to B) ويبدأ المحرك بالتباطؤ تحت تأثير العزم الكهرومغناطيسي (T_{stop}), إلى أن يصل إلى (C) ($w=0$) فيتم عندها فصل المحرك عن المصدر وإدخال كابح ميكانيكي لمنع حركة المحرك في الاتجاه المعاكس).

- $I_a = \frac{-V_a - E_a}{R_{a\Sigma}} < 0 \Rightarrow I_a$ will chang it's direction
- $I_{a\,stop.} = 2I_{a\,s.c} >> I_n \Rightarrow$ خطر ← لذلك يتم إضافة R على التوالي مع المنتج
- I_n (التيار المقنن)

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SUBJECT DC motors

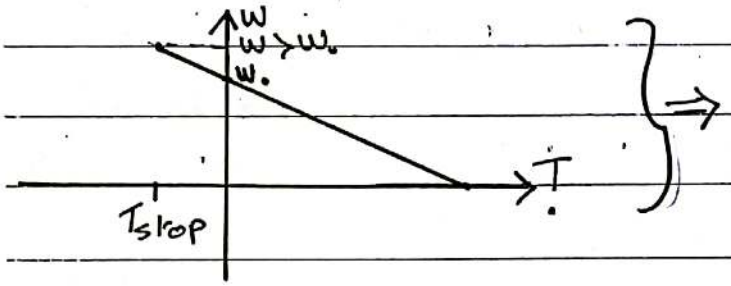
SECTION methods of transient operation

(3) Regenerative braking

- at $w = w_o \Rightarrow V_a = E_a$
- at $w > w_o \Rightarrow E_a > V_a \Rightarrow I_a < 0 \Rightarrow P_i < 0$ (from motor to source)

الشرط في التوليد العاكس هو أنه تكون القوة الدافعة الكهربائية المتولدة على طرفي المنتج > جهد الشبكة :-

$(E_a > V_a)$ happens when $\rightarrow w > w_o$

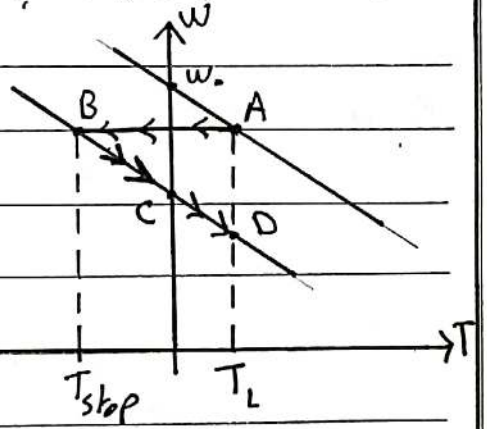


⇒ * زيادة السرعة للمحرك عبر انزال الحمل.

* يمكن أيضاً تحقيق حالة الكبح عبر خفض السرعة للمحرك عن طريق تقليل جهد المنتج.

فينتقل المحرك من خاصية السرعة الأكبر إلى خاصية السرعة الأقل (عبر الانتقال إلى الربع الثاني).

between (B & C) ⇒ يعمل المحرك كمولد ويعيد الطاقة للشبكة.



* قبل D: يكون $\frac{dw}{dt} < 0 \leftarrow T_M < T_L$

أي أن المحرك يتباطأ إلى أن يستقل عند D.

mech char for reg bra by reducing voltage

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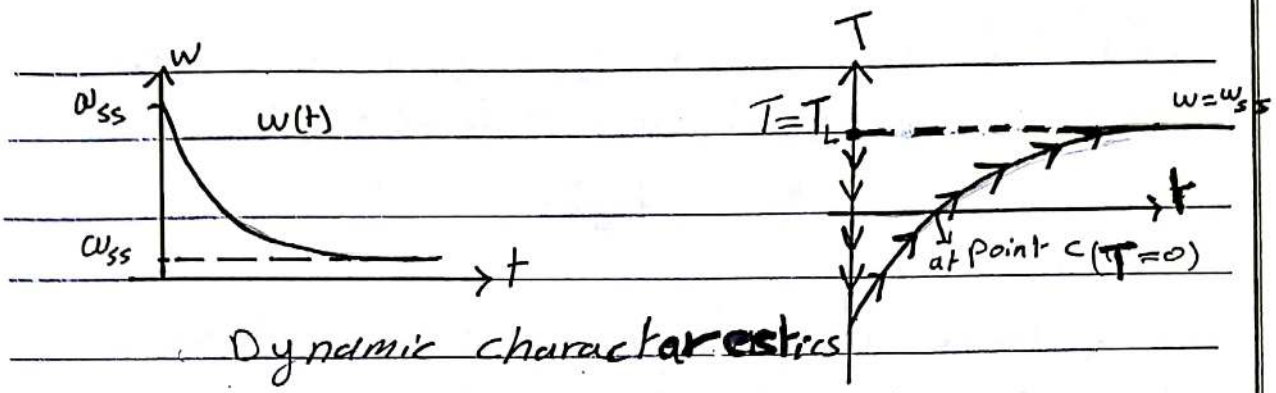
SUBJECT DC motors

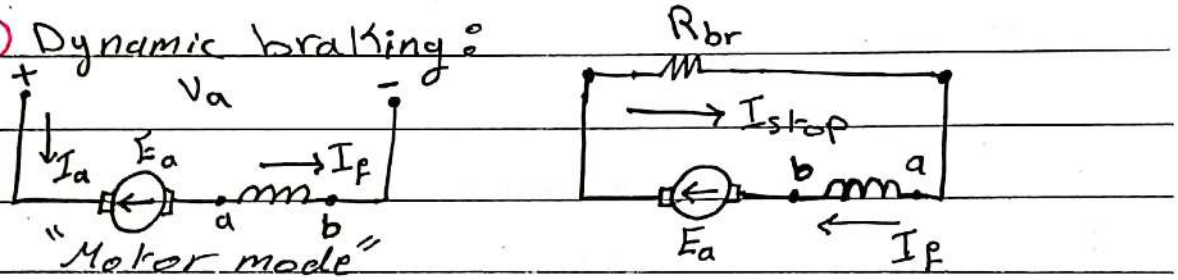
SECTION methods of transient operations



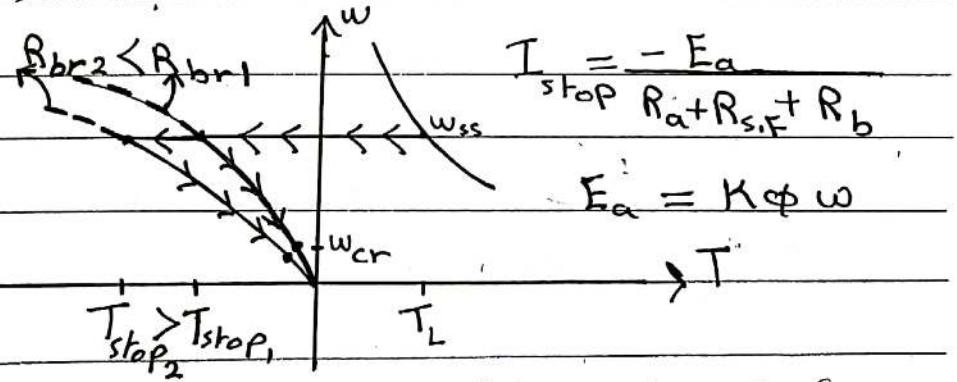
Dynamic characteristics

* Special considerations when braking series DC motors

① Dynamic braking:



A "Dynamic braking with self excitation" ← يتم فيها استخدام المحرك نفسه كمولد دون حاجة لمصدر خارجي



$$I_{stop} = \frac{-E_a}{R_a + R_{s.F} + R_b}$$

$$E_a = K\phi\omega$$

العزم عكسي لأنه I يعتمد على ϕ.

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SUBJECT DC motors

SECTION Transient operation methods of

ينتقل المحرك من A إلى الربع الثاني ثم يبدأ بالتباطؤ ويستمر على خاصية التوقف "الديناميكي":-

تباطؤ ($\omega\downarrow$) $\rightarrow E_a\downarrow \rightarrow I_a\downarrow \rightarrow \phi\downarrow$

يستمر التوقف الديناميكي إلى أنه تصل السرعة للقيمة الحرجة (ω_{cr}) وعندها لا تتحقق شروط الكبح الذاتي أي تنتهي عملية التوقف الديناميكي ويستمر بعدها المحرك بالتوقف الميكانيكي أي أنه تصل لسرعة إلى الصفر.

$\omega_{cr} < \omega < \omega_{ss} \Rightarrow$ dynamic braking

$0 < \omega < \omega_{cr} \Rightarrow$ mechanical braking

* Conditions for dyn braking:

1- I_f should remain in the same direction

تغيير اتجاه I_a والحفاظ على I_f على اتجاهه.

2- Limit I_{br} by inserting R_{br}.

3- The motor should have ϕ_{res}, $E_a >$ voltage drop

$\therefore$ voltag drop $= I_{br} R_\varepsilon$

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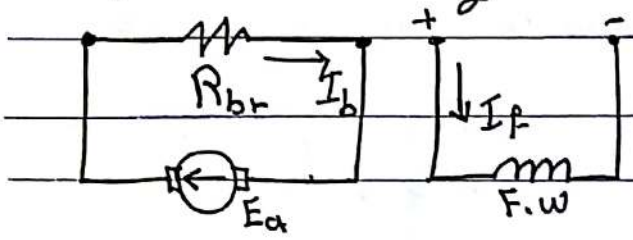
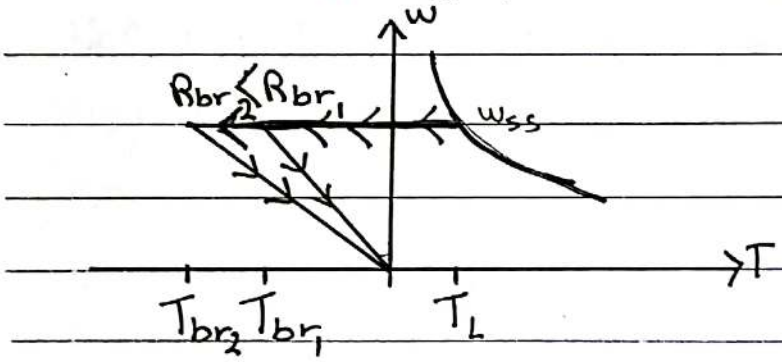
① Dynamic braking with separately excitation:

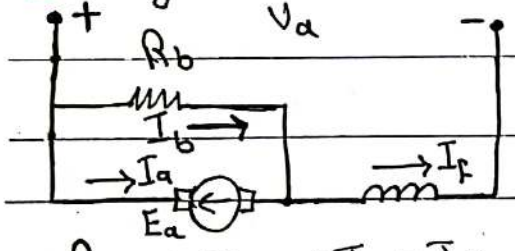


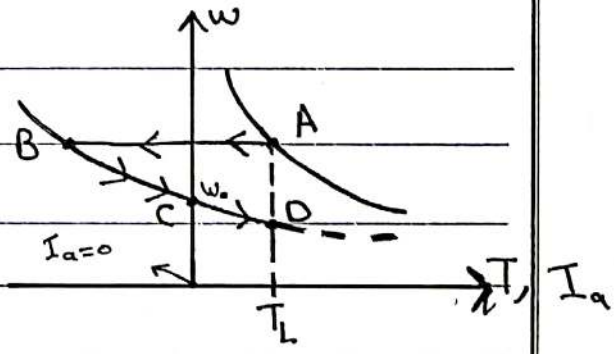
* Conditions: 1- I_f in the same direction

2- $I_b \ll I_{an}$



② Regenerative braking:





- $R_b \to \infty \Rightarrow I_a = I_f$
- R_b = finite value $\Rightarrow I_f = I_a + I_b \Leftrightarrow I_f \uparrow \Rightarrow \phi \uparrow \Rightarrow w \downarrow$
- at C $\Rightarrow$ no-load $\Rightarrow I_a = 0 \Rightarrow I_f = I_b \Rightarrow \phi \neq 0 \Rightarrow w_o \neq \infty$

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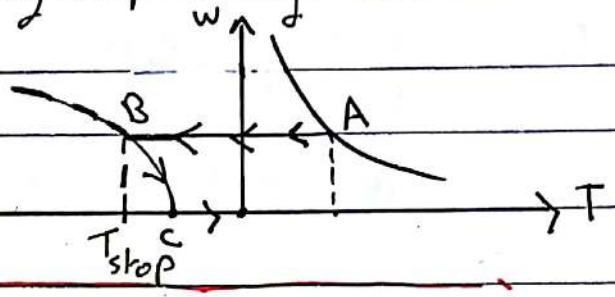
③ plugging braking:
.chang in polarity of armature only



EX: DC separately excited motor, has Ratings:
220 V, 970 rpm, 100 A, $R_a = 0.05\,\Omega$,
plugging braking from $n_{ss} = n_p = 1000$ rpm,
Find: ① $R_{plugging} = ??$ where $I_{plug} = 2 I_a$
② $T_{plugging} = ??$
③ $T = ??$ where $w = 0$

Sol: ① $E_{an} = V_{an} - I_{an} R_a = 215\,V$

$E_a \alpha n \Rightarrow E_a|_{at\ n_p = 1000} = E_{an} * \frac{n_p}{n_n} = 215 * \frac{1000}{970} = 221.65\,V$

for plugging operation $\Rightarrow I_{plug} = \frac{-V_a - E_a|_{n=1000}}{R_a + R_{plug}}$

$I_{plug} = 2I_a = 200 = \frac{-220 - 221.65}{0.05 + R_{plug}}$

$R_{plug} = 2.16\,\Omega$

$\Rightarrow$

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② $T = K\phi I_a$, $\omega_p = \frac{2\pi n_p}{60} = 104.72$

$E_a|_{np} = 221.65 = K\phi \omega_p$

$K\phi = \frac{221.65}{104.72} = 2.116$

$T_p = 2.116 * (2*100) = 423.3$ N.m

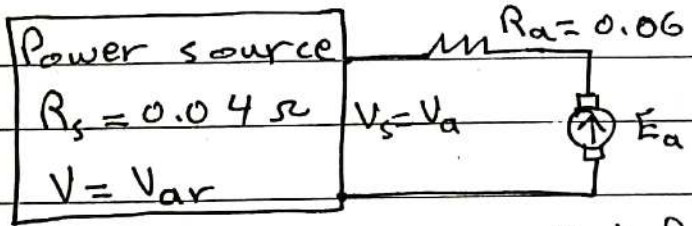
③ at $w=0 \Rightarrow E_a = K\phi w = 0 \Rightarrow I_a = \frac{V_a - E_a}{R_a + R_{plug}}$

$I_a = 99.55$A

$T \propto I_a \Rightarrow T|_{at w=0} = T_{plug} * \frac{I_{(w=0)}}{I_{(wp=104.72)}}$

$T_{(w=0)} = 423.3 * \frac{99.55}{200} = 210.7$ N.m

Ex: DC separately excited motor, has ratings: 220V, 200A, 800 rpm, $R_a = 0.06\ \Omega$



Find (V_s), knowing that for regenerative braking:

$T_{br} = 80\% T_n$

$n_{br} = n_{ss} = 600$ rpm

⇒

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SUBJECT DC motors

SECTION static char of 3-phase IM

Sol:- $T \propto I_a \Rightarrow I_{br} = 0.8 I_n = 160\ A$

$E_{an} = V_{an} - I_{an} R_a = 208\ V$

$E_a \propto n \Rightarrow \frac{E_{a(br)}}{E_{a(n)}} = \frac{n_{br}}{n_n} \Rightarrow E_{a(br)} = 156\ V$

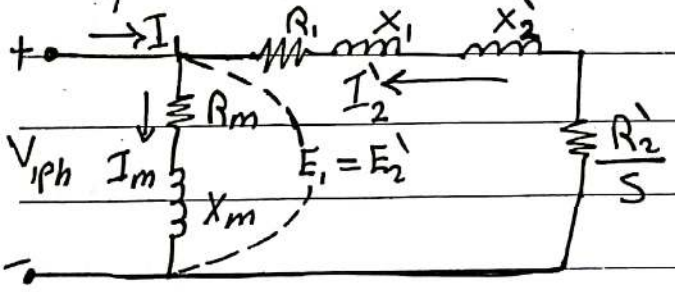
$I_{br} = \frac{E_{a(br)} - V_{a(br)}}{R_a + R_s} \Rightarrow V_{a(br)} = V_s = 140\ V$

Static char of 3 phase induction motor:-

1- Electromechanical char ($I = f(\omega)$).

2- Mechanical char ($T = f(\omega)$)

Equivalent sheem for T type:-



1 ~ means stator 2 ~ means for rotor

2' ~ mens for rotor (reffered to stator)

* في المحركات الصغيرة يتم إهمال Im لأنه تقريباً = صفر

- $R_1 = c_1 r_1$, R_1 is the equivalent resistance of stator.
- $X_1 = c_1 x_1$, r_1 // // real resistance // //.
, X_1 // // equivalent inductance // //.
, x_1 // // real inductance // //.

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SUBJECT DC motors

SECTION static char of 3-phase IM

- $R_2' = C_1^2 r_2'$
- $X_2' = C_1^2 x_2'$

- $R_m = r_m + r_1$ (For magnetic resistance)
- $X_m = x_m + x_1$ (For magnetic inductance)

- $\left(C_1 = 1 + \frac{X_1}{X_m}\right) \Rightarrow$ transformation ratio

- $s = \frac{n_s - n_r}{n_s} = \frac{\omega_s - \omega_r}{\omega_s}$

$\omega_s - s\omega_s = \omega_r \Rightarrow \boxed{\omega_s (1-s) = \omega_r}$

- $n_s = \frac{60 f_1}{P/2} = \frac{120 f}{P}$ ($f \rightarrow$ frequency, $P \rightarrow$ poles) , $\omega_s = 2\pi n_s$

If $I_m =$ zero $\Rightarrow I_1 = I_2'$ (rms value)

$$I_2' = \frac{V_{1ph}}{\sqrt{X_{s.c}^2 + \left(R_1 + \frac{R_2'}{s}\right)^2}}$$ ⇐ From the eq T circuit

↳ electro mechanical char $\left(I_2' = f(s) \text{ or } I_2' = f(\omega)\right)$

* بتعويض قيم مختلفة للانزلاق (s) في العلاقة نحصل على الخاصية الكهروميكانيكية.

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SUBJECT AC motors

SECTION static char of 3-phase IM

s	$-\infty$	~~~~	0	~~~	1	~~~~	$+\infty$
I_2'							

$w_r = (1-s)\, w_s$

↳ $s = 0$ (no load state)

↳ $w_r = w_s$

↳ $\frac{R_2'}{s} = \frac{R_2'}{0} = \infty \Rightarrow I_2' = 0$ (مقاومة عالية جداً فلا تمر منها تيار)

↳ $s = 1$ (short circuit)

↳ $w_r =$ zero

↳ $\frac{R_2'}{s} = R_2' \Rightarrow$ أقل مقاومة ممكنة $I_{2st}' = I_{2s.c}'$

$$I_{2st}' = \frac{V_{1ph}}{\sqrt{X_{s.c}^2 + (R_1 + R_2')^2}}$$

↳ $s = \pm\infty \Rightarrow \frac{R_2'}{s} = 0 \Rightarrow$

$$I_2' = \frac{V_{1ph}}{\sqrt{X_{s.c}^2 + R_1^2}}$$

↳ when $s < 0 \Rightarrow (I_{2\,max}')$

(يعمل المحرك كمولد) $R_1 + \frac{R_2'}{s} = 0 \Rightarrow$

$$I_{2\,max}' = \frac{V_1}{X_{s.c}}$$

$\Rightarrow s = -\frac{R_2'}{R_1}$

$I_2'\big|_{s<0} > \lim_{s \to \pm\infty} I_2'$

← هي أكبر قيمة ممكنة للتيار وأكبر من التيار عند ($s = \pm\infty$)

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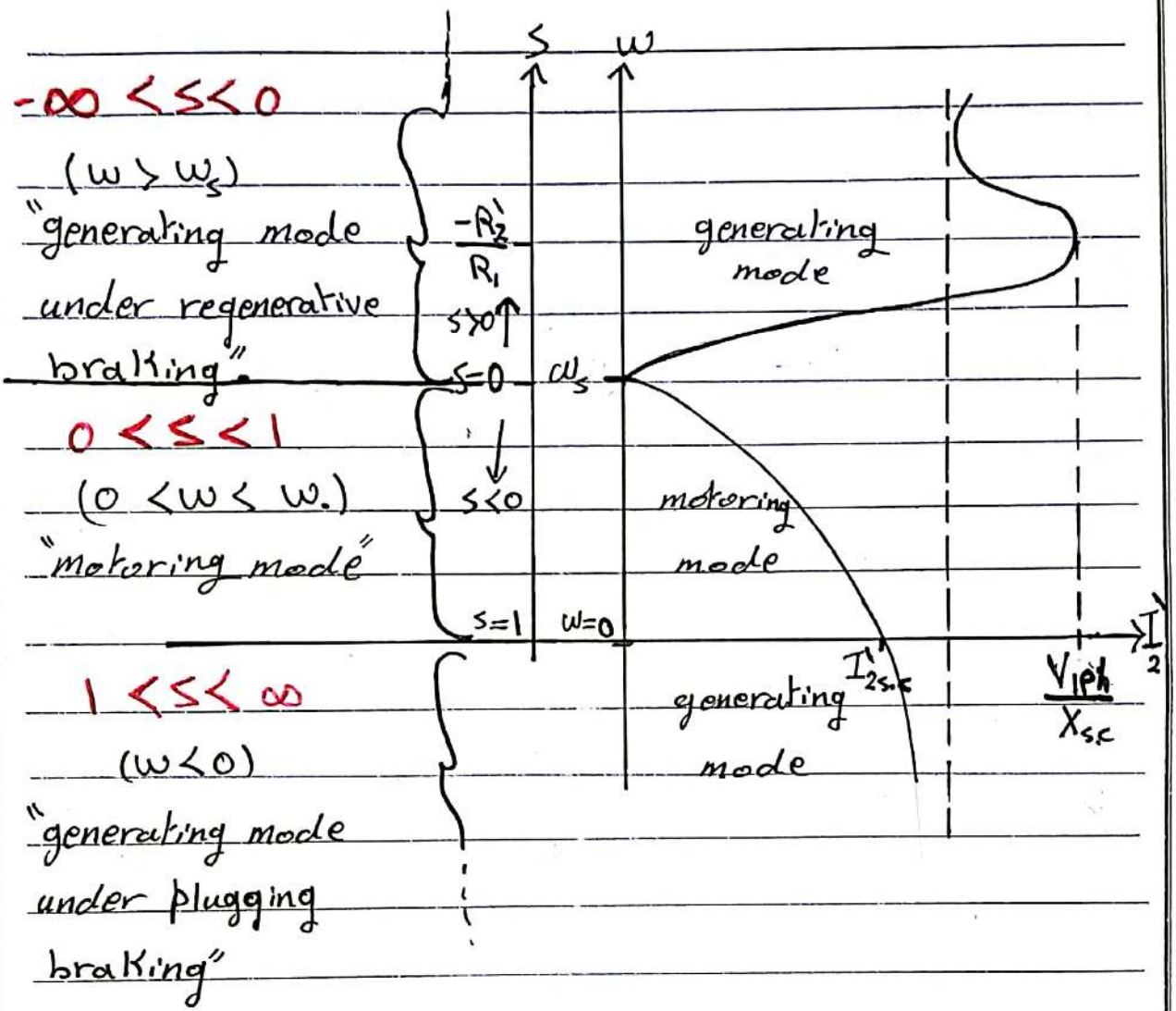
SUBJECT AC motors

SECTION Static char of 3-phase IM



(Electro mechanical char for induction motor in both motoring and generating mode)

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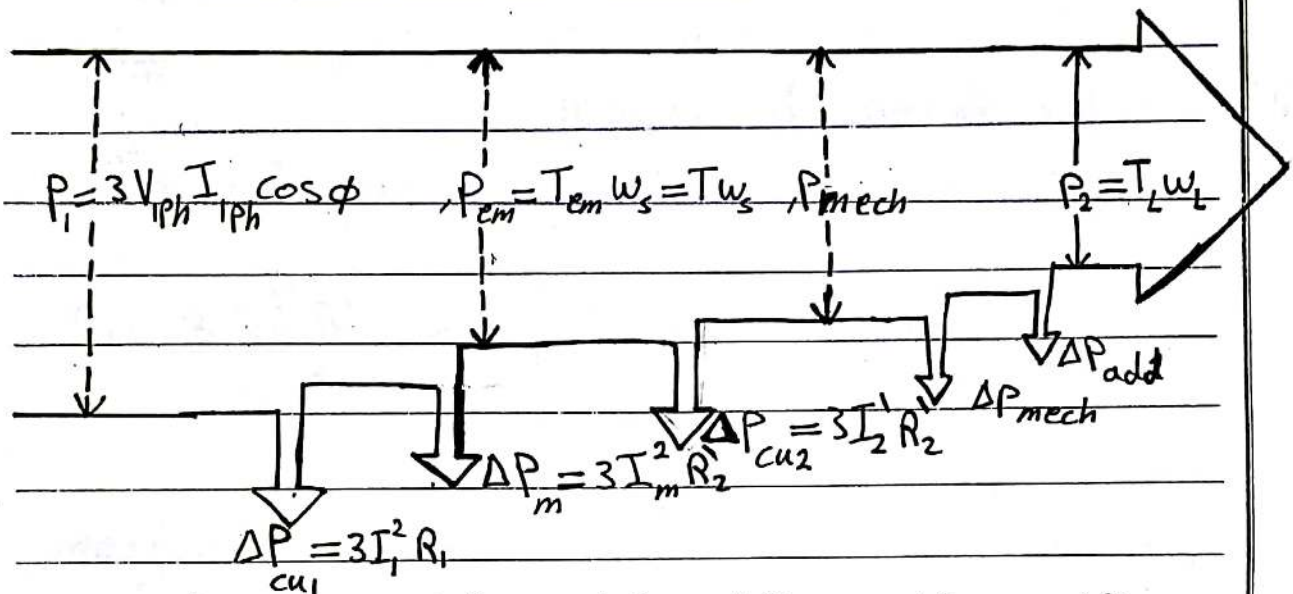
SUBJECT AC motors

SECTION Static char of 3-phase I.M

Induction Ac motor power Flow

يوضح تسلسل استهلاك القدرة في دارة المحرك الحثي.



- مجموع الفقد $(\Sigma\Delta P) = \Delta P_{cu_1} + \Delta P_m + \Delta P_{cu_2} + \Delta P_{mech} + \Delta P_{add}$
  - ΔP_{cu_1}: cupper losses in stator
  - ΔP_{cu_2}: cupper losses in rotor
- $P_2 = P_1 - \Sigma\Delta P$
- $\Delta P_{mech} + \Delta P_{add} \cong 0.015 P_{in} \cong$ zero $\Rightarrow P_{mech} = P_2$
- $\Delta P_{mech} \neq P_{mech}$
  - ΔP_{mech}: mech losses
  - P_{mech}: mech power
- $T = T_{em} + \Delta T \cong T_{em}$
- $P_{em} = \Delta P_{cu_2} + P_{mech} = \Delta P_{cu_2} + P_2$
- $P_{em} = 3I_2'^2(R_2')/s$

$\Delta P_{cu_2} = 3I_2'^2 R_2' = sP_{em}$ $\quad P_{mech} = P_{em} - P_{cu} = \left(\frac{1}{s} - 1\right) 3I_2'^2 R_2 = (1-s)P_{em}$

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SUBJECT AC motors

SECTION Static char of 3-phase IM

$\Delta P_{cu2} = P_{em} - P_2$, $T_{em} = T$

$\Delta P_{cu2} = T_{em}\omega_s - T\omega_r$

$\Delta P_{cu2} = T(\omega_s - \omega_r)$, $s = \frac{\omega_s - \omega_r}{\omega_s}$

$\Delta P_{cu2} = T s \omega_s$ ⇒ this means that

(ΔP_{cu2}) is a function of (s). ($\Delta P_{cu2} = f(s)$)

called slip losses

$3 I_2'^2 R_2' = T s \omega_s$

$T = \frac{3 I_2'^2 R_2'}{\omega_s s}$ ← sub in $I_2' = \frac{V_{1ph}}{\sqrt{X_{s.c}^2 + (R_1 + \frac{R_2'}{s})^2}}$

$T = \frac{3 V_{1ph}^2 R_2'}{\omega_s s \left(X_{s.c}^2 + (R_1 + \frac{R_2'}{s})^2\right)}$

$I \propto V_{ph}$

$T \propto V_{ph}^2$

mech char ⇐

$T = f(\omega)$

$T = f(s)$

- At $s = 0 \Rightarrow T = 0$ & $\omega_r = \omega_s$
- At $s = 1 \Rightarrow T_{starting} = T_{s.c} = \frac{3 V_{1ph}^2 R_2'}{\omega_s (X_{s.c}^2 + (R_1 + R_2')^2)}$ & $\omega_r = 0$
- To Find max values ⇒ $\frac{dT}{ds} = 0 \Rightarrow s_{1,2} = \frac{\pm R_2'}{\sqrt{R_1^2 + X_{s.c}^2}}$ ⇓ the critical slips at which T_{max} accoure

$T_{max} = \frac{3 V_{1ph}^2}{2\omega_s (R_1 \pm \sqrt{R_1^2 + X_{sc}^2})}$ ← sub in T equa

+ ⇝ For generating mode

− ⇝ For motoring mode

⇒ $T_{max\ motor} < T_{max\ gen}$

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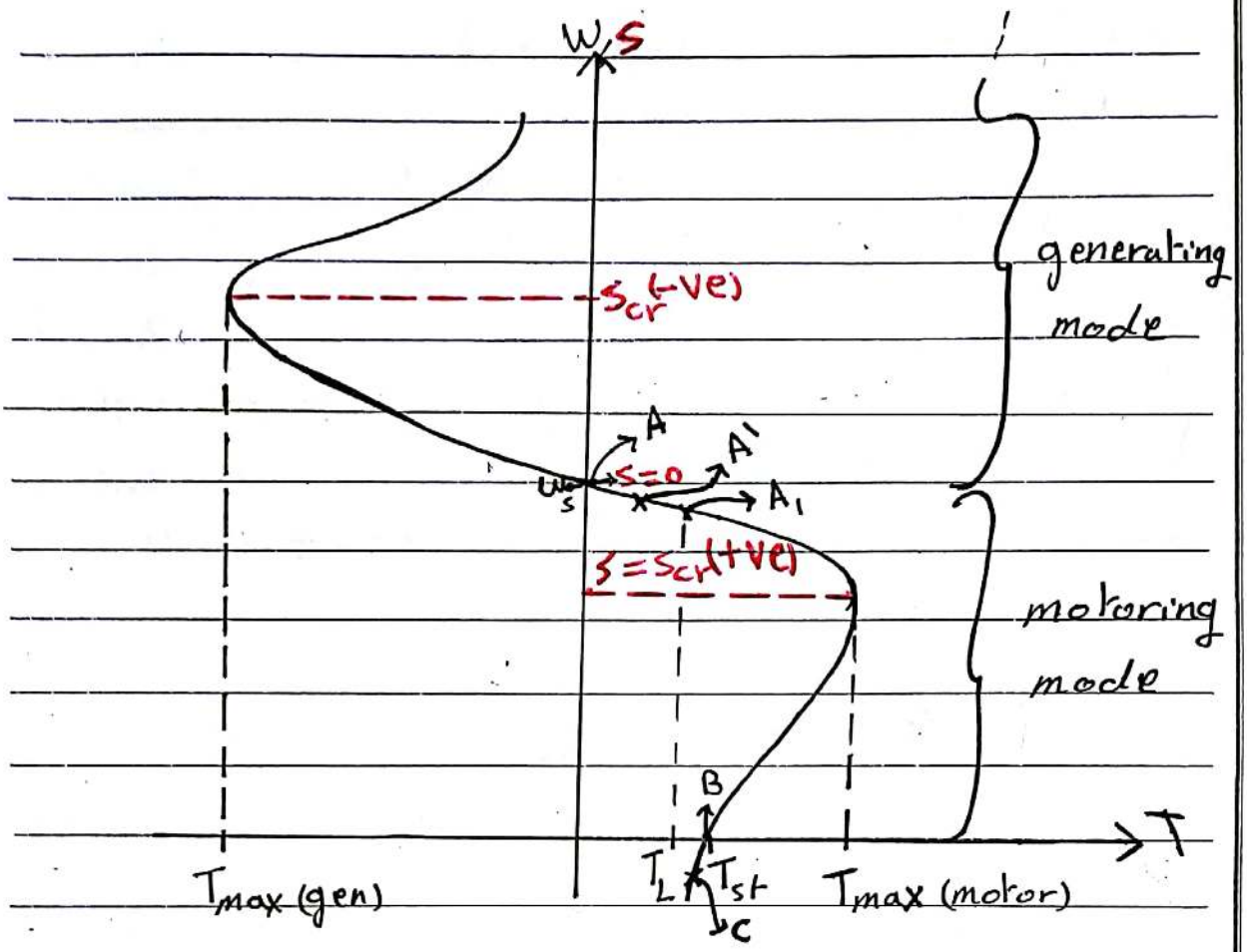
SUBJECT AC motors

SECTION static char of 3-phase IM



"mechanical char for induction motor in both motoring and generating modes"

- $A - A_1$ ⇒ the linear part
- $A_1 - B$ ⇒ القطع لزائد

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SUBJECT AC motors

SECTION static char of 3-phase IM

* يمكن كتابة العزم بدلالة (S_{cr} & T_{max}) كالآتي:

$$T = \frac{2\,T_{max}\,(1 + a\,S_{cr})}{\frac{S}{S_{cr}} + \frac{S_{cr}}{S} + 2a\,S_{cr}} \qquad , a = \frac{R_1}{R_2'}$$

* For industrial (في المحركات ذات القدرات المتوسطة والعالية)

↳ $R_1 \approx 0$

$$R_1 \approx 0 \Rightarrow a \approx 0 \Rightarrow T = \frac{2\,T_{max}}{\frac{S}{S_{cr}} + \frac{S_{cr}}{S}}$$

$$\text{and} \Rightarrow S_{cr} = \pm \frac{R_2'}{X_{s.c}}$$

$$\text{and} \Rightarrow T_{max} = \frac{3\,V_{1ph}^2}{2\,\omega_s\,X_{s.c}}$$

A when $S > S_{cr} \Rightarrow \frac{S_{cr}}{S} \approx 0$

$$\text{so} \Rightarrow T = \frac{2\,T_{max}}{\frac{S}{S_{cr}}} = \frac{B}{S} \Rightarrow$$ العلاقة التي تمثل القطع الزائد في المنحنى

$$T \propto \frac{1}{S}$$

B when $S_{cr} > S \Rightarrow S = (0 - 0.3)\,S_{cr} \Rightarrow \frac{S}{S_{cr}} \approx 0$

$$\text{so} \Rightarrow T = \frac{2\,T_{max}}{\frac{S_{cr}}{S}} = A * S \Rightarrow$$ تمثل الجزء الخطي من المنحنى

$$T \propto S$$

$$B(\text{constant}) = \frac{2\,T_{max}}{\frac{1}{S_{cr}}} \quad \& \quad A(\text{constant}) = \frac{2\,T_{max}}{S_{cr}}$$

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SUBJECT AC motors

SECTION Static char of 3-phase IM

* under normal conditions:

- $S_n = 0.03 - 0.05$
- $S_{cr} = 0.18 - 0.25$

* under rated conditions:

$$T = T_n = \frac{2\,T_{max}}{\frac{S_n}{S_{cr}} + \frac{S_{cr}}{S_n}}$$

* $\lambda = \frac{T_{max}}{T_n} \Rightarrow T_{max} = \lambda T_n$

* $S_{cr} = S_n \left[\lambda \pm \sqrt{\lambda^2 - 1}\right]$

Ex: 3 phase induction motor has:

50 hp, 440 V, 50 Hz, $T_{max} = 250\% T_n$

$S_{cr} = 10\%$, $R_1 = 0$, $\Delta P_{mech} = 0$ (rotational losses)

* Find: [1] n_n, [2] P_{cu2}, [3] T_{st}

sol:

1) $S_{cr} = S_n \left(\lambda \pm \sqrt{\lambda^2 - 1}\right)$

$0.1 = S_n \left(2.5 \pm \sqrt{(2.5^2 - 1)}\right) \Rightarrow S_n = 0.021$, ~~$-0.25$~~

$S_n = \frac{n_s - n_{rn}}{n_s}$, $n_s = \frac{120 f}{P} = \frac{120 * 50}{4} = 1500$ rpm

$0.021 = \frac{1500 - n_n}{1500} \Rightarrow n_n = 1468.5$ rpm

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or we can find S_n from $\Rightarrow T_n = \dfrac{2T_{max}}{\frac{S_n}{S_{cr}} + \frac{S_{cr}}{S_n}}$

$$\frac{T_n}{T_{max}} = \lambda = \frac{2}{\frac{S_n}{S_{cr}} + \frac{S_{cr}}{S_n}}$$

2) $P_{em} = \Delta P_{cu_2} + P_2$

$3 I_2'^2 R_2' S = 3 I_2'^2 R_2' + 3 I_2'^2 R_2' \frac{1-S}{S} \Rightarrow \Delta P_{cu_2} = S P_{em}$

and $\Rightarrow P_2 = P_{em}(1-S)$

$\dfrac{\Delta P_{cu_2}}{P_2} = \dfrac{S}{1-S}$, $P_2 = 50$ [redacted] $* 746$ watt

$\Delta P_{cu_{2_n}} = \dfrac{P_2 S_n}{1 - S_n} = 800$ w

3) $T_{st}\big|_{s=1} \Rightarrow T_{st} = \dfrac{2T_{max}}{\frac{1}{S_{cr}} + \frac{S_{cr}}{1}} = 1200$ N.m

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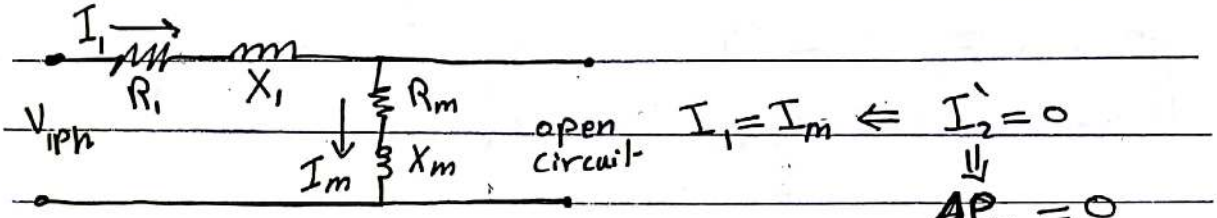
Operating force modes of induction motors
(same as in DC motors)

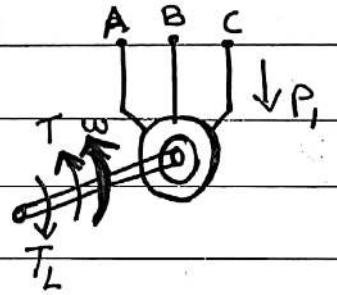
1 No-load mode (A):

$\omega = \omega_s$, $T = 0$, $s = 0$



- $P_1 = P_2 + \Sigma\Delta P$
- $P_2 = T\omega = 0$ $\Big] \Rightarrow \eta = 0$
- $\Sigma\Delta P = \Delta P_{cu_1} + \Delta P_m + \Delta P_{cu_2} + \Delta P_{mech} + \Delta P_{add}$

$\Rightarrow P_1 = \Sigma\Delta P = \Delta P_{cu_1} + \Delta P_m$



عملياً بسبب الاحتكاك يوجد عزم قليل
فتعمل المحرك عند النقطة (A')

2 Motoring mode (from A to B):

$\omega_s > \omega > 0$ $\omega_s \Leftarrow$ سرعة المجال الثابت

$0 < T < T_{s.c}$

$0 < s < 1$

* يعمل المحرك عملياً لفترات طويلة عند
($T_n, \omega_n, \Sigma\Delta P_n$) rated values
* يمكن أنه يعمل عند $T_L < T_n$ (Linear part) $\Leftarrow \downarrow\eta$
* يمكن أن يعمل $T_L > T_n$ لفترات زمنية قصيرة $\Leftarrow$ تسخن المحركات
$\lambda < \lambda_n$ & وتتلف..

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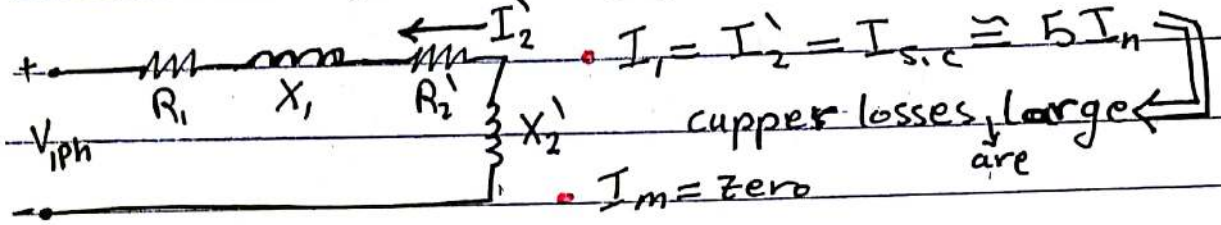
In motoring mode $\Rightarrow P_1 = P_2 + \Sigma\Delta P$
& $P_2 = T\omega \neq$ zero

3] short circuit mode (starting) (B) :
$\omega = 0$, $T = T_{s.c}$, $s = 1$



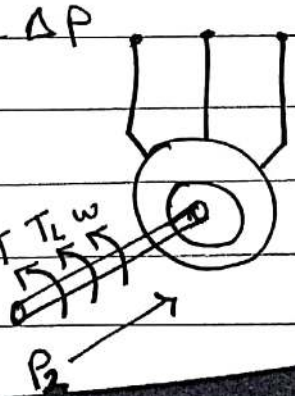
$I_1 = I_2' = I_{s.c} \cong 5 I_n$
cupper losses are large
I_m = zero

$P_2 = T\omega$ = zero
$\Sigma\Delta P = \Delta P_{cu1} + \Delta P_{cu2} = 3 I_{s.c}^2 (R_1 + R_2')$
$P_1 = P_2 + \Sigma\Delta P = 3 I_{s.c}^2 (R_1 + R_2')$

يعمل المحرك هنا في حالة عدم تمكنه من الاقلاع أو زيادة الحمل والحمل لفترات طويلة عند هذه الحالة يسبب تلف المحرك.

4] Regerative mode (D)
$\omega > \omega_s$
$T < 0 \Rightarrow (P_2 = T\omega) < 0 \Rightarrow$ (P_2 إذن يعكس) ينعكس اتجاه
$s > 0$ ($P_1 < 0$) ← ينعكس اتجاه إذن

$-P_1 = -P_2 + \Sigma\Delta P \Rightarrow P_2 = P_1 + \Sigma\Delta P$



motor operate as gen in parallel to source

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SUBJECT AC motors

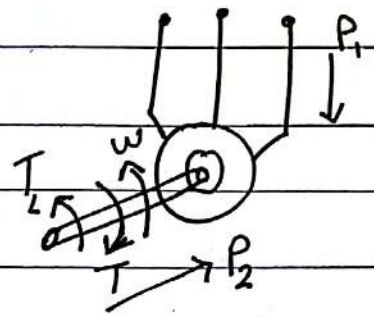
SECTION operating force mode of I.M / synch motor

5 plugging mode (C) :

$\omega < 0 \Rightarrow (P_2 = T\omega) < 0$

$T < T_{s.c}$

$s > 1$



motor operate as gen

in series to source

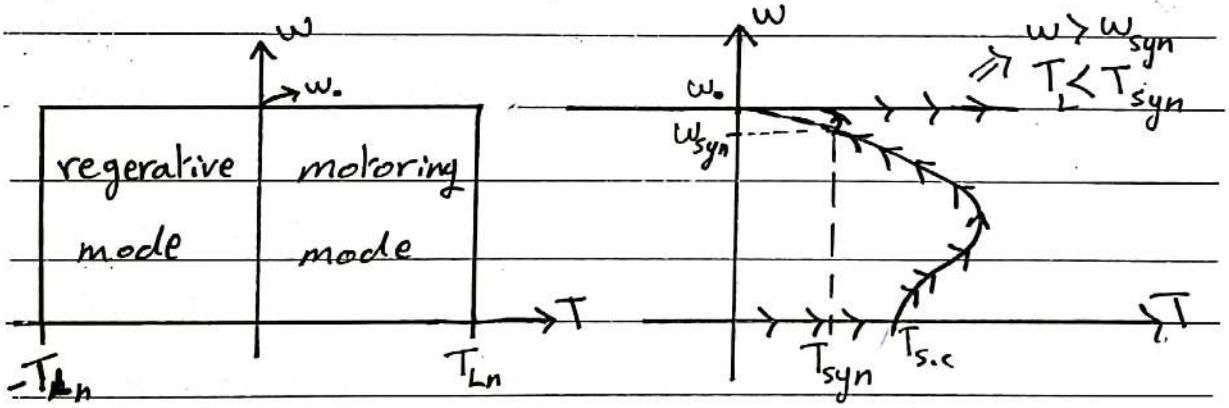
$+P_1 = -P_2 + \Sigma\Delta P \Rightarrow \Sigma\Delta P = P_1 + P_2$

Synchronous motor ($s=0$, $\omega=\omega_s$)

↳ stator (3 ϕ windings feed by 3ϕ AC power source)

↳ rotor → AC windings

↳ DC windings (excitation)



$\omega_{syn} = (0.9 - 0.95)\,\omega_s$

عند سرعة وعزم التوافق (ω_{syn} & T_{syn}) تم وصل ملفات التهييج مع مصدر DC ⇐ فينتقل المحرك تلقائيا إلى التوافق.

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SUBJECT AC motors

SECTION Synchronous motor

يعمل المحرك المتزامن بسرعة ثابتة عند الأحمال المختلفة لكن عند زيادة الحمل عن حد معين تخرج المحرك عن التزامن ويستخدم في: - الضخات، الضواغط، آلات الطرد المركزي، المبردات.

* لتحديد استقرارية عمل المحرك المتزامن يتم ايجاد العلاقة بين E & F -

$R_1 \approx 0 \Rightarrow \Delta P_{cu1} = 0 \Rightarrow P_1 = 3V_1 I_1 \cos\phi$

$P_1 = P_{em} + P_{cu1} + P_m$

$so \Rightarrow P_1 = P_{em} = T\omega_s$

$$T = \frac{3V_1 I_1}{\omega_s} \cos\phi = \frac{3V_1 E_1}{\omega_s X_1} \sin\delta$$

when $\delta = 90°$

$\sin\delta = 1 \Rightarrow T = T_{max} = \frac{3V_1 E_1}{\omega_s X_1}$

$so \Rightarrow T = T_{max} \sin\delta$

(δ) is the power or torque angle whic is the angle between E & δ

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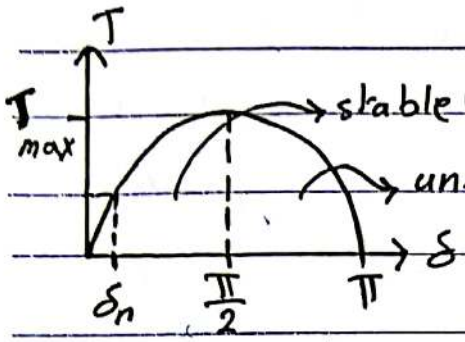
SUBJECT AC motors

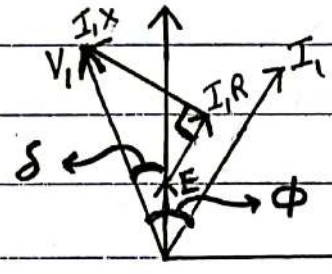
SECTION Synchronous motor / Transient operation of AC



يعني بالحدود بين الثوابت

$\delta_n = 20° - 30°$

"$T = f(\delta)$"



δ : angle between E & V_1 .

ϕ : angle between I_1 & V_1 .

$$\lambda = \frac{T_{max}}{T_n} = \frac{1}{\sin\delta_n} = 2-3$$

* Transient operation (starting, braking, & reversing) of AC motors:

II Starting of induction motors (I M) :-

static methods of starting → direct
→ indirect (electrical)

(A) Direct method :

used in motors of small power ratings and unlimited power supply.

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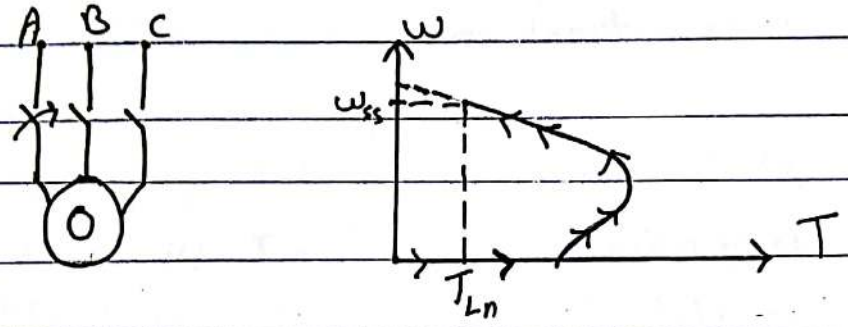
يتم توصيل الـstator على الشبكة بدون أي وسائل وسيطة.



(B) Indirect electrical method:
used in motors with medium & larg power rating ⟹ I_{st} is larg & limited powe supply

$$I_2' = \frac{V_{1ph}}{\sqrt{(R_1 + \frac{R_2'}{S})^2 + (X_1 + X_2')^2}} \quad , \quad I_{2\,s.c}' = \frac{V_{1ph}}{\sqrt{(R_1 + R_2')^2 + (X_1 + X_2')^2}}$$

$\frac{I_{s.c}'}{2} >> I_{2n}' \Rightarrow \varepsilon\Delta P >> \varepsilon\Delta P_n \Rightarrow$ خطر تلف

يتم توصيل ملفات الـstator على الشبكة عبر وسائل لتقليل تيار البدء (I_{st})

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SUBJECT AC motors

SECTION Transient operating of AC motor

* I_{st} can be reduced by:

1) inserting R or L or both to stator circuit.

2) reducing voltage

↳ using Auto-transformer – V (Var) & F (const).

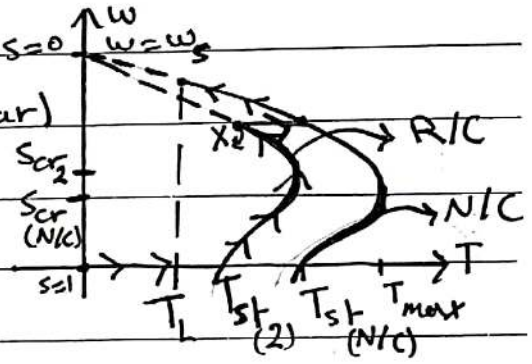
↳ using Y–Δ stator connection.

↳ using AC voltage regulaters.

① Inserting R or L or both to stator circuit.

[R_{add} or L_{add} ⇒ I_{st}↓ ⇒ T↓ ⇒ S_{cr}↓]

- N/C: (normal char)
- R/C: with R_{add} (regulated char)
- $S_{cr_2} < S_{cr(N/C)}$



* disadvantages of this method:

power loss in stator circuit is increased due to R↑ (L is used to reduce P_{losses}) ⇒ η↑

* advatages of this method:

easy to use and low cost

* at point X: (resistance should be removed then) motor continue in running on the N/C.

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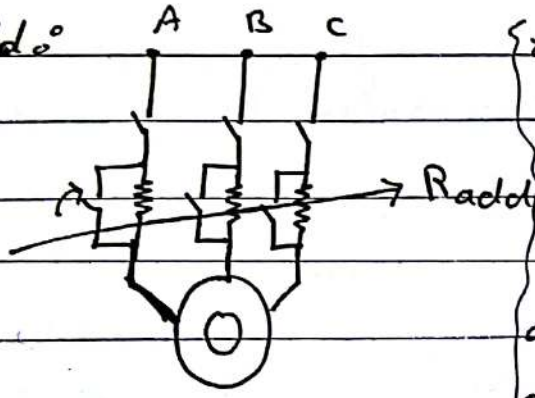
SUBJECT AC motors

SECTION Transient operation of AC motor



*we make switching to remove R_{add} at (X), which is before $W_{ss(R/C)}$, because if it close, then it will continue operating on R/C.

* How to select R_{add} or X_{add}?

depending on ratings & reduction in $I'_{2(s.c)}$.

ratings: $V_{1ph(n)}$, $I_{1(n)}$, $n_{(n)}$, $\cos\phi_{(n)}$, $P_{2(n)}$

λ, $\Rightarrow I'_{2s.c\ (R/C)} = ?$

calculations:

$$Z_{s.c(N/C)} = \frac{V_{1ph}}{I_{s.c(N/C)}}$$

$$Z_{s.c(R/C)} = \frac{V_{1ph}}{I_{s.c(R/C)}}$$

$$\Rightarrow \frac{Z_{s.c(N/C)}}{Z_{s.c(R/C)}} = \frac{I_{s.c(R/C)}}{I_{s.c(N/C)}} < 1$$

$$R_{add} = \sqrt{\left(\frac{Z_{s.c(N.C)}}{\alpha}\right)^2 - X^2_{s.c(N.C)}} - R_{s.c(N/C)}$$

$$X_{add} = \sqrt{\left(\frac{Z_{s.c(N/C)}}{\alpha}\right)^2 - R^2_{s.c(N/C)}} - X_{s.c(N/C)}$$

$$Z_{s.c} = \sqrt{R^2_{s.c} + X^2_{s.c}}$$

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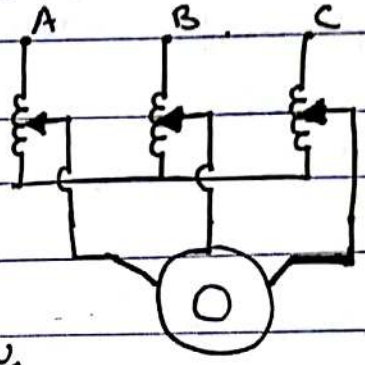
② Reducing voltage ($V_{1ph} < V_{1ph(n)}$) :-

a) using auto transformer :

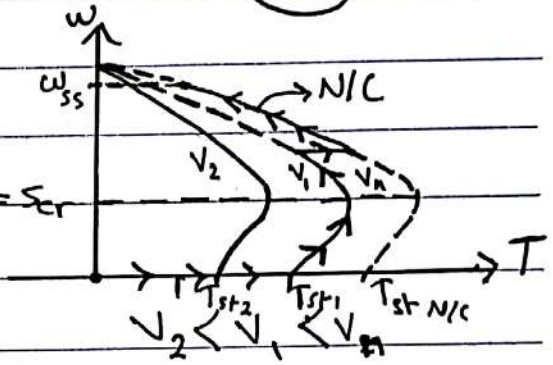


- $I'_{2s.c}$ is reduced by the same ratio as V_{1ph} is reduced.
- T_{st} is reduced by $(V_{1ph})^2$

$V_{1p} \downarrow 50\%$

$I_{2s.c} \downarrow 50\%$

$T_{st} \downarrow 75\%$

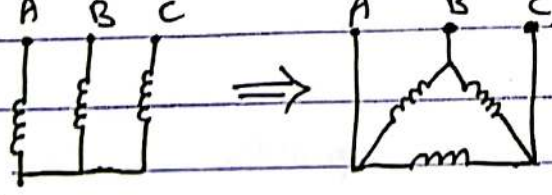


* we increas the voltage before (w_{ss}) $\equiv (V_{1p} < V_{1ph\,n})$, so that it accelerate on the N/C until it reaches w_{ss}.

"mech char for different values of stator voltage"

أي أنه المحرك يتعلق على الجهد المنخفض ثم يتم رفعه قبل أنه يصل لنقطة الاستقرار حتى يستمر على الخاصية الطبيعية للمحرك (N/C) عند الجهد المقنن (V_{1phn}).

b) using star-delta connection



$V_\Delta = \sqrt{3}\, V_Y$

$T_\Delta = 3\, V_Y$

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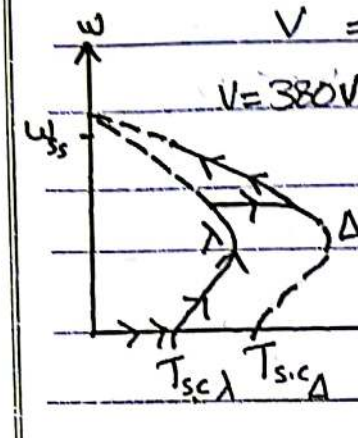
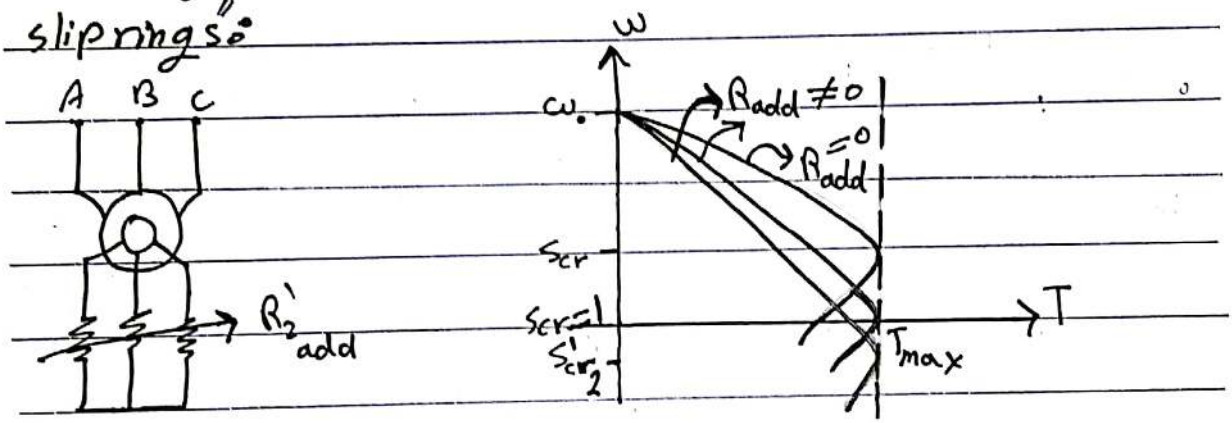


يتم اقلاع المحرك على خاصية (λ) ⇐ V = 220V
ثم يتم تحويل التوصيلة بالقابس إلى Δ ⇐ V = 380V
وهنا نقلل I_{st} بمقدار ($\sqrt{8}$)
وتقل T_{st} بمقدار (3)

(disadv° reduction of voltage is a fixed ratio)

*Indirect starting of wound rotor motors:
"inserting R to the rotor circuit (+ with R_2') through slip rings"



من قوانين T, T_{max}, s_{cr} السابقة :-
↳ عند اضافة مقاومة على التوالي مع R_2 (مقاومة rotar) ⇐ s_{cr} تزداد
↳ // // // // ⇐ T_{st} تزداد ⇐ running time will be reduced
↳ // // // // ⇐ T_{max} لا يتأثر

* we keep adding R until $T_{st} = T_{max} \Rightarrow s_{cr} = 1$

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* مما سبق نلاحظ أنه قيمة Scr لا تتأثر عند تغير الجهد ولكن تتأثر بإضافة مقاومات مع دائرة الروتر أو الستيتور.

$$S_{cr} = \frac{R_2'}{\sqrt{R_1^2 + X_{s.c}^2}}$$

, R_1 "stator resistance"

, R_2' "in rotor resistance"

EX: Three phase IM ratings

15 KW , 1465 rpm , 220 V (phase) , 50 Hz

29.3 A , $\lambda = 2.3$, $\cos\phi = 0.4$

Find: ① R_{1add} , ② X_{1add} , Knowing that:

$I_{s.c(R/C)} = 0.5\, I_{s.c(N/C)}$

$I_{s.c(N/C)} = 7\, I_n$

sol:

$$Z_{s.c(N/C)} = \frac{V_{1ph(n)}}{I_{s.c(N/C)}} = \frac{220}{7 * 29.3} = 1.08\ \Omega$$

$$R_{s.c(N/C)} = Z_{s.c(N/C)} \cos\phi_{(n)} = 0.43\ \Omega$$

$$X_{s.c(N/C)} = \sqrt{Z_{s.c(N/C)}^2 - R_{s.c(N/C)}^2} = 1\ \Omega$$

For $I_{s.c(R/C)} = 0.5\, I_{s.c(N/C)} \Longrightarrow \alpha = 0.5$

so $\Rightarrow X_{1add} = 1.12\ \Omega$

$R_{1add} = 1.5\ \Omega$

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SUBJECT AC motors

SECTION Transient operation of AC motor

Ex: $R_1 = 3\ \Omega$, $R_2' = 2\ \Omega$, $X_{s.c} = 10\ \Omega$

V_{1ph} is reduced by 10%

a) Change in starting torque = ?

b) $R_{2add}' = ?$

$V_{1ph(st)} = 0.9 V_{1ph(n)}$

a) $T_{st} \propto V_{1ph}^2$

$$\frac{T_{st(n)}}{T_{st(s)}} = \left(\frac{V_{1ph(n)}}{V_{1ph(n)} * 0.9}\right)^2$$

$T_{st} = 0.81\ T_{st(n)}$

b) $T_{st} = T_{max}$

$$1 = \frac{2 + R_{2add}'}{\sqrt{(3)^2 + (10)^2}} \Rightarrow R_{2add}' = 7.54\ \Omega$$

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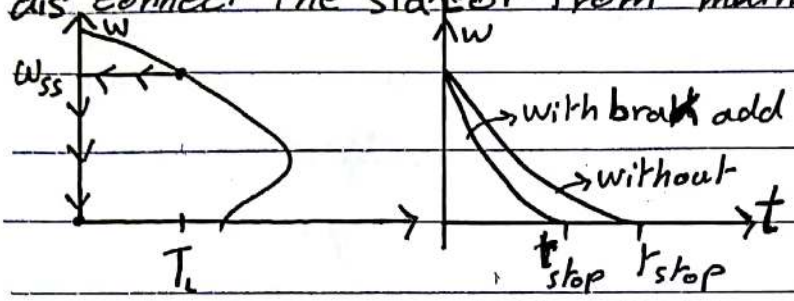
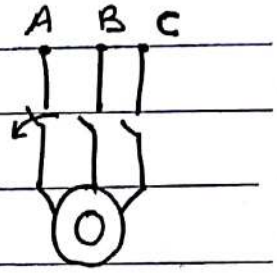
2 Braking of 3 phase I.M.

A Mechanical braking

dis connect the stator from main supply





braking under load inertia

$$T - T_L = J_\varepsilon \frac{dw}{dt} \Rightarrow \boxed{dt = J_\varepsilon \frac{dw}{T - T_L}}$$

$I=0 \rightarrow \phi=0$

$\rightarrow T=0$

↳ without additional brak $\left(t_{stop} = J_\varepsilon \frac{w_L}{T_L}\right)$

↳ with $\left(t_{stop} = J_\varepsilon \frac{w_L}{T_{Br} - T_L}\right)$

B Electrical braking methods

1- plugging braking (change the sequence of any 2 phases)

عكس طورين من أطراف العضو الساكن stator

لذلك يعكس المحرك اتجاه دورانه ← يعمل على خاصية المحرك في الربع ٣ مروراً بالربع ٢ ، وعندها $w=0$ يتم فصل stator عن المصدر $\phi<0 \Rightarrow T<0$

$$S_p = \frac{-w_. - w_{ss}}{-w_.} = 2 - S_n \approx 2 \quad \text{(plugging slip)}$$

$$w_{ss} = w_. (1 - S)$$

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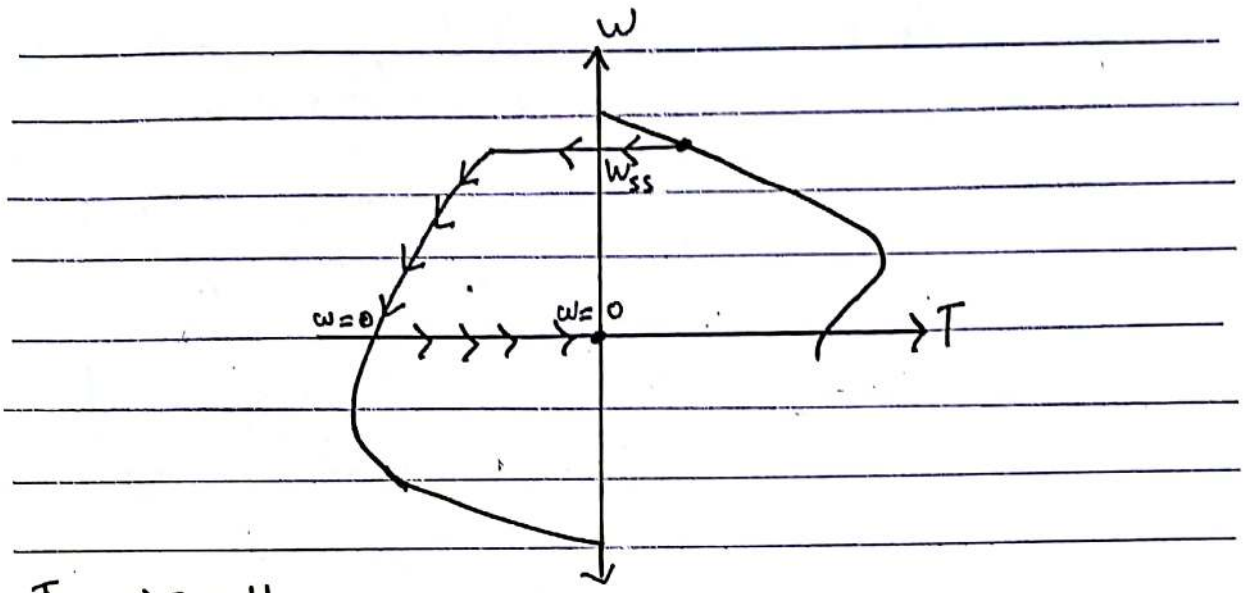
SUBJECT AC motors

SECTION Transient operation of AC motor



T_{br} → small

For wound rotor ⇒ R_2' add → I_{br} ↓ → T_{br} ↓ → T_{start} ↑

Ex: 400V(L-L), Δ-connected, 6 poles, 50Hz

$R_1 = R_2' = 1\ \Omega$, $X_1 = X_2' = 2\ \Omega$

plugging braking - from $n = n_{ss} = 950$ rpm

① $T_{br} = ?$, $T_{2b}' = ?$

② $R_{2add}' = ?$, $I_b' = 1.5\ I_{2n}'$, $T_b = ?$

Sol:

$$n_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000 \text{ rpm} \Rightarrow w_s = 104.72 \text{ rad/s}$$

$$s_n = \frac{n_s - n_n}{n_s} = 0.05$$

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$$\bullet\ I'_{2n} = \frac{V_{1ph(n)}}{\sqrt{\left(R_1 + \frac{R'_2}{S_n}\right)^2 + X_{s.c}^2}} = 10.8\ A$$

$$\bullet\ T_n = \frac{3}{\omega_s} \cdot I'^2_{2n} \cdot \frac{R'_2}{S_n} = 66.83\ N.m$$

① I_n plugging (For $n = n_n$)

$S_p = 2 - S_n = 1.95$

$$I'_{2b} = \frac{V_{1ph(n)}}{\sqrt{\left(R_1 + \frac{R'_2}{S_p}\right)^2 + X_{s.c}^2}} = 54\ A$$

$T_p = 42.84\ N.m$

with R'_2 add

$$I'_{2p} = \frac{V_{1ph}}{\sqrt{\left(R_1 + \frac{R'_2 + R'_{add.}}{S_p}\right)^2 + X_{s.c}^2}}$$

$\Rightarrow R'_{2add} = 3.73\ \Omega$

$$T_{b(p)} = \frac{3}{\omega_s}\left(1.5\ I'_2\right)^2\left(\frac{R_2 + R'_{add}}{S_p}\right) = 95.35\ N.m$$

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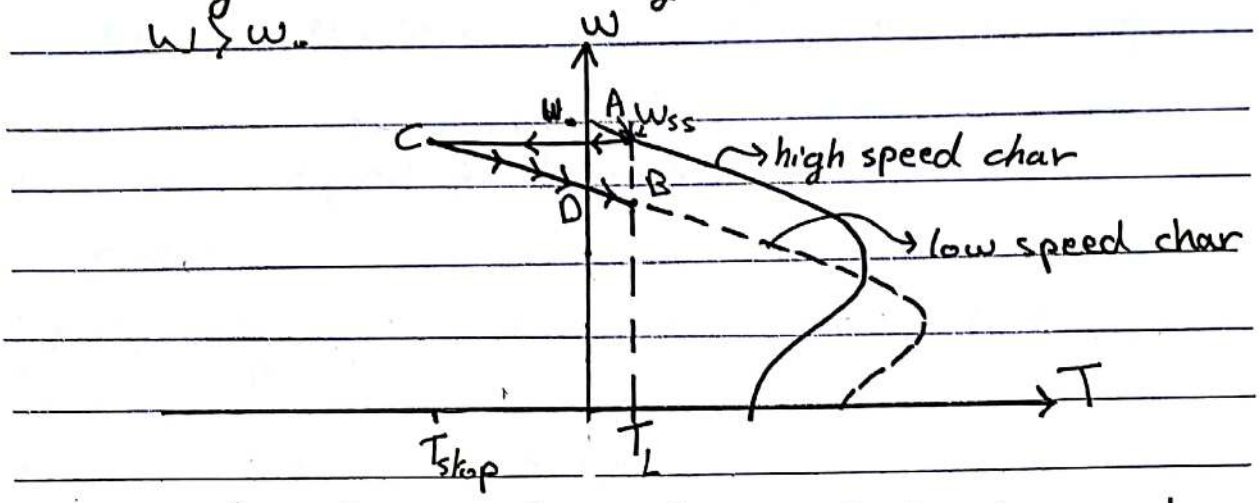
2- Regenerative braking



* during transforming from H-speed to L-speed motor operates as generator in parallel with source in seconed quadrant. Return to Kinetic energy (mech power) in form of elec power to source, (C-D).

* In regenerative braking, w_{ss} goes from the H-speed to L-speed, but this can't done directly from A to B, so it goes trough A → C → D → B

C − D: regenerative char

D − B: motor mode

steady state operation on reg braking char, linear protection

$T_{max(L)} = T_{max(H)}$

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SUBJECT AC motors

SECTION Transient operation of AC motor

Ex: 400(L-L) V, λ connected, 3 phase, 6 poles 50 Hz, $R_1 = R_2' = 1\ \Omega$, $X_1 = X_2' = 2\ \Omega$, for reg braking

Find: ① T_{max}, Range of speed for

② speed at $T = 100$ N.m

Sol:

① $n_n = \frac{120 f}{P} = 100 rpm \Rightarrow \omega_s = 104.72$ rad/s

$$S_{cr} = \frac{-R_2'}{\sqrt{R_1^2 + X_{s.c}^2}} = -0.242$$

$$I_2' = \frac{V_{1ph}}{\left(R_1 + \frac{R_2'}{S_{cr}}\right)^2 + X_{s.c}^2} = 45.5\ A$$

At $s = s_{cr} \Rightarrow T = T_{max} = \frac{3 I_2'^2 R_2'/S_{cr}}{\omega_s} = -244.5$ N.m

max holding torque = 244.5 N.m

$\omega_{cr} = \omega_s (1 - S_{cr}) = 130$ rad/s

$[104.72 \leq \omega \leq 130] \Rightarrow$ range of speed

② $T_m = T_L = -100$ N.m $= \frac{-3 V_{1ph}^2 R_2'/S}{\omega_s \left(\left(R_1 + \frac{R_2'}{S}\right)^2 + X_{s.c}^2\right)}$

$S_1 = -0.957$ ✗

$S_2 = -0.063$ ✓

$\omega = \omega_s (1 - S_2) = 1.063 * 104.72 = 111.32$ rad/s

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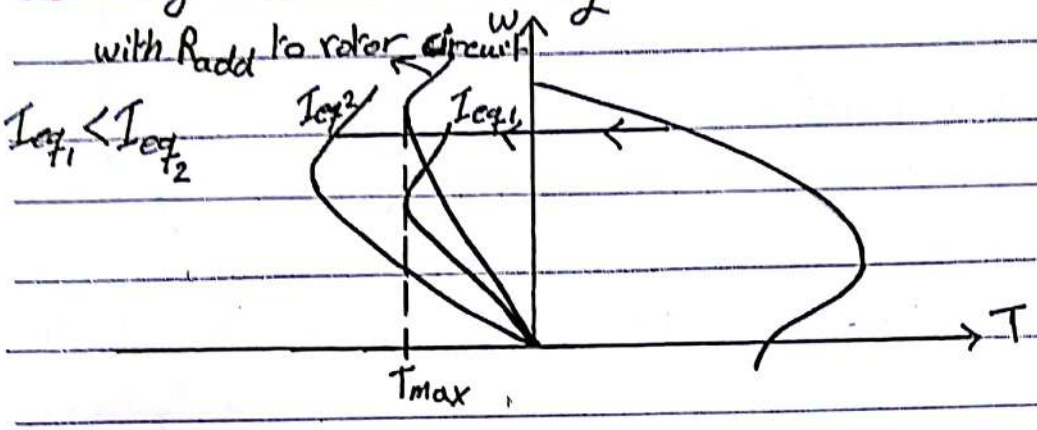
3- Dynamic braking

with Radd to rotor circuit

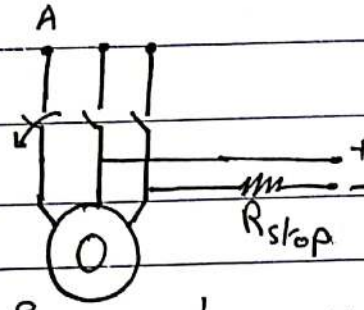


[1] For induction motors, squiral cage

main supply is disconnected, at the same we insert DC current to any two phases.

(slip is important to preduce torque)

يتم فصل التغذية عن stator وتغذيه بأي فازين بـ DC



يتم تحويل الطاقة الحركية إلى طاقة كهربائية تستهلك على شكل حرارية في ملفات Rotor

[2] For synchronous motors

disconnect the stator & short circuit it wit R_{stop} -

يوجد DC مسبقاً لذلك يتم فصل تغذية AC عن stator وقصر دارته مع ابقاء مصدر التهييج موصول مع ملفات التهييج

⇒

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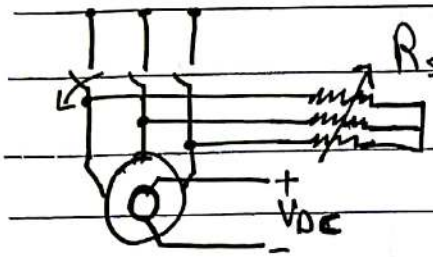
SUBJECT AC motors

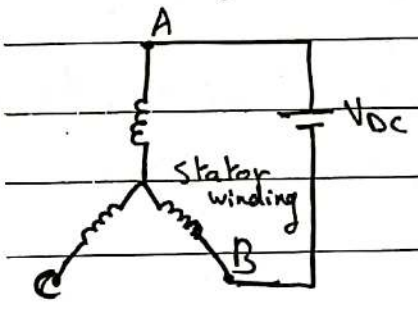
SECTION Transient operation of AC motor



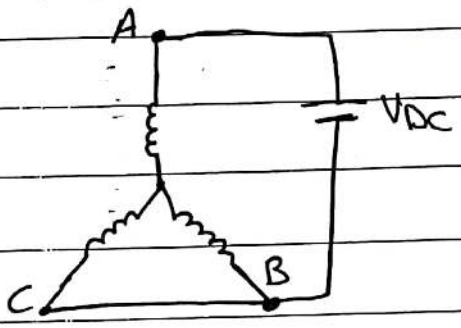
DC current ← ينشأ مجال مغنطيسي ثابت

يعمل عزم توقف (Tbr)

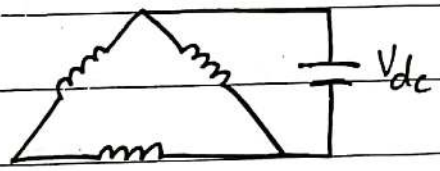
* DC dynamic braking:



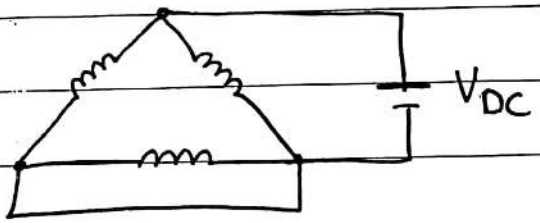
2-led connection (λ)



3-led connection (λ)

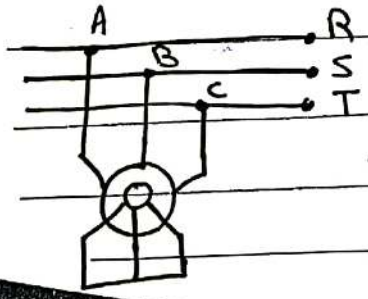


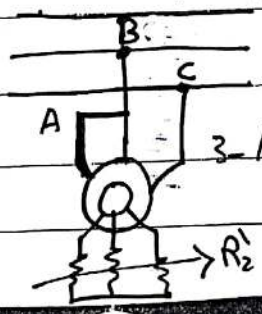
2-led connection (Δ)



3-led connection (Δ)

* AC dynamic braking: ⇒ this method is not commonly useed.





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SUBJECT Automatic Control

SECTION Introduction

Automatic control of DC & AC motors :-

- Control devices :-
  - Design the power circuit + control circuit of electric drive.
  - To perform switching operation → we need control principle → have to know sequence of control (switching) → used method for starting, braking, reversing → setting calibration of control devices

- principles :-
  1- Time principle .
  2- speed // .
  3- Voltage // (E_m) .
  4- displacement // (θ) .
  5- current // .

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SUBJECT Automatic control

SECTION Introduction

* Time principle:

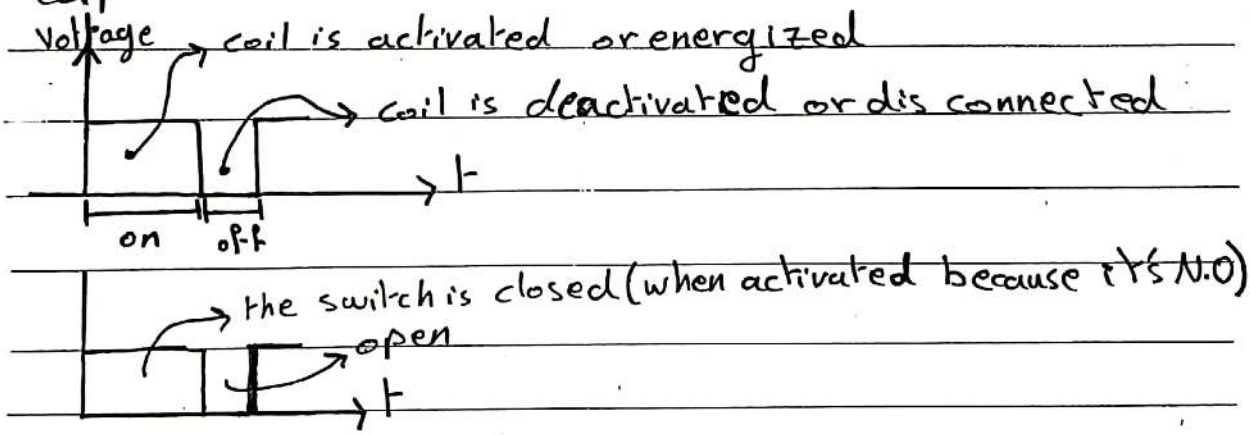
use electro magnetic devices "contactors & relays" that depend on time.

- contacts → normaly open (N.O)
  → normaly closed (N.C)

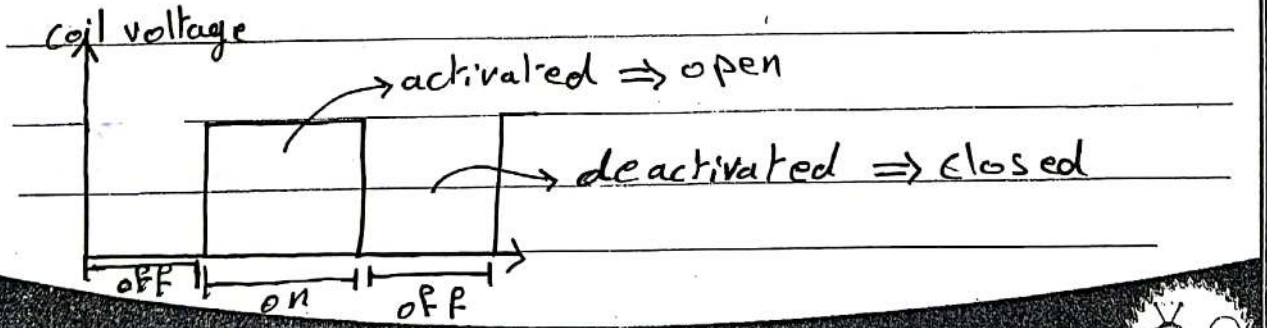
عند وصول التغذية لأطراف الملف يتكون مجال مغناطيسي وقوة جاذبة كهربائية تؤدي إلى جذب أو عدم جذب بين التلامسات ويكون الوضع الطبيعي للتلامسات (disactive)

* تتحمل التلامسات تيار عالي ← تستخدم في power circuit

* For N.O contact:



* For N.C contact:



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SUBJECT Automatic control

SECTION 3 Introduction

• Relays :

Time relay (timer) — T.R —

1) with time delay on :

يتغير وضع التلامسات بعد فترة من الزمن (نفقد النظر عن وضعها الطبيعي (open or closed) ، وتعود مباشرة لوضعها الطبيعي عند فصل التغذية.

↳ For N.O ⇒ يغلق تبعاً حيز زمني ⇒

ولكنه يفتح بشكل لحظي.

↳ For N.C ⇒ يفتح تبعاً حيز زمني ⇒

ويغلق بشكل لحظي

* لاحظ أن السهم في حالة (time delay on) يشير إلى عكس اتجاه ما تقوم به التلامسات عند تغذيتها.

مثلاً N.O ⇐ عند التغذية يجب أن يغلق ⇐ لذلك السهم للأعلى.

( )

2) with time delay off

عند وصل التغذية ⇐ يتغير وضع التلامسات فوراً.

عند فصل التغذية ⇐ تعود التلامسات للوضع الطبيعي بعد فترة من الزمن...

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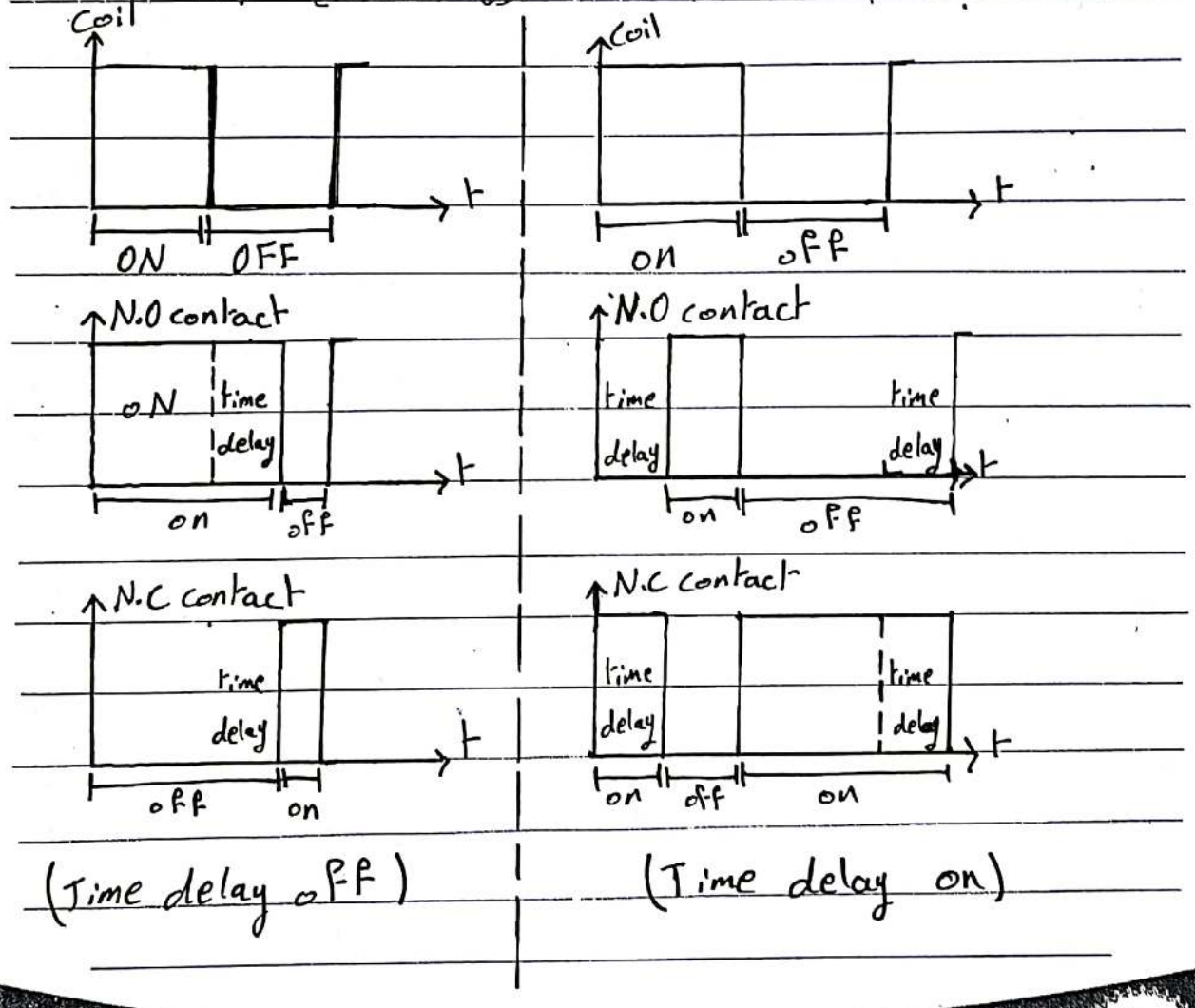
SUBJECT Automatic control

SECTION Introduction

→ for N.O ⇒ يغلق بشكل لحظي ⇒
ويفتح بشكل زمني

for N.C ⇒ يفتح بشكل لحظي ⇒
ويغلق تبأخير زمني

* لاحظ أن السهم في حالة (with delay off) يشير إلى نفس اتجاه التلامسات عند وصول التغذية لها.



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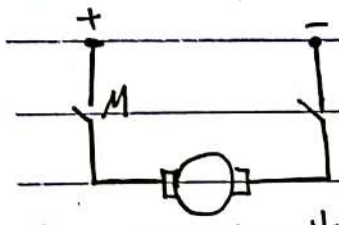
SUBJECT Automatic control

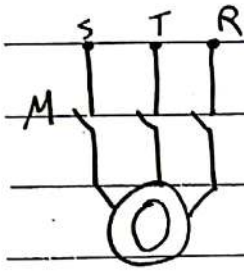
SECTION Examples

Ex: Direct start & Mechanical braking, For sep DC motor & I.M.

1) the power circuit.
2) the control circuit.
3) the static & dynamic char.

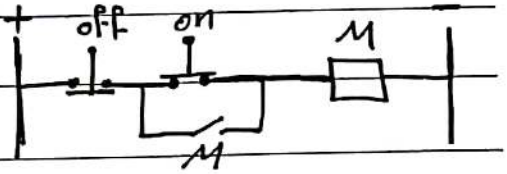


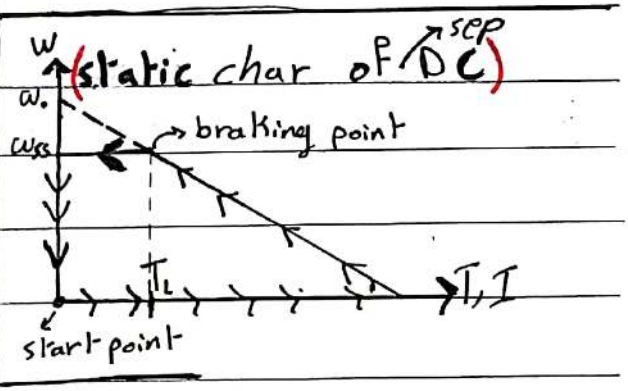
(power circuit of DC motor)



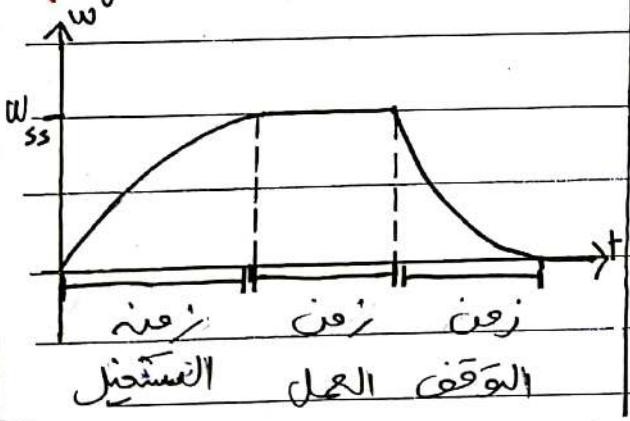
(power circuit of IM)

(control circuit For both)

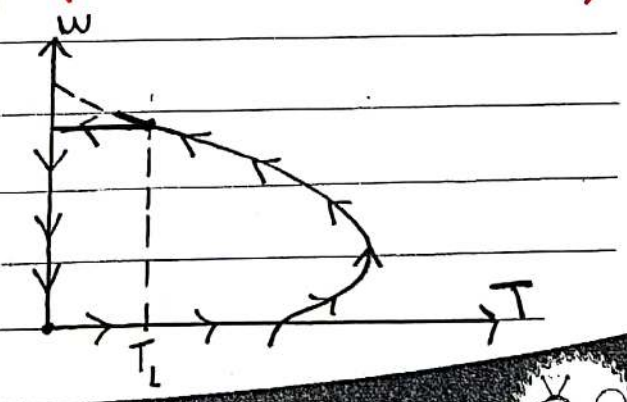




(dynamic char)



(static char of AC induction motor mech)



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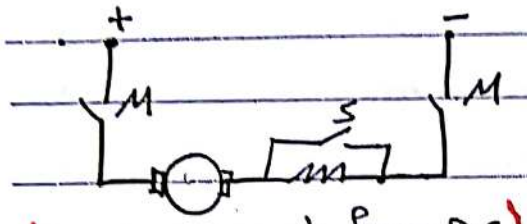
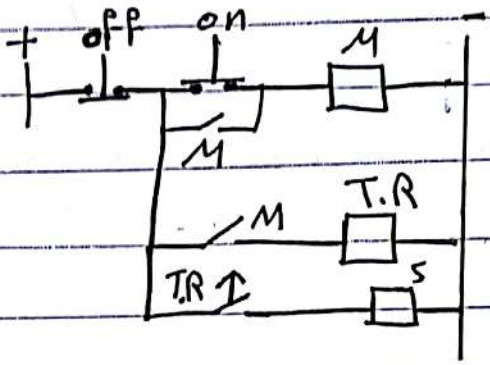
SUBJECT Automatic control

SECTION Examples

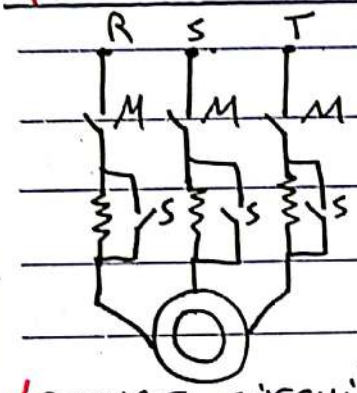
EX: One step rheostatic start according to time principle and mechanical braking.



(power circuit for DC)

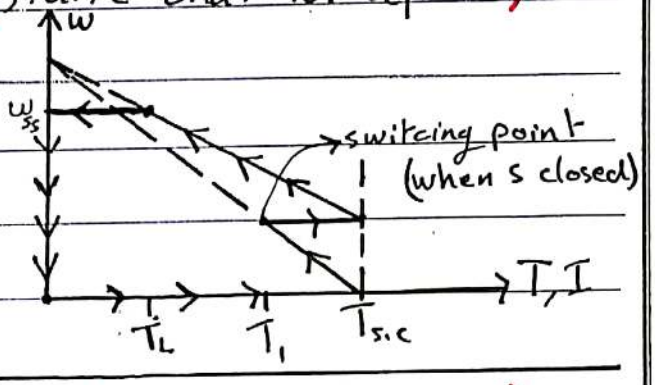


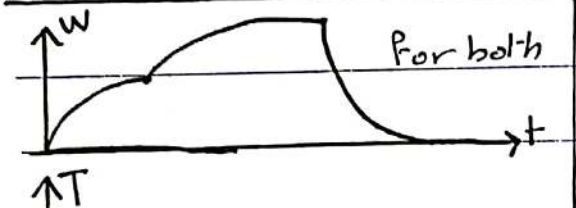
(control circuit)

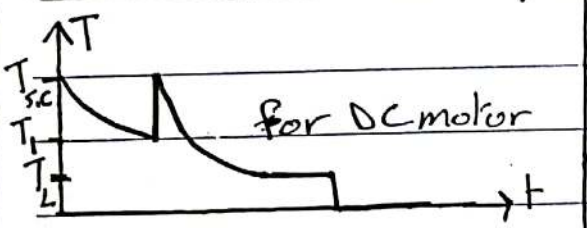


(power circuit for AC)

(static char for sep DC)

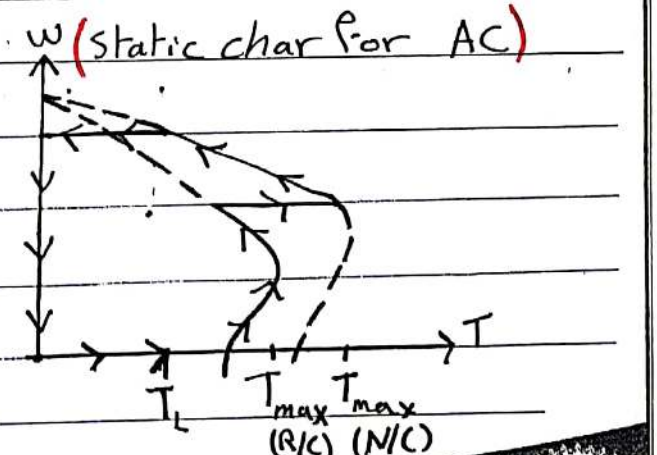






(Dynamic char)

(static char for AC)



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SUBJECT Automatic control

SECTION Example

Ex: two step rheostatic start & dynamic braking

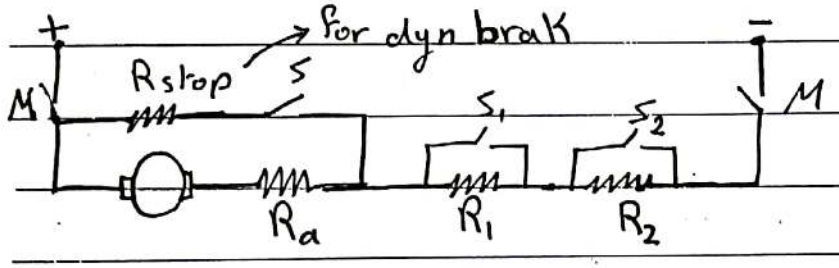
Dynamic braking

↳ in DC ⇒ we short circuit the arm with R and disconnect the source

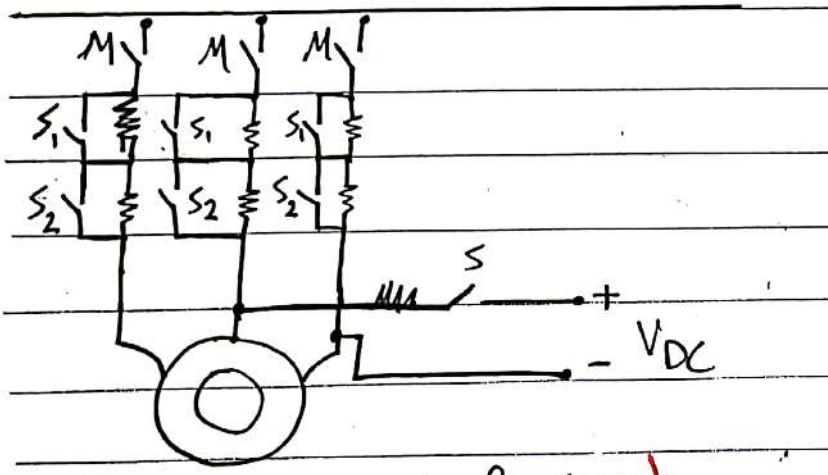
↳ in AC: synch motor ⇓ disconnect stator & short circuit it with R & keep DC on rotor

or induction motor (IM) ⇓ disconnect stator & insert DC to any two phases

Ex on DC & 3-phase IM:



(power circuit of DC)



(power circuit of IM)

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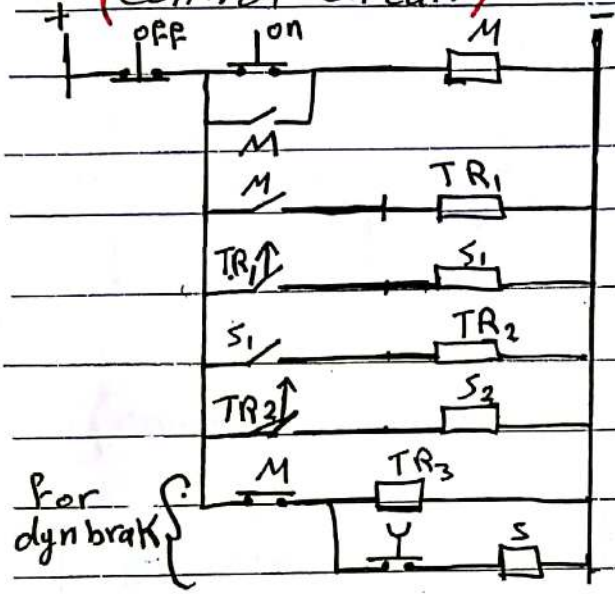
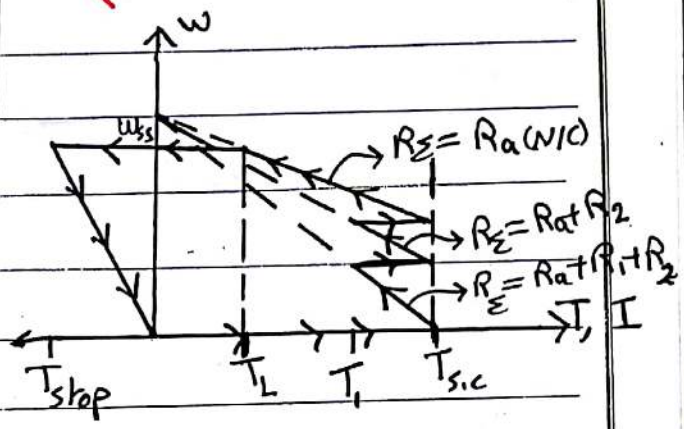
SUBJECT Automatic control

SECTION Examples

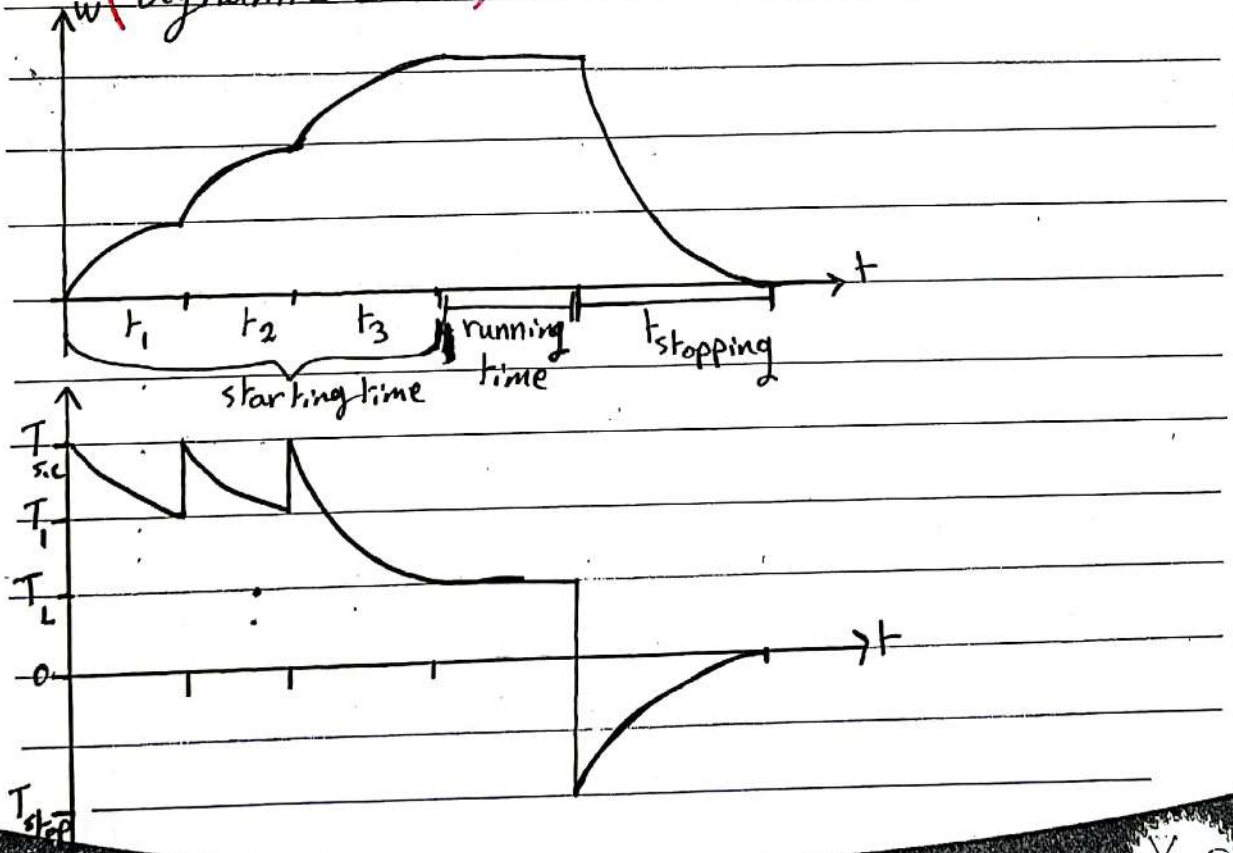
(control circuit)



(static char of DC sep or shunt motor)



(dynamic char)



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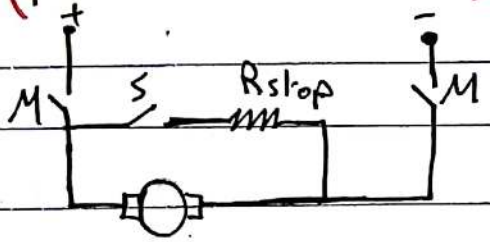
SUBJECT Automatic control

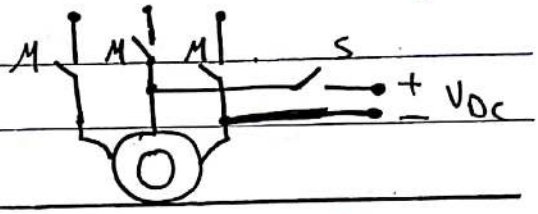
SECTION Examples

Ex° Direct start & dyn. braking (according to "t" prin)

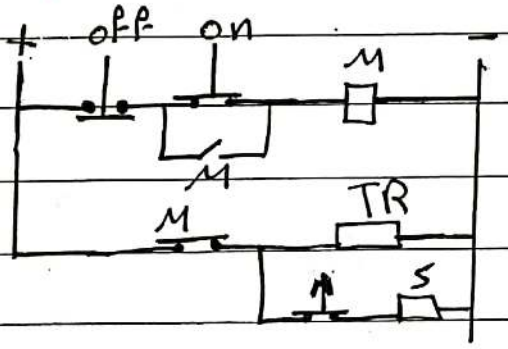
(power circuit of DC)



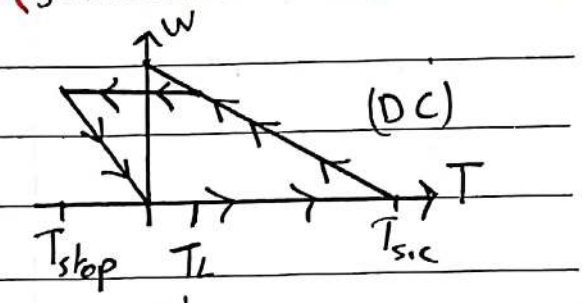
(power circuit of I M)

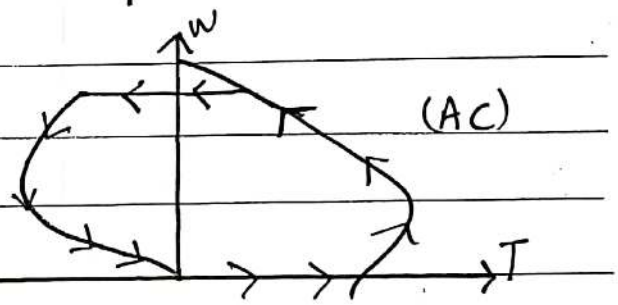


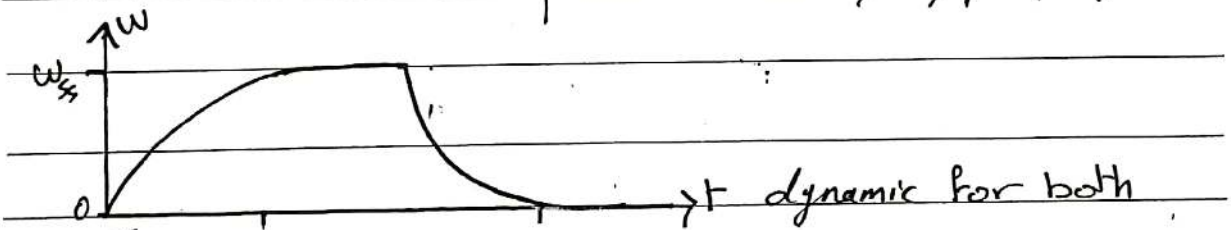
(control circuit)

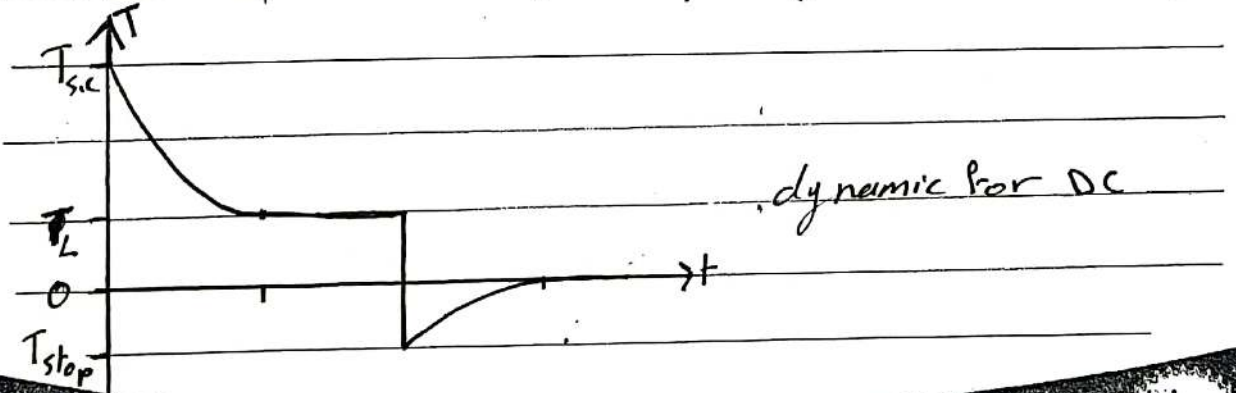


(static char)









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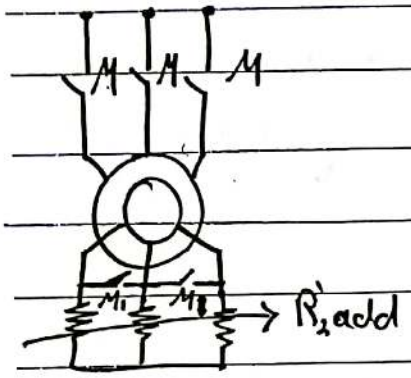
SUBJECT Automatic control

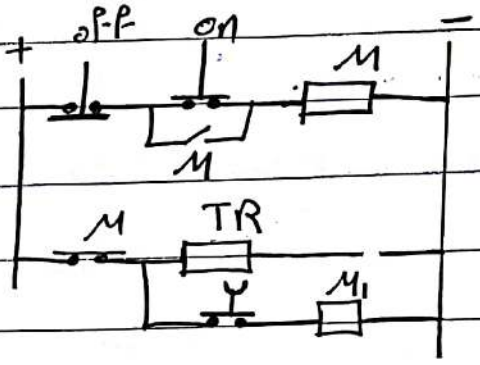
SECTION Examples

Ex° Rheostatic start of wound rotor 3-phase IM:

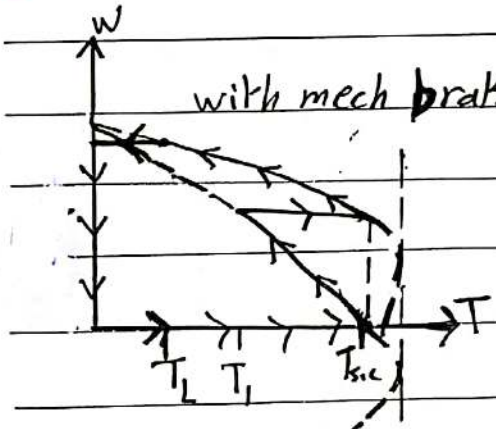
(In wound rotor IM ⇒ R is added to the rotor circuit)



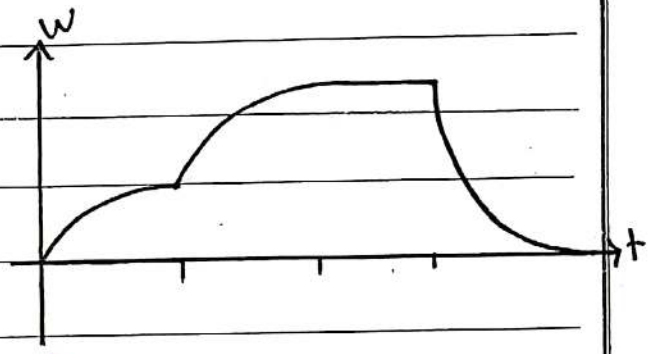
(power circuit)



(control circuit)



(static char-mechanical)



(dynamic char)

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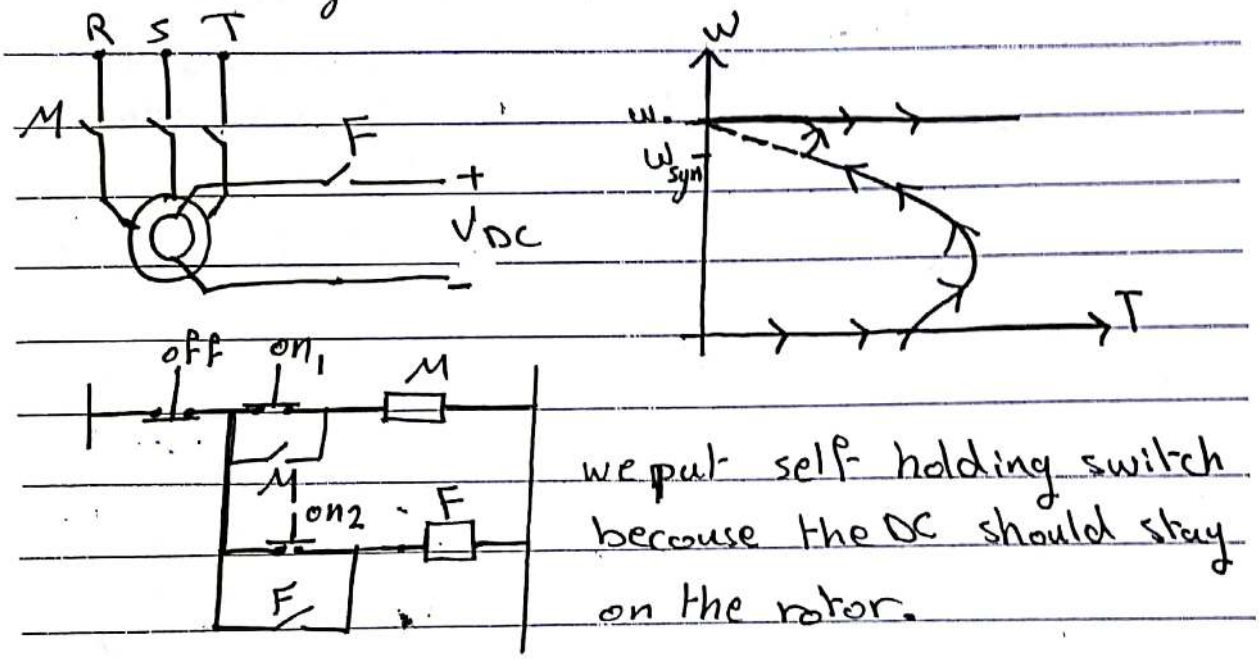
SUBJECT Automatic control

SECTION Examples

AC synchronous motor (SM) :-

manual start and control of (SM)
same as manual start of IM, but when "w" reaches w_{syn} ⇒ we add a DC source to rotor.



we put self holding switch because the DC should stay on the rotor.

on1 ⇐ نقل التغذية للكونتاكتور M ويتم وصل الجهد على stator فينتقل الحمل على خاصية العزم المتزامن ⇐ بعد فترة قصيرة ⇐ يصل المحرك للسرعة التوافقية (w_{syn}) فيتم الضغط على on2 ⇐ يتم الـ rotor مع تغذية DC وبذلك يتم ادخال المحرك التوافقي في خاصية التوافق.

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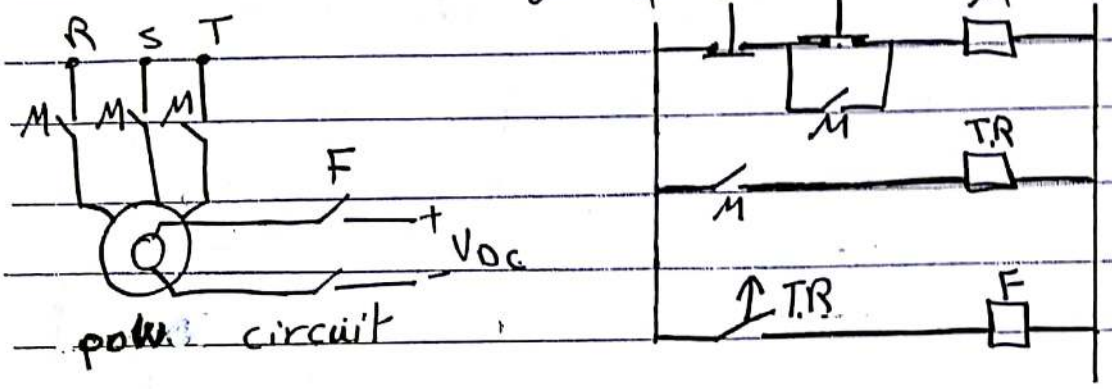
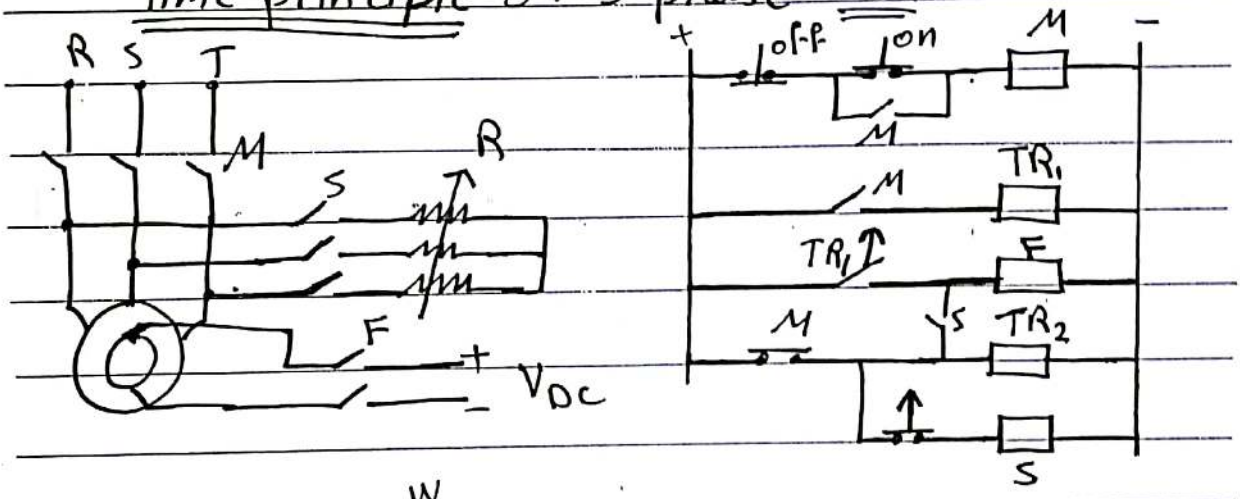
SUBJECT Automatic control

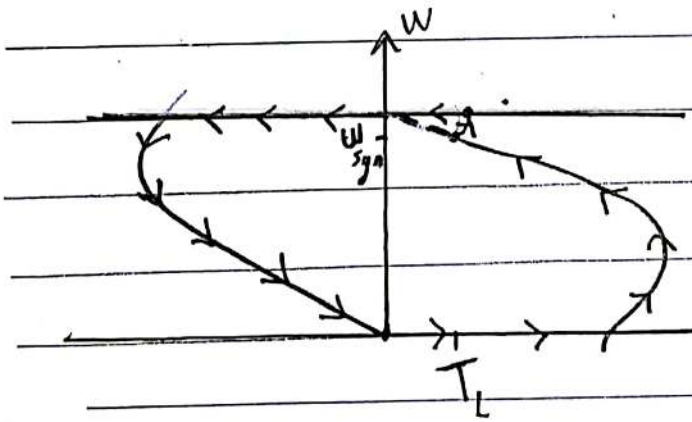
SECTION Examples

* Automatic (according to time principle) starting and mech braking



Ex° start & dynamic braking according to time principle of 3-phase SM :°





1) ON ↓ → M ↑ (closed) ⇒ TR_1 (active)

2) F ↑ (closed)

3) off → M (open) ⇒ TR_2 (active)

4) S (off)

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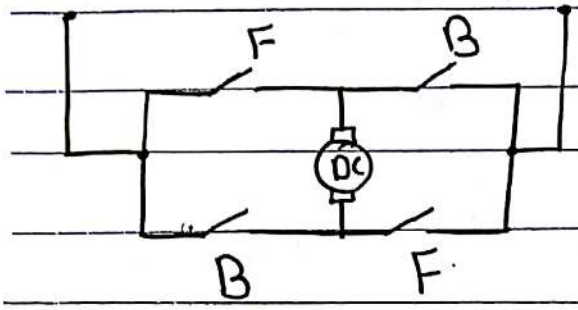
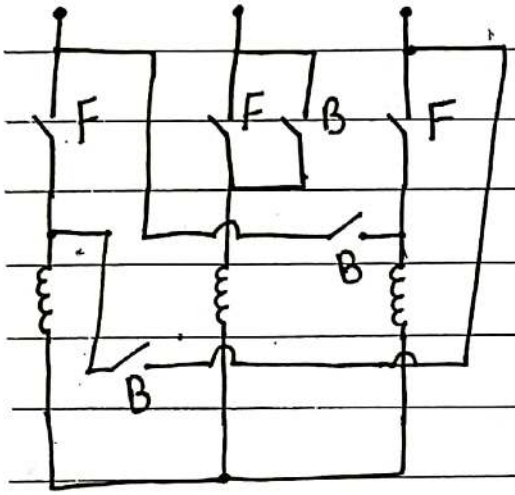
SUBJECT Automatic control

SECTION Examples

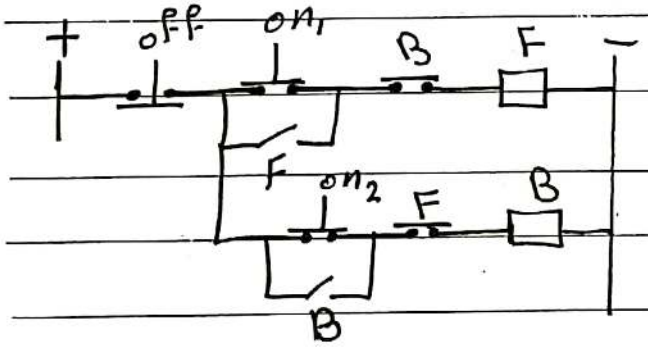
* Manual reverse:



F: Forward
B: backward



⇒ For TM



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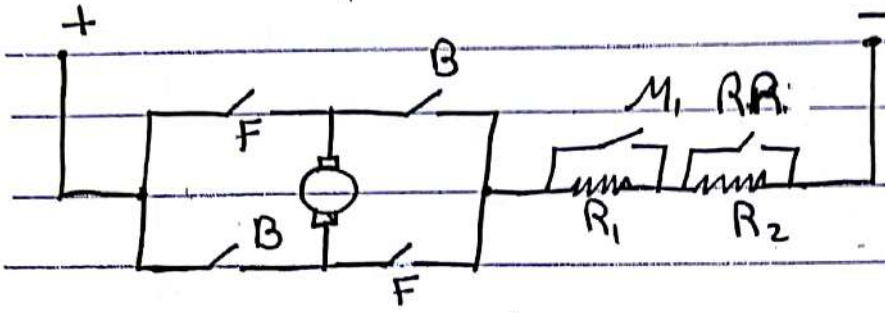
SUBJECT Automatic control

SECTION Examples

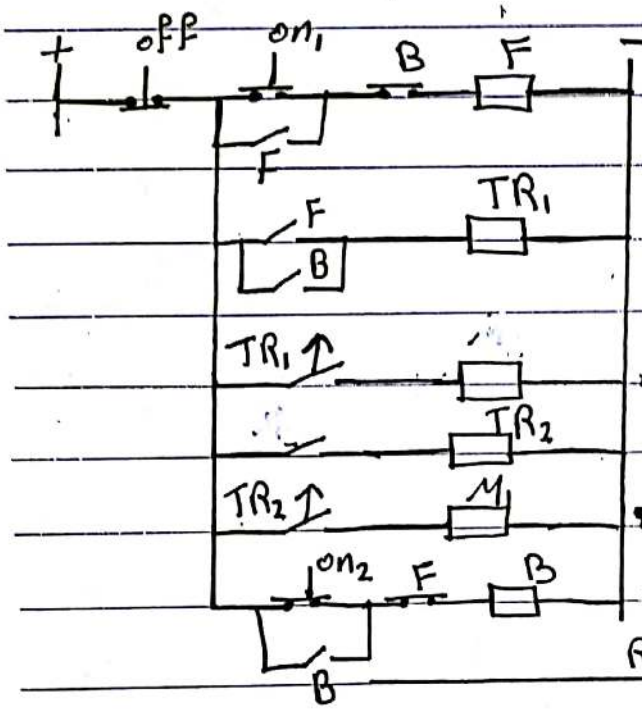
* Automatic reverse:-

For D.C. sep excited:



(power circuit)

RR: reversing relay



(control circuit)

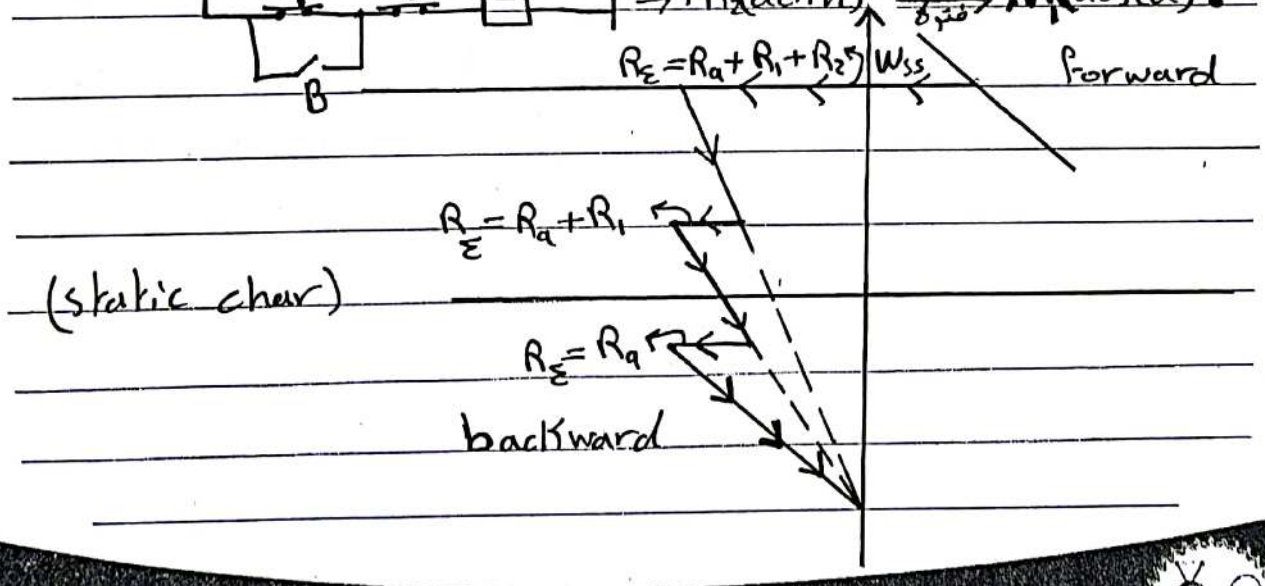
$on_2 \downarrow \rightarrow B \uparrow \rightarrow V_a$ change it's polarity $\rightarrow I_a$ changes direction

* $on_1 \downarrow \rightarrow F \uparrow$ (closed) & B open

$\Rightarrow TR_1$ (active) $\xrightarrow{\text{بعد}}$

$\Rightarrow RR$ (closed) $\rightarrow R_2$ removed $\rightarrow$

$\rightarrow TR_2$ (active) $\xrightarrow{\text{بعد}}$ M (closed).



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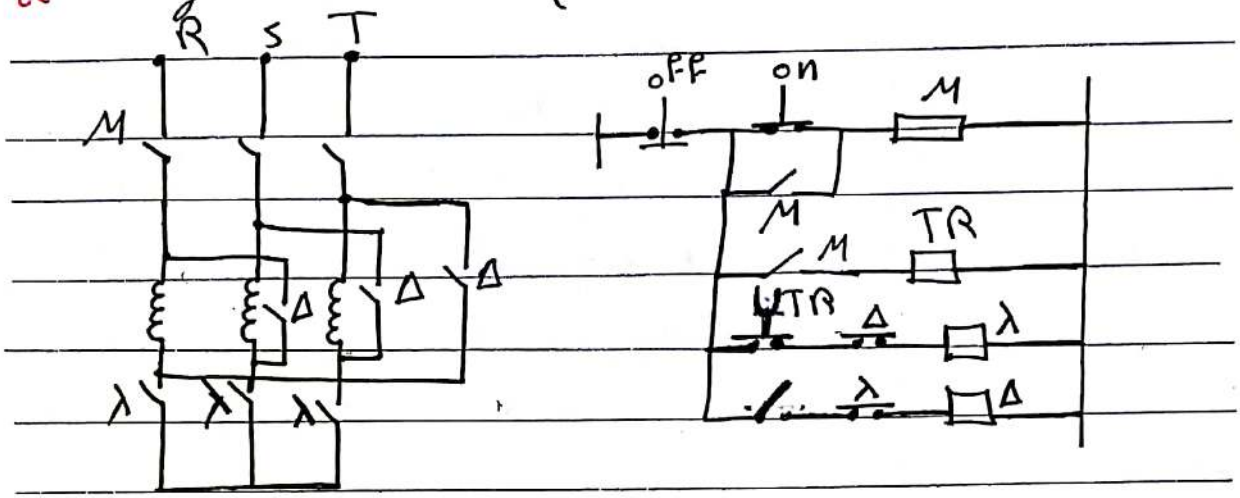
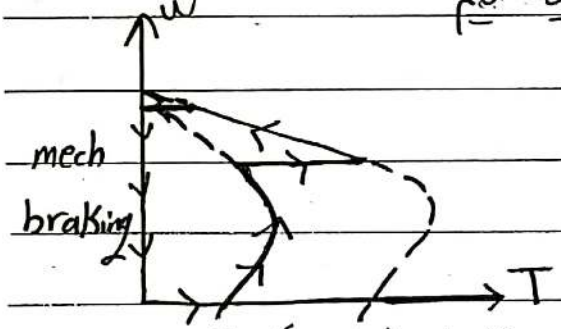
SUBJECT Automatic control

SECTION Example

* change connection (From λ to Δ) start :



* الخطوة التي يفقد فيها λ تغذية يتم تغذية Δ



← TR (activated) ← M ← on ← تقل التغذية للكونتاكتور

← يبدأ TR بعد زمن التعويل اللازم وتقل التغذية لملفات Stator ويعمل المحرك على توصيلة λ ← عند انتهاء زمن المرحل TR يتم فصل ملف كونتاكتور (λ) وتوصيل ملف كونتاكتور Δ.

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SUBJECT Speed control.

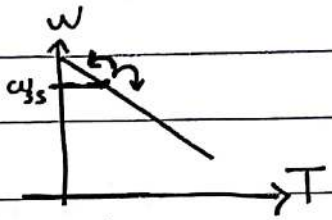
SECTION Introduction

Methods of speed control of electric drive:

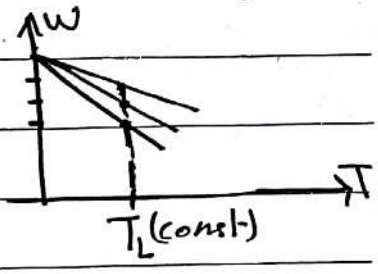
- DC Drives

what is speed control?

1) shift in s.s operating point (ω_1) due to load disturbance



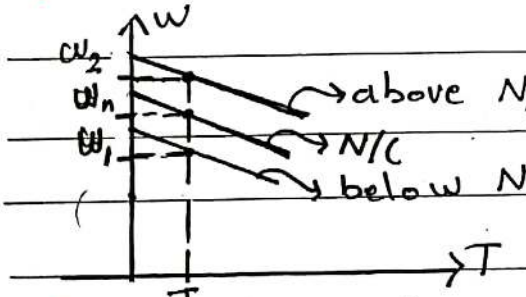
2) change in operating speed at the same load



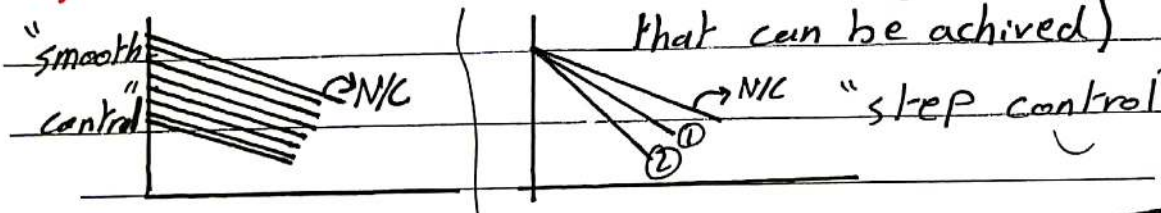
speed control: a shift or transition from one T/s char to another T/s char at the same load.

* Factor effecting speed control:

1) direction of speed control (position of new char)



2) Smoothness of control (number of regulated char that can be achived)



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3) Range of speed control

$R = \frac{W_{max}}{W_{min}}$, $W_{min} \leq W \leq W_{max}$

W_{max} : depeneds on the method of speed control

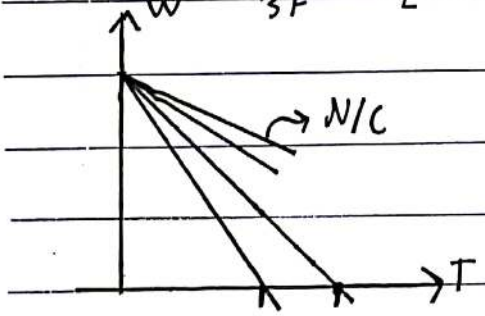
↳ For case of "below N/C" $\Rightarrow W_{max} = W_n$

↳ For case of "above N/C" $\Rightarrow$ • theoretically : $W_{max} \to \infty$

• practically : $W_{max} \approx 2W_n$

W_{min} : defined by holding point (T_L)

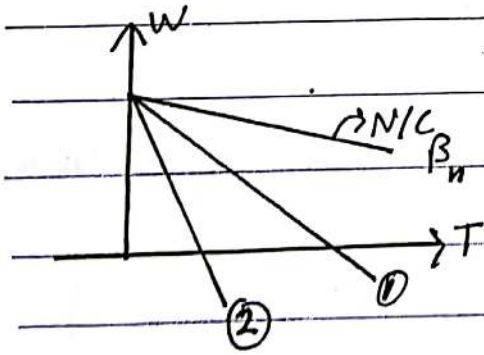
$T_{st} = T_L \Rightarrow W = 0$



if ($T_L > T_{st}$) $T_{st} = T_L$

motor ⇓ does not start

4) steady state stability



① $\Rightarrow \beta_1 = \frac{\Delta T}{\Delta W} < \beta_n$

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5) efficiency (η)

6) cost, maintanance, size ... etc

* speed control methods :-

1) Mechanical speed control method:
using gears ⇒ most commonly used because of:
1- gears are additional load on motors.
2- less reliability.
3- large size.
4- inaccurate.
5- step control.

2) Electrical speed control methods:

* For sep DC motor

- $w = \frac{V_a}{K\phi} - \frac{I_a R_a}{K\phi} = w_o - \Delta w$ (electro-mech)

- $w = \frac{V_a}{K\phi} - \frac{T R_a}{(K\phi)^2} = w_o - \Delta w$ (mech or T/S)

$\beta = \frac{\Delta T}{\Delta w}$

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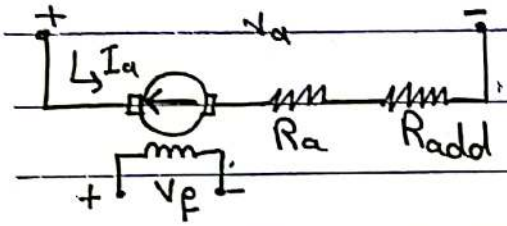
SUBJECT Speed control

SECTION speed control of sep DC motor

At $T_L = T$ = constant (S.S operation) we can chang: ① total resistance in armature circuit.
② terminal (armature) voltage (V_a).
③ magnetic flux (ϕ)

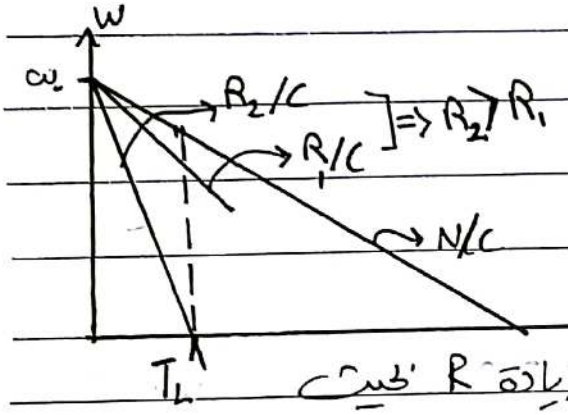
① Radd (increase R_E of arm circuit)



من المعادلات السابقة لمحرك sep نستنتج أنه بزيادة R على التوالي مع محاورة الشتغل:-

$$\omega_{N/C} = \omega_{(R/C)}$$

char is below N/C & B ⟸ تقل السرعة ⟸ $\Delta W_{(R/C)} > \Delta W_{(N/C)}$



$\downarrow I_{st}$ & T_{st}

$R = \frac{W_{max}}{W_{min}} = \frac{W_n}{W_R}$

ϕ is constant

في حال زيادة R كبير

يصبح $T_{st} < T_L$ ⟸ لا يقلع المحرك

this method have low cost, easy to use, there's an increas of power loss due (R).

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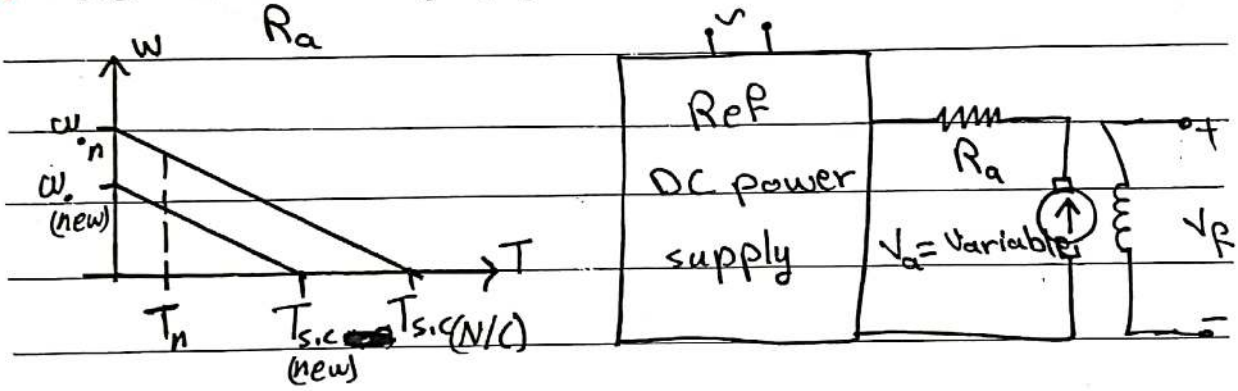
② $V_a < V_{an}$ (decrease V_a)

with $K\phi$ constant & Radd = zero $\Rightarrow R_\Sigma = R_a$

$V_a > V_{an} \Rightarrow$ خطر

- $\omega_o = \frac{V_{an}}{(K\phi) \rightarrow const} \Leftrightarrow (V_a \downarrow \Rightarrow \omega_o \downarrow)$
- $\Delta\omega = \frac{I_a R_a}{K\phi} \Rightarrow$ no change $\Delta\omega_{N/C} = \Delta\omega_{new} \Rightarrow \beta_{N/C} = \beta_{new}$
- $I_{s.c} = \frac{V_a}{R_a} \Leftrightarrow (V_a \downarrow \Rightarrow I_{s.c} \downarrow)$
- $T_{s.c} = \frac{V_a K\phi}{R_a} \Leftrightarrow (V_a \downarrow \Rightarrow T_{s.c} \downarrow)$



③ $\phi < \phi_n$ with $V_a = V_{an}$ (const) & $R_{a\Sigma} = R_a$ ($R_{add} = 0$)

عكس ← $R_f \neq 0$ اضافة R على التوالي مع ملفات التيار مع صغر، المقاومة $\Rightarrow \phi < \phi_n$

على V_f ثابت.

عكس ← V_f تخفض $(V_f < V_{fn}) \Rightarrow \phi < \phi_n$

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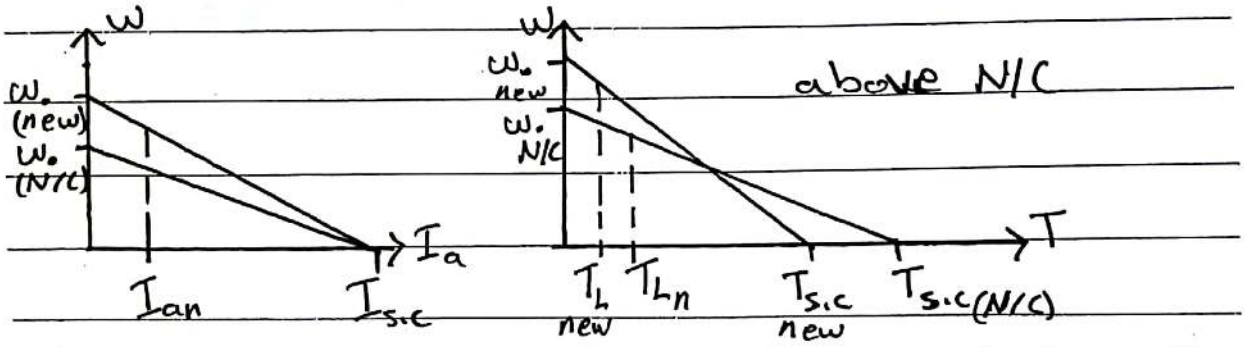
SUBJECT Speed control

SECTION speed control of sep DC motor

- $\omega_. = \frac{V_a}{K\phi} \Rightarrow (\phi\downarrow \rightarrow \omega_.\uparrow)$
- $I_{s.c} = \frac{V_a}{R_a} \Rightarrow I_{s.c}{}_{N/C} = I_{s.c}{}_{new}$
- $T_{s.c} = \frac{V_a K\phi}{R_a} \Rightarrow (\phi\downarrow \rightarrow T_{s.c}\downarrow)$
- $\Delta\omega\downarrow \Rightarrow \beta_{new} < \beta_{N/C} \Rightarrow$ disadvantage



$\omega_{.N/C} T_{Ln} = (\omega_. T_L)_{new} = constant$ القدرة ثابتة

$\phi\downarrow$ by 20% $\Rightarrow \phi_2 = 80\%\,\phi_1$

$\frac{\omega_{.2}}{\omega_{.1}} = \frac{\phi_1}{\phi_2} \Rightarrow \omega_{.2} = \frac{1}{0.8}\omega_{.1}$, $\omega_. \propto \frac{1}{\phi}$

$\frac{\Delta\omega_a}{\Delta\omega_1} = \left(\frac{\phi_1}{\phi_2}\right)^2 \Rightarrow \Delta\omega_2 = 1.5625\,\Delta\omega_1$, $\Delta\omega \propto \frac{1}{(\phi)^2}$

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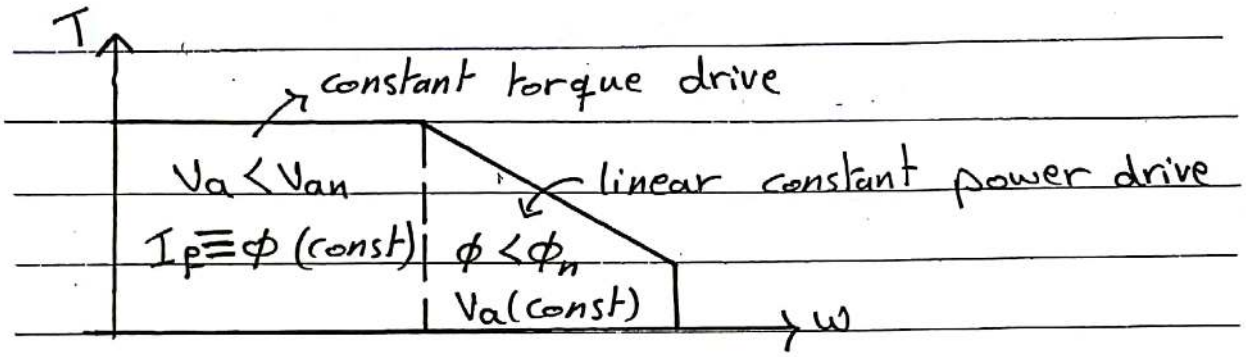
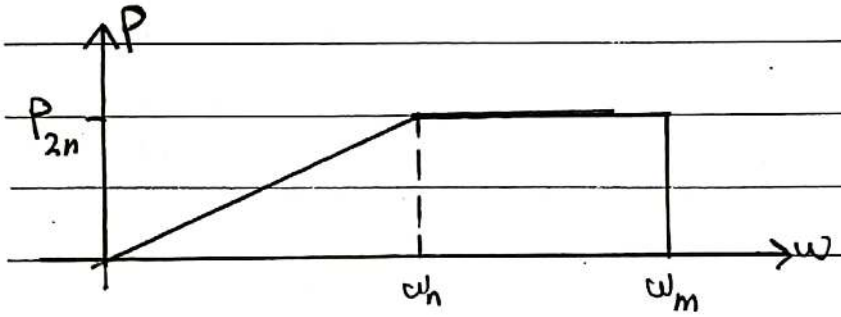
④ Combained

For the same motor, speed can be controlled by :

* V_a = Variable ($V_a < V_{an}$) ⇒ below N/C
* $I_f \equiv \phi$ = Variable ($\phi < \phi_n$) ⇒ above N/C





* $T_{max} = I_{max} K\phi_{max}$

$\omega_n < \omega \leq \omega_{max}$, $\phi \leq \phi_n$

$P_2 = \omega_1 T \Rightarrow P_2 = P(\omega)$

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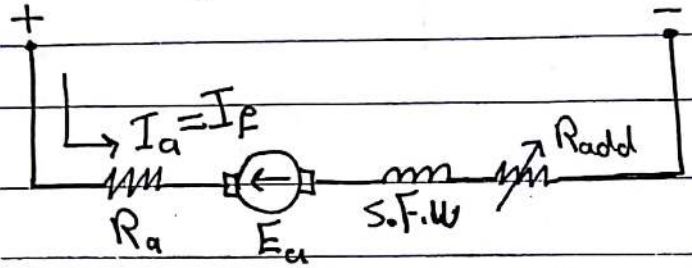
SUBJECT Speed control

SECTION Speed control of series DC motor

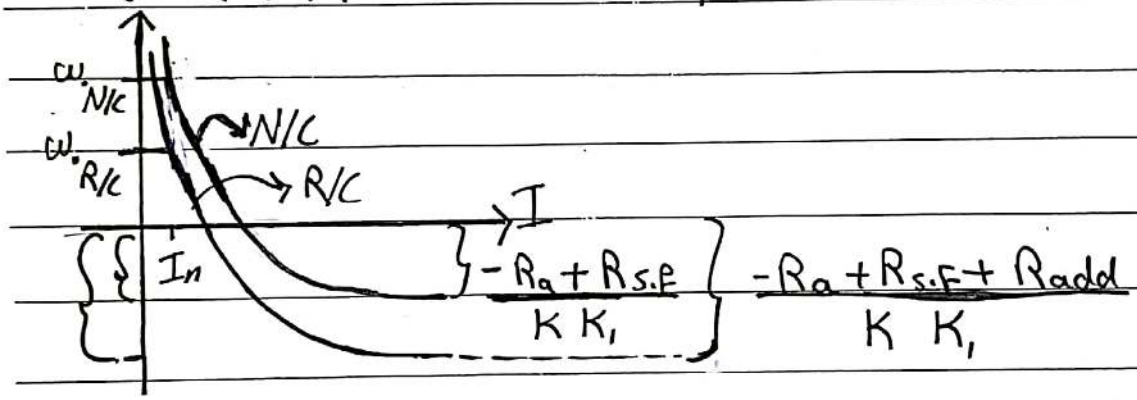
* speed control for DC series motors:

① adding (R_{add}) to the arm circuit



$$w = \frac{V_{an}}{K.K_1.I_a} - \frac{R_a + R_{s.F} + R_{add}}{K.K_1}$$

$$w = \frac{V_{an}}{\sqrt{K.K_1.T}} - \frac{R_a + R_{s.F} + R_{add}}{K.K_1}$$



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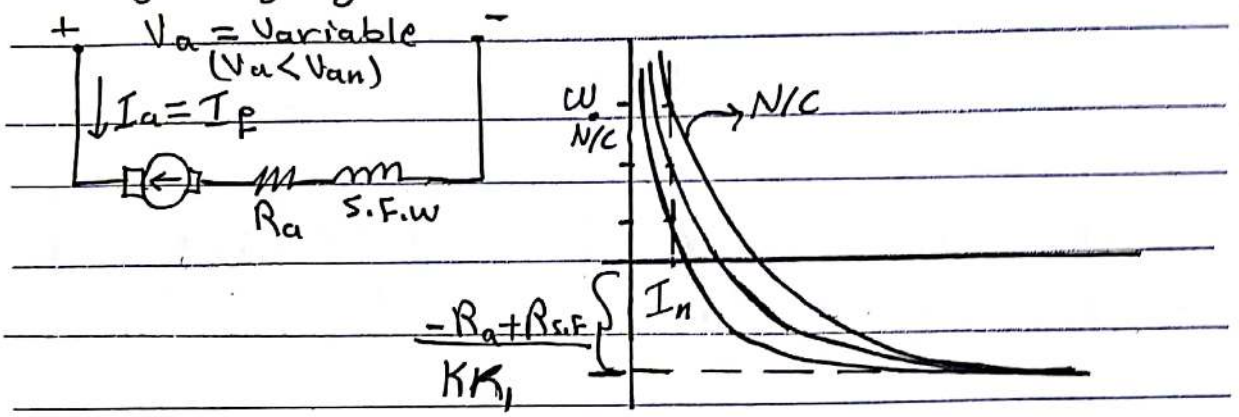
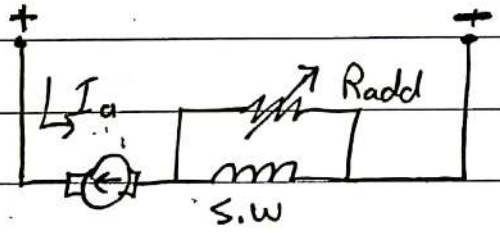
② by varying V_a



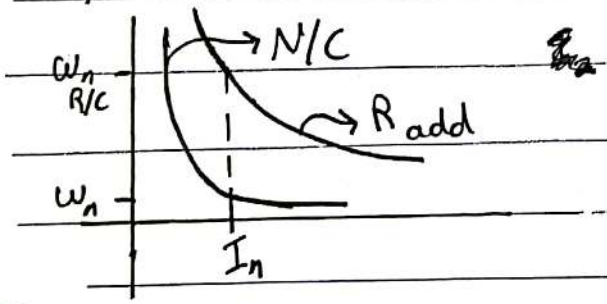
③ by varying $\phi \Rightarrow$ through adding R in parallel to the series Field winding (S.F.W) $\Rightarrow \phi < \phi_n$



$I_a = I_{S.W} + I_{R_{add}} \Rightarrow I_{S.F} \neq I_a$ but $I_{S.F} < I_a$

$\Rightarrow \phi < \phi_n$

$$R_{a\Sigma\,(eq)} = \left(R_a + \frac{R_{S.F} \,//\, R_{add}}{R_{S.F} + R_{add}}\right)_{R/C} < (R_a + R_{S.F})_{N/C} \Rightarrow \beta \uparrow$$



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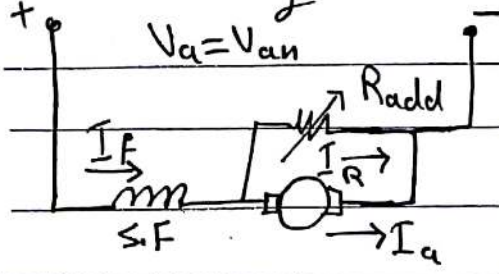
④ by adding Radd in parallel to the armature winding :-

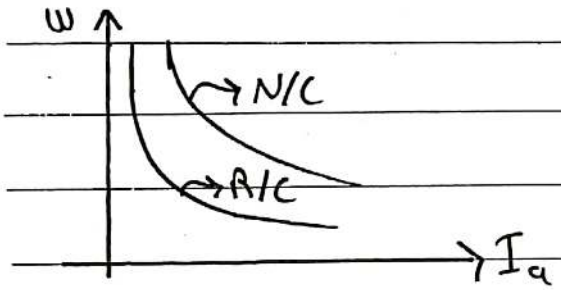


$I_F = I_a + I_R$

if I_a const ⇒ when Radd ↓ W ← ↑ ϕ ← ↑ I_F

$$\left(R_{eq}_{R/C} = R_{S.F} + \frac{R_a}{1 + \frac{R_a}{R_{add}}}\right) < \left(R_{eq}_{N/C} = R_{S.F} + R_a\right)$$

ΔW ↓ ⇒ β ↑



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SUBJECT Speed control

SECTION speed control of 3phase IM

* Speed control of 3 phase induction motor(IM):

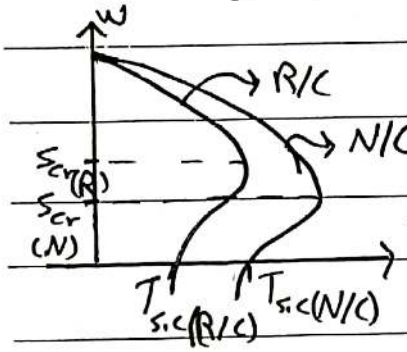
$$T = \frac{3 V_{ph}^2 \, R_2'/s}{\omega_s \left(\left(R_1 + \frac{R_2'}{s}\right)^2 + (X_1 + X_2')^2 \right)}$$

① By effecting the stator circuit (no of poles)
(R_1 add ↑, X_1 add ↑, V_{1ph} ↓, f_1 = var, P = var)

② By effecting the rotor circuit
R_2' add ↑, X_2' add ↑, slip recovery control s)

* Commonly used methods:

1) Inserting R_1 add (to stator)



* ineffecient control
* step control
* vary narrow rang & speed control ⇒ η ↓ ⇒ ↑ cu-cu loss

2) Inserting X_1 add (to stator)
has the same effect & char, but power losses are reduced ↓.

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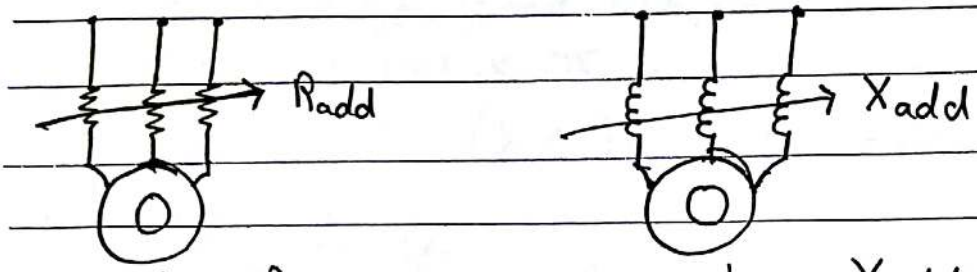
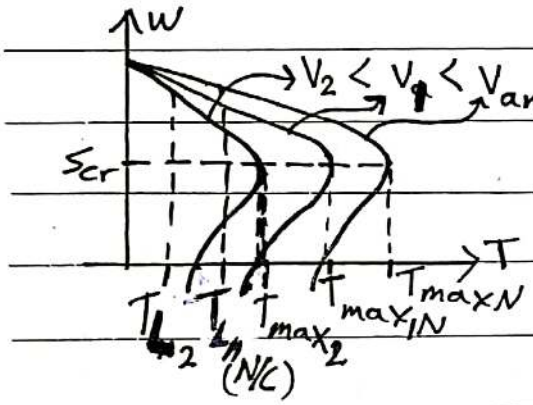


inserting Radd inserting Xadd

3) Reducing $V_a \Rightarrow (V_a < V_{an})$
using autotransformer or power semiconductor devices (inverters):



- $\lambda = const = \frac{T_{max}}{T_n}$
- $T_{max} \downarrow \Rightarrow T_{L_n} \downarrow$

- this method is used for smoth control.
- has good effeciency (η)
- β is reduced, because S (slip) increases due to
- has limited rang of speed.

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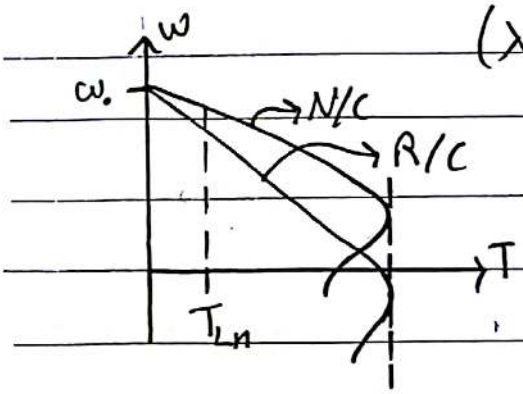
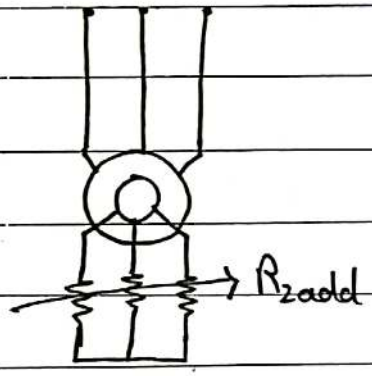
4) By inserting R_{2add} (to rotor)
used in wound rotor



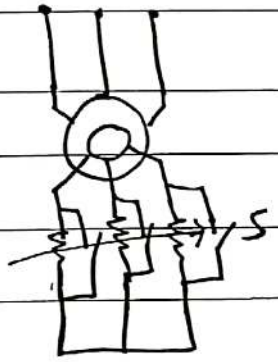
(λ & T_{max}) are const $\Rightarrow T_{Ln}$ const

$S_{cr} \uparrow \Rightarrow$ slip losses $\uparrow \Rightarrow$ ineff-icient (in rotor circuit)

$\Delta w \uparrow \Rightarrow w \downarrow$ & BL



To improve this method we use switches on the rotor & we control them by the duty cycle



$$\text{duty cycle} = \frac{T_{on}}{T_{on}+T_{off}}$$

$T_{on} = V_{ar}$

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5) By changing no of poles $\Rightarrow (\omega_s = \text{variable})$

$$\left.\begin{aligned} \omega_s &= \frac{2\pi f_1}{P/2} \\ n_s &= \frac{60 f_1}{P/2} \end{aligned}\right\} \Rightarrow \omega_s \,\&\, n_s \propto \frac{1}{P}$$

* used methods to chang no of poles:

a) 2-speed stator windings by changing the connection of stator windings (λ or Δ)

chang in P only by 2 or $\frac{1}{2}$

$\Downarrow$ $6 \rightarrow 12$ $\quad$ $\Downarrow$ $12 \rightarrow 6$

$\omega_s = 2\omega_{sn}$ or $\omega_s = \frac{1}{2}\omega_{sn}$

b) 2-connection

* Connection for constant power control: ($P_2 \approx$ const)

($\lambda\lambda \rightarrow \Delta$ or $\Delta \rightarrow \lambda\lambda$)

* Connection for constant torque control:

($\lambda \rightarrow \lambda\lambda$ or $\lambda\lambda \rightarrow \lambda$)

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$*(\Delta \rightarrow \lambda\lambda) \Rightarrow \#P \downarrow \Rightarrow \omega_s \uparrow$

(number of poles $\equiv$ no of P $\equiv \#P$)

$$\omega_{s\lambda\lambda} = 2\omega_{s\Delta}$$

$$\frac{T_{max\Delta}}{T_{max\lambda\lambda}} = \frac{3V_{ph\Delta}^2 / 2\omega_{s\Delta} X_{sc\Delta}}{3V_{ph\lambda\lambda}^2 / 2\omega_{s\lambda\lambda} X_{sc\lambda\lambda}} = \frac{V_{ph\Delta}}{V_{ph\lambda\lambda}} = 3$$

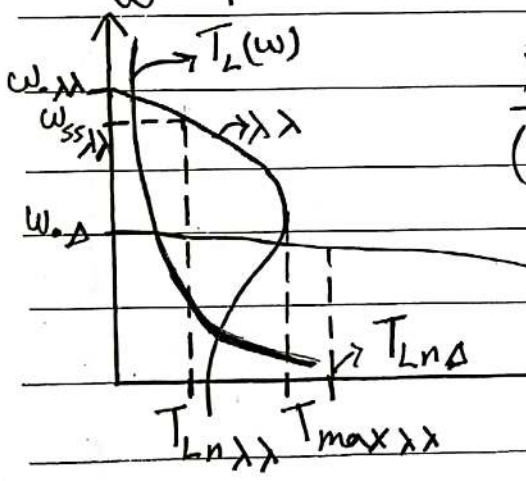
for λ (const) $\Rightarrow \frac{T_{max\Delta}}{T_{Ln\Delta}} = \frac{T_{max\lambda\lambda}}{T_{Ln\lambda\lambda}}$

$$T_{Ln\lambda\lambda} = \frac{T_{max\lambda\lambda}}{T_{max\Delta}} T_{Ln\Delta}$$

$$T_{Ln\lambda\lambda} = \frac{1}{3} T_{Ln\Delta}$$

$P_2 = \omega T_L = \text{const}$

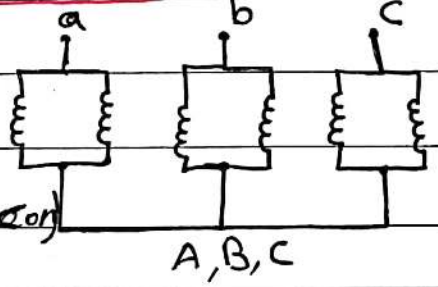
$(T_L \alpha \frac{1}{\omega})$

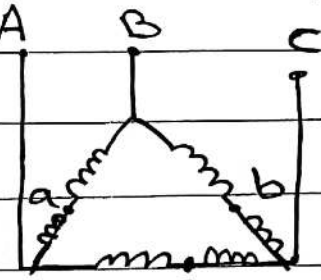


P = 6

$\lambda\lambda$ connection

(High speed con)





(low speed con) Δ connection

P = 12

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SUBJECT speed control

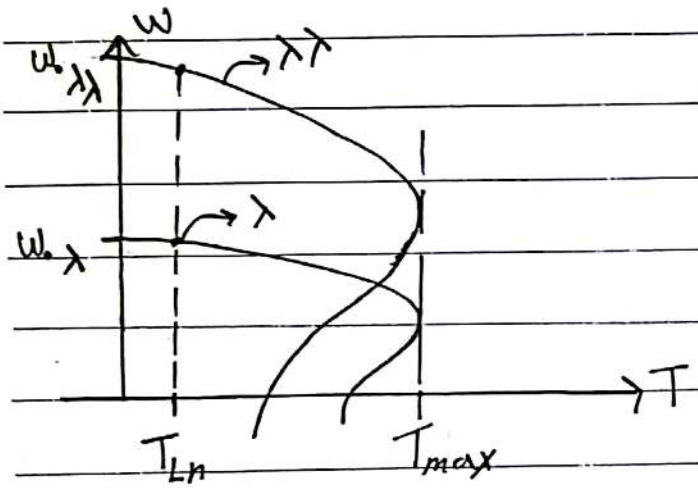
SECTION speed control of 3-phase IM

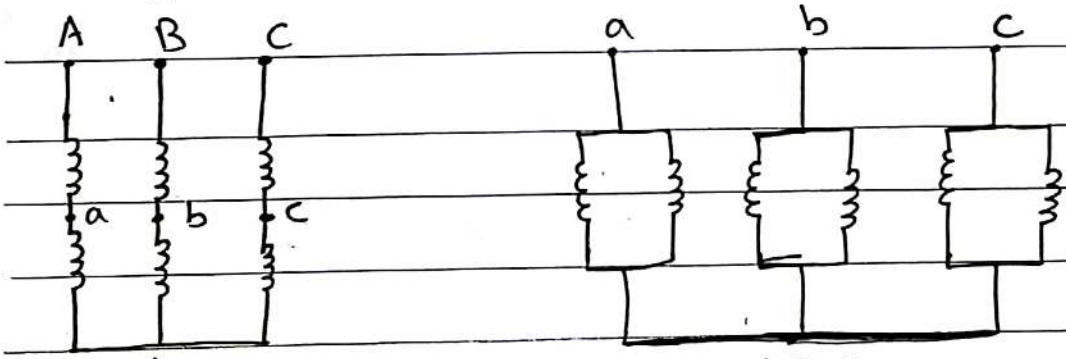
$* (\lambda \longrightarrow \lambda\lambda) \Rightarrow \# P \downarrow \Rightarrow \omega_s \uparrow \equiv \boxed{\omega_{s\lambda\lambda} = 2\omega_{s\lambda}}$

$$\frac{T_{max\lambda}}{T_{max\lambda\lambda}} = \frac{V^2_{ph\lambda} / 2\omega_{s\lambda} X_{sc\lambda}}{V^2_{ph\lambda\lambda} / 2\omega_{s\lambda\lambda} X_{sc\lambda\lambda}} = 1$$

so $\Rightarrow \boxed{T_{max\lambda} = T_{max\lambda\lambda}}$

λ & T_{max} (are const) $\Rightarrow \boxed{T_{Ln}\text{ (constant)}}$





λ
$P = 12$
low speed connection

ABC
$\lambda\lambda$
$P = 6$
High speed con

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6) By Frequency (F) - voltage (Va) control :

- $e_1 = w \frac{d\phi}{dt}$ (قوة دافعة كهربائية) ... (1)
- $\phi = \phi_{max} \sin(wt)$ (التغير اللحظي) ... (2)
- $w = 2\pi f_1$ ... (3)

sub 3 & 2 in 1 $\Rightarrow e_1 = 2\pi f_1 \phi_{max} w \cos wt$

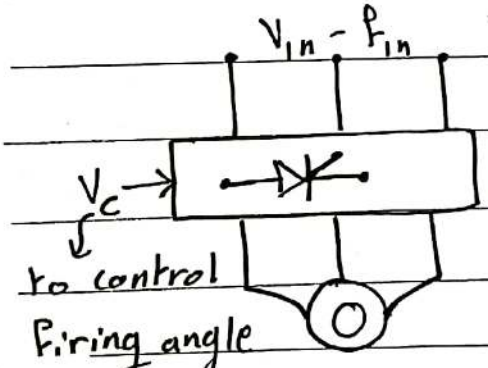
$E_1 = 4.44 \phi_{max} f_1 w$

$E_1 = K \phi f_1$

$V_1 = E_1 + R_1 I_1$

$R_1 \approx 0 \Rightarrow V_1 = E_1 = K \phi f_1 \Rightarrow V_1 = f(f_1)$

↳ (A) V (var) & f (const) = f_{in} } limited application (disadvantage)
↳ (B) V (const) = V_{an} & f (var). }
↳ (C) V (var) & f (var) } widely used



using static converter
(power electronic devices)

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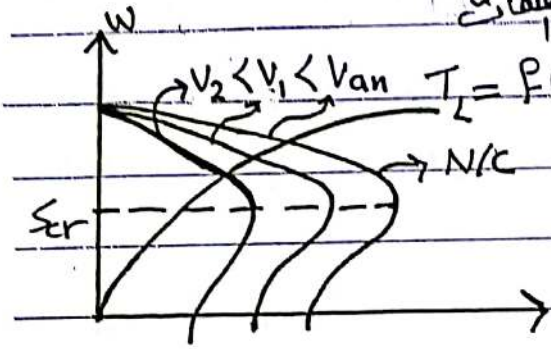
SUBJECT speed control

SECTION speed control of 3-phase IM

(A) V (var) & f- (const) = f_{in}

Variable voltage control (VVC)

- $V \uparrow > V_{.n} \Rightarrow \phi \uparrow \Rightarrow$ خطر مغنطة و إشباع
- $V \downarrow \leq V_{an} \Rightarrow \phi \downarrow \Rightarrow T \downarrow$ اذا كان عزم الحمل ثابت يؤدي لزيادة التحميل وارتفاع حرارة الملفات



$V \leq V_{.n}$ & $T \propto V_{1ph}^2$

(B) V (const) = V_{an} & f- (var)

Variable frequency control (VFC)

$f_{min} \leq f_1 \leq f_{max}$

$f_{min} \approx (5-10)\% f_{in}$

$f_{max} \approx 200\% f_{in}$ ⇒ because usually there's no speed faster than 2*(nominal speed) (for mech considerations)

$f_1 \propto \frac{1}{\phi}$

[sam results as (A)]

$f_1 \uparrow \rightarrow w_{.} \uparrow \rightarrow X_{s.c} \uparrow \rightarrow I'_2 \downarrow \rightarrow T \downarrow \rightarrow T_{max} \downarrow$

$\rightarrow$ ($f_1 \propto \frac{1}{\phi}$ $\rightarrow \phi \uparrow \rightarrow$ saturation

$\phi \downarrow \rightarrow T \downarrow \rightarrow$ over loading (over heating)

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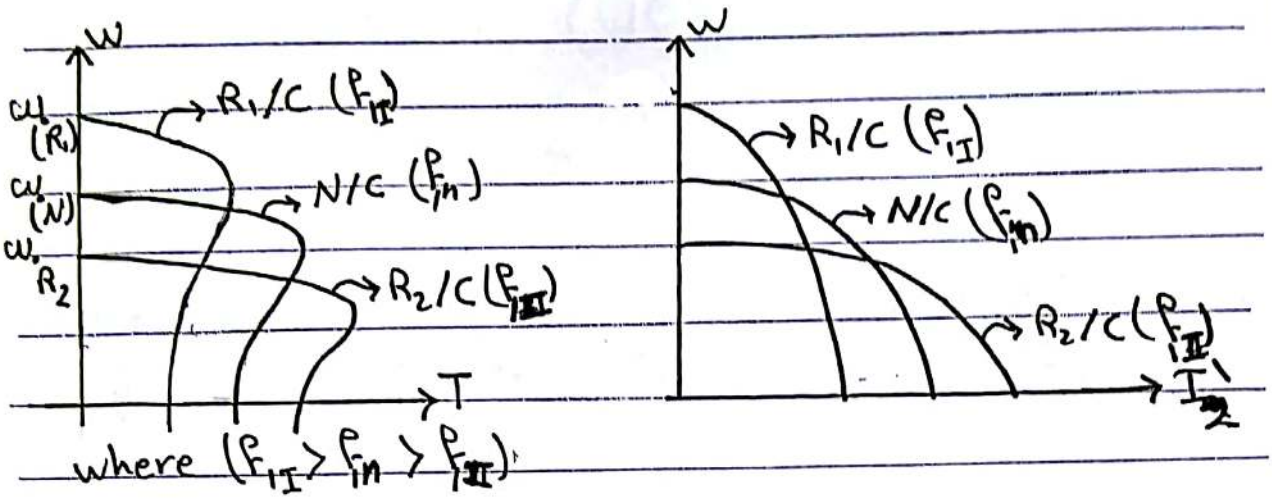


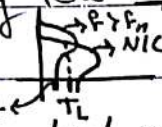
where $(f_{1I} > f_{in} > f_{1II})$

* For $f_1 > f_{in}$ (very high frequency)

$f_1 \propto w_s \uparrow \rightarrow T_{max} \downarrow$ & $T_{s.c} \downarrow$ & $S_{cr} \downarrow$ & $I'_{2s.c} \downarrow$

if T_L const & T_{st} is very low $\Rightarrow$ motor will not be able to start

$(T_{st} < T_L)$



*used for partially loaded motors (not fully loaded)

* For $f_1 < f_{in}$ (very low frequency)

its limitation is that the current is very high causing an overheat.

* Solution is $\Rightarrow$ VVVF control using V/f control principle

V/f: variation of voltage & frequency at the same time of a certain relationship depending on type of load.

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ⓒ V(var) & f(var) (V/f control) :

$\lambda = \text{const}$

$$\lambda = \frac{T_{max}}{T_L} = \frac{3V_{ph}^2}{2\omega_s\left(R_1 + \sqrt{(R_1^2 + X_{s.c}^2)}\right)} \cdot \frac{1}{T_L}$$

$R = 0 \Rightarrow \lambda = \frac{3V_{ph}^2}{2\omega_s X_{s.c}} \cdot \frac{1}{T_L}$, $\omega_s = \frac{4\pi f}{p}$, & $X_{s.c} = 2\pi f_1 L_{s.c}$

$$\lambda = \frac{3V_{ph}^2}{2\left(\frac{4\pi f_1}{p}\right)(2\pi f_1 L_{s.c})} \cdot \frac{1}{T_L}$$

$$\lambda = A\,\frac{V_{1ph}^2}{f_1^2 T_L} \quad , A = \text{is all constants}$$

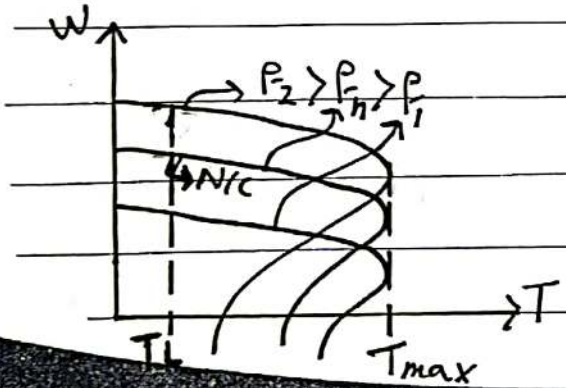
$$\lambda(\text{const}) \Rightarrow A\frac{V_{1ph}^2}{f_1^2 T_L} = A\frac{V_{1ph(n)}^2}{f_{1n}^2 T_{Ln}} \Rightarrow \boxed{\frac{V_1}{f_1} = \frac{V_1}{f_{1n}}\sqrt{\frac{T_L}{T_{Ln}}}}$$

$$T_L = T_{Ln}\left(\frac{\omega}{\omega_n}\right)^x$$

1) $x = 0 \Rightarrow T_L = T_{Ln} \Rightarrow \boxed{\frac{V_1}{f_1} = \frac{V_{1n}}{f_{1n}} \Rightarrow \text{const}}$

For const load drive

في نقل البضائع والركاب



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2) $x=-1 \Rightarrow T_L = T_{Ln}\frac{\omega_n}{\omega} \Rightarrow T \propto \frac{1}{\omega}$

(آليات الانتاج)

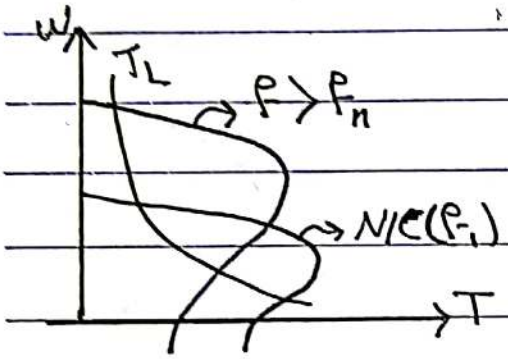
عزم الحمل عكسي مع السرعة

(const power drive)

$$\Rightarrow \frac{V_1}{f_1} = \frac{V_{1n}}{f_{1n}}\sqrt{\frac{T_L}{T_{Ln}}}$$

$$\frac{V_1}{f_1} = \frac{V_{1n}}{f_{1n}}\sqrt{\frac{\omega_n}{\omega}} = \frac{V_{1n}}{f_{1n}}\sqrt{\frac{f_{1n}}{f_1}} \Rightarrow \frac{V_1}{f_1} = \frac{V_{1n}}{\sqrt{f_1 f_{1n}}} \Rightarrow$$

$$\Rightarrow \boxed{\frac{V_1}{\sqrt{f_1}} = \frac{V_{1n}}{\sqrt{f_{1n}}} = \text{const}}$$



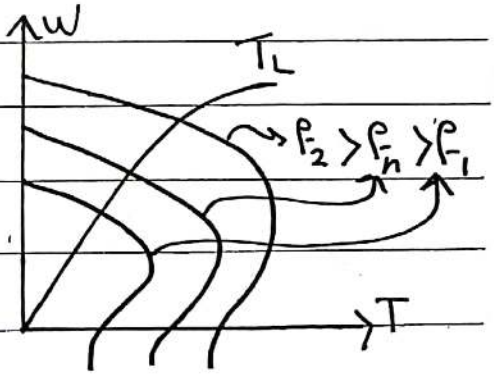
3) $x=2 \Rightarrow T_L = T_n\left(\frac{\omega}{\omega_n}\right)^2 \Rightarrow T_L \propto \omega^2$

عزم الحمل طردي مع مربع السرعة

المراوح والمضخات

$$\Rightarrow \frac{V_1}{f_1} = \frac{V_{1n}}{f_{1n}} * \frac{\omega}{\omega_n} = \frac{V_{1n}}{f_{1n}}\frac{f_1}{f_{1n}}$$

$$\Rightarrow \frac{V_1}{V_{1n}} = \frac{f_1^2}{f_n^2} \Rightarrow \boxed{\frac{V_1}{f_1} = \text{const}}$$



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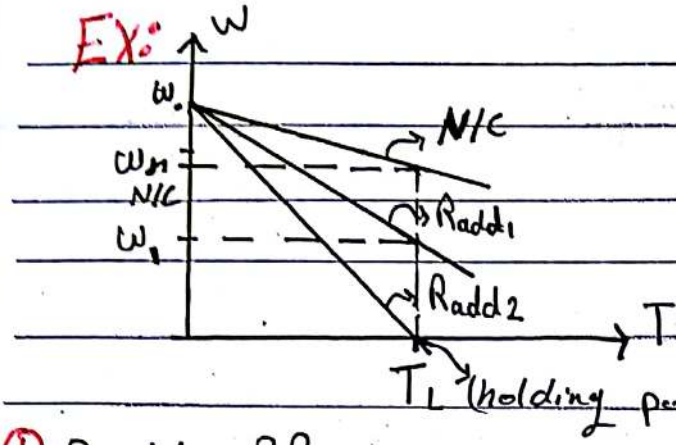
Speed control of 3-phase IM

Ex: 

T_L (holding point) $\Rightarrow T_L = T_{st}$

① $R_{add_1} = ??$
$(w_n \rightarrow w_1)$

② At holding point $R_{add_2} = ?$

1) $w_1 = \frac{V_{an}}{(K\phi)_n} - T_L \frac{R_a + R_{add_1}}{(K\phi)_n^2} \Rightarrow R_{add_1} =$

2) $w = 0$
$w_1 = \Delta w$
$\Rightarrow \frac{V_{an}}{(K\phi)_n} = T_L \frac{R_a + R_{add_2}}{(K\phi)_n^2} \Rightarrow R_{add_2} =$

Ex: DC shunt motor has
150 V, 1200 rpm, $R_a = 1\,\Omega$, $R_f = 150\,\Omega$
T_L = constant
$I = 10$ A
Find: ① $R_{add} = ?$ when $w_2 = 50\%\, w_n$
② $\Delta P_{rot} = 100$ W, $\eta_n = ?$, $\eta_2 = ?$

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Speed control of 3-phase IM

Sol:-

1) $I_a = I - I_f$, $I_f = \frac{V_a}{R_f}$

$I_{am} = 10 - \frac{150}{150} = 9\ A$

$E_n = (K\phi)_n \omega_n = V_{an} - I_{an} R_a$

$E_2 = (K\phi)_n \omega_2 = V_{an} - I_{a_2}(R_2 + R_{add})$

$\frac{E_n}{E_2} = \frac{\omega_n}{\omega_2} = \frac{n_n}{n_2} = \frac{2}{1} = \frac{150 - (9*1)}{150 - 9(1 + R_{add})}$

$R_{add} = 7.83\ \Omega$

2) $P_{in} = V_a I = 150 * 10 = 1500\ W$

$\Sigma \Delta P = \Delta P_f + \Delta P_{arm} + \Delta P_{rot}$

$\Sigma \Delta P_n = I_f^2 R_f + I_{an}^2 R_n + 100 = 331\ W$

$\Sigma \Delta P_2 = 150 + 9^2(1 + 7.83) + 100 = 965.23\ W$

$\eta_n = \frac{P_{in} - \Sigma \Delta P_n}{P_{in}} = 77.93\ \%$

$\eta_2 = \frac{P_{in} - \Sigma \Delta P_2}{P_{in}} = 35.65\ \%$

R_{add} for holding condition : $R_{add} = \frac{V_{an}}{I_{an}} - R_a = 15.67\ \Omega$

Don't miss the change.. Be part of it !

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